Consider the y-momentum equation for two-dimensional, steady flow, obtained from Equation (15.19b) as (neglecting the normal stress τ_{yy})

$$\rho u\frac{\partial v}{\partial x}+\rho v\frac{\partial v}{\partial y}=-\frac{\partial p}{\partial y}+\frac{\partial}{\partial x}\left[\mu\left(\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\right)\right] \tag{17.24}$$

In terms of the nondimensional variables, Equation (17.24) becomes

$$\rho' u'\frac{\partial v'}{\partial x'}+\rho' v'\frac{\partial v'}{\partial y'}=-\frac{1}{\gamma M_\infty^2}\frac{\partial p'}{\partial y'}+\frac{1}{\mathrm{Re}_\infty}\frac{\partial}{\partial x'}\left[\mu'\left(\frac{\partial v'}{\partial x'}+\frac{\partial u'}{\partial y'}\right)\right] \tag{17.25}$$

The order-of-magnitude equation for Equation (17.25) is

$$O(\delta)+O(\delta)=-\frac{1}{\gamma M_\infty^2}\frac{\partial p'}{\partial y'}+O(\delta^2)\left[O(\delta)+O\left(\frac{1}{\delta}\right)\right] \tag{17.26}$$

From Equation (17.26), we see that $\partial p'/\partial y' = O(\delta)$ or smaller, assuming that $\gamma M_\infty^2 = O(1)$. Since δ is very small, this implies that $\partial p'/\partial y'$ is very small. Therefore, from the y-momentum equation specialized to a boundary layer, we have

$$\frac{\partial p}{\partial y}=0 \tag{17.26a}$$

Equation (17.26a) is important; it states that at a given x station, *the pressure is constant through the boundary layer in a direction normal to the surface*. This implies that the pressure distribution at the outer edge of the boundary layer is impressed directly to the surface without change. Hence, throughout the boundary layer, $p = p(x) = p_e(x)$.

It is interesting to note that if M_∞^2 is very large, as in the case of large hypersonic Mach numbers, then from Equation (17.26) $\partial p'/\partial y'$ does not have to be small. For example, if M_∞ were large enough such that $1/\gamma M_\infty^2 = O(\delta)$, then $\partial p'/\partial y'$ could be as large as $O(1)$, and Equation (17.26) would still be satisfied. Thus, for very large hypersonic Mach numbers, the assumption that p is constant in the normal direction through a boundary layer is not always valid.

Consider the general energy equation given by Equation (15.26). The nondimensional form of this equation for two-dimensional, steady flow is given in Equation (15.33). Inserting $e = h - p/\rho$ into this equation, subtracting the momentum equation multiplied by velocity, and performing an order-of-magnitude analysis similar to those above, we can obtain the boundary-layer energy equation as

$$\rho u\frac{\partial h}{\partial x}+\rho v\frac{\partial h}{\partial y}=\frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right)+u\frac{\partial p}{\partial x}+\mu\left(\frac{\partial u}{\partial y}\right)^2 \tag{17.27}$$

The details are left to you.

In summary, by making the combined assumptions of $\delta \ll c$ and $\mathrm{Re} \geq 1/\delta^2$, the complete Navier-Stokes equations derived in Chapter 15 can be reduced to

simpler forms which apply to a boundary layer. These boundary-layer equations are

$$\boxed{\textit{Continuity:} \qquad \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0} \tag{17.28}$$

$$\boxed{x\ \textit{momentum:} \qquad \rho u \frac{\partial u}{\partial x} + \rho v \frac{\partial u}{\partial y} = -\frac{dp_e}{dx} + \frac{\partial}{\partial y}\left(\mu \frac{\partial u}{\partial y}\right)} \tag{17.29}$$

$$\boxed{y\ \textit{momentum:} \qquad \frac{\partial p}{\partial y} = 0} \tag{17.30}$$

$$\boxed{\textit{Energy:} \qquad \rho u \frac{\partial h}{\partial x} + \rho v \frac{\partial h}{\partial y} = \frac{\partial}{\partial y}\left(k \frac{\partial T}{\partial y}\right) + u \frac{dp_e}{dx} + \mu \left(\frac{\partial u}{\partial y}\right)^2} \tag{17.31}$$

Note that, as in the case of the Navier-Stokes equations, the boundary-layer equations are nonlinear. However, the boundary-layer equations are simpler, and therefore are more readily solved. Also, since $p = p_e(x)$, the pressure gradient expressed as $\partial p/\partial x$ in Equations (17.23) and (17.27) is reexpressed as dp_e/dx in Equations (17.29) and (17.31). In the above equations, the unknowns are u, v, ρ, and h; p is known from $p = p_e(x)$, and μ and k are properties of the fluid which vary with temperature. To complete the system, we have

$$p = \rho RT \tag{17.32}$$

and

$$h = c_p T \tag{17.33}$$

Hence, Equations (17.28), (17.29), and (17.31) to (17.33) are five equations for the five unknowns, u, v, ρ, T, and h.

The boundary conditions for the above equations are as follows:

At the wall: $\qquad y = 0 \quad u = 0 \quad v = 0 \quad T = T_w$

At the boundary-layer edge: $\qquad y \to \infty \quad u \to u_e \quad T \to T_e$

Note that since the boundary-layer thickness is not known a priori, the boundary condition at the edge of the boundary layer is given at large y, essentially y approaching infinity.

17.4 HOW DO WE SOLVE THE BOUNDARY-LAYER EQUATIONS?

Examine again the boundary-layer equations given by Equations (17.28) to (17.31). With these equations, are we still in the same "soup" as we are with the complete Navier-Stokes equations, in that the equations are a coupled system

of nonlinear partial differential equations for which no general analytical solution has been obtained to date? The answer is partly yes, but with a difference. Because the boundary-layer equations are simpler than the Navier-Stokes equations, especially the boundary-layer y-momentum equation, Equation (17.30), which states that at any axial location along the surface the pressure is constant in the direction normal to the surface, there is more hope of obtaining meaningful solutions for the flow inside a boundary layer. For almost one hundred years, engineers and scientists have "nudged" the boundary layer equations in many different ways, and have come up with reasonable solutions for a number of practical applications. The most complete and authoritative book on such solutions is by Hermann Schlichting (Reference 42).

In Chapters 18 and 19 we will discuss some of these solutions—their technique and some practical results. Solutions of the boundary layer equations can be classified into two groups: (1) classical solutions, some of which date back to 1908, and (2) numerical solutions obtained by modern computational fluid dynamic techniques. In the subsequent chapters, we will show examples from both groups. In addition to this subdivision based on the solution technique, boundary-layer solutions also subdivide on a *physical basis* into laminar boundary layers and turbulent boundary layers. This subdivision is natural for the reasons discussed in Section 15.2—the nature of turbulent flow is quite different than that of laminar flow. Indeed, for certain types of flow problems, *exact* solutions have been obtained for laminar boundary layers. To date, *no* exact solution has been obtained for turbulent boundary layers, because we still do not have a complete understanding of turbulence, and hence all turbulent boundary-layer solutions to date have depended on the use of some type of approximate model of turbulence. These contrasts will become more clear when you read Chapter 18, which deals with laminar boundary layers, and Chapter 19, which deals with turbulent boundary layers. You will see that laminar boundary-layer theory is well in hand, but turbulent boundary-layer theory is not. In some sense, what a pity that nature always tends toward turbulent flow, and hence the vast majority of practical boundary-layer problems deal with turbulent boundary layers—if it were the other way around, the engineering calculation of skin friction and aerodynamic heating at the surface would be much simpler and more reliable.

Finally, we note what is meant by a "boundary-layer solution." The solution of Equations (17.28) to (17.31) yields the velocity and temperature profiles throughout the boundary layer. However, the *practical* information we want is the solution for τ_w and $\dot{q}_w$, the *surface* shear stress and heat transfer, respectively. These are given by

$$\tau_w = \mu_w \left(\frac{\partial u}{\partial y} \right)_w \tag{17.34}$$

and

$$\dot{q}_w = k_w \left(\frac{\partial T}{\partial y} \right)_w \tag{17.35}$$

where the subscript w denotes conditions at the wall. *Question:* Where do values of $(\partial u/\partial y)_w$ and $(\partial T/\partial y)_w$ come from? *Answer:* From the velocity and

temperature profiles obtained from a solution of the boundary layer equations, where the profiles evaluated at the wall provide both $(\partial u/\partial y)_w$ and $(\partial T/\partial y)_w$. Hence in order to obtain the values for τ_w and $\dot{q}_w$ along the wall, which are usually considered the engineering results of most importance, we first have to solve the boundary-layer equations for the velocity and temperature profiles throughout the boundary layer, which by themselves usually are of lesser practical interest.

17.5 SUMMARY

Return to the road map given in Figure 17.2, and make certain that you feel at home with the material represented by each box. The highlights of our discussion of boundary layers are summarized as follows:

The basic quantities of interest from boundary-layer theory are the velocity and thermal boundary-layer thicknesses, δ and δ_T, respectively, the shear stress at the wall, τ_w, and heat transfer to the surface, $\dot{q}_w$. In the process, we can define two additional thicknesses: the displacement thickness

$$\delta^* \equiv \int_0^{y_1} \left(1 - \frac{\rho u}{\rho_e u_e}\right) dy \qquad \delta \leq y_1 \to \infty \tag{17.3}$$

and the momentum thickness

$$\theta \equiv \int_0^{y_1} \frac{\rho u}{\rho_e u_e} \left(1 - \frac{u}{u_e}\right) dy \qquad \delta \leq y_1 \to \infty \tag{17.10}$$

Both δ^* and θ are related to decrements in the flow due to the presence of the boundary layer; δ^* is proportional to the decrement in mass flow, and θ is proportional to the decrement in momentum flow. Moreover, δ^* is the distance away from the body surface through which the outer inviscid flow is displaced due to the boundary layer. The body shape plus δ^* defines a new effective body seen by the inviscid flow.

By an order-of-magnitude analysis, the complete Navier-Stokes equations for two-dimensional flow reduce to the following boundary-layer equations:

Continuity:
$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0 \tag{17.28}$$

x momentum:
$$\rho u \frac{\partial u}{\partial x} + \rho v \frac{\partial u}{\partial y} = -\frac{dp_e}{dx} + \frac{\partial}{\partial y}\left(\mu \frac{\partial u}{\partial y}\right) \tag{17.29}$$

y momentum:
$$\frac{\partial p}{\partial y} = 0 \tag{17.30}$$

(*continued*)

Energy: $$\rho u\frac{\partial h}{\partial x}+\rho v\frac{\partial h}{\partial y}=\frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right)+u\frac{dp_e}{dx}+\mu\left(\frac{\partial u}{\partial y}\right)^2 \tag{17.31}$$

These equations are subject to the boundary conditions:

At the wall: $y=0 \quad u=0 \quad v=0 \quad h=h_w$

At the boundary-layer edge: $y\to\infty \quad u\to u_e \quad h\to h_e$

Inherent in the above boundary-layer equations are the assumptions that $\delta \ll c$, Re is large, and M_∞ is not inordinately large.

CHAPTER 18

Laminar Boundary Layers

Lamina—A thin scale or sheet. A layer or coat lying over another.
Funk and Wagnalls Standard Desk Dictionary, 1964

PREVIEW BOX

Here we get down to the brass tacks of actually solving the boundary-layer equations and obtaining practical formulas for the calculation of skin friction and aerodynamic heating to the surface. The solution, however, depends on whether the flow in the boundary layer is laminar or turbulent. And what a big difference there is! The difference is so important that we have separate chapters for each case—the present chapter dealing with laminar boundary layers, and Chapter 19 dealing with turbulent boundary layers.

Laminar boundary layers are easier to deal with, and that is why we treat them first. The analysis of a laminar boundary layer is more theoretically "pure" than that for a turbulent boundary layer. You will feel intellectually comfortable with this chapter, so sit back and enjoy this intellectual experience.

18.1 INTRODUCTION

Within the panoply of boundary-layer analyses, the solution of *laminar* boundary layers is well in hand compared to the status for turbulent boundary layers. This chapter is exclusively devoted to laminar boundary layers; turbulent boundary layers is the subject of Chapter 19. The basic definitions of laminar and turbulent flows are discussed in Section 15.1, and some characteristics of these flows are illustrated in Figures 15.5 and 15.6; it is recommended that you review that material before progressing further.

The roadmap for this chapter is given in Figure 18.1. We will first deal with some well-established classical solutions that come under the heading of

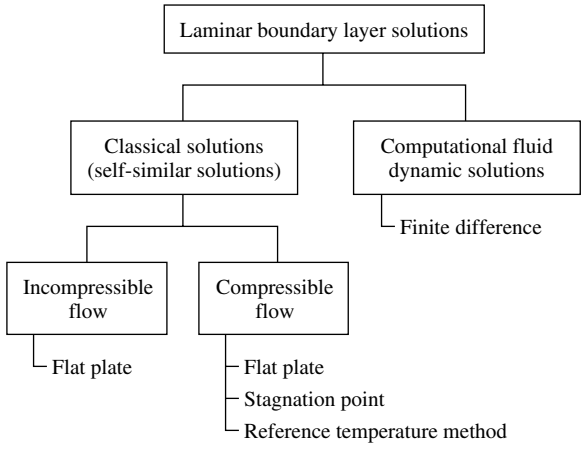


Figure 18.1 Road map for Chapter 18.

self-similar solutions, a term that is defined in Section 18.2. In this regard, we will deal with both incompressible and compressible flows over a flat plate, as noted on the left side of our roadmap in Figure 18.1. We will also discuss the boundary-layer solution in the region surrounding the stagnation point on a blunt-nosed body, because this solution gives us important information on aerodynamic heating at the stagnation point—vital information for high-speed flight vehicles. As part of the classical solution of compressible boundary layers, we will discuss the reference temperature method—a very useful engineering calculation that makes use of classical incompressible boundary-layer results to predict skin friction and aerodynamic heating for a compressible flow. Then we will move to the right side of the roadmap in Figure 18.1 and discuss some more modern computational fluid dynamic solutions to laminar boundary layers. Unlike the classical self-similar solutions, which are limited to a few (albeit important) applications such as flat plates and the stagnation region, these CFD numerical solutions deal with the laminar boundary layer over bodies of arbitrary shapes.

Note: As we progress through this chapter, we will encounter ideas and results that are already familiar to us from our discussion of Couette flow in Chapter 16. Indeed, this is one of the primary reasons for Chapter 16—to introduce these concepts within the context of a relatively straightforward flow problem before dealing with the more intricate boundary-layer solutions.

18.2 INCOMPRESSIBLE FLOW OVER A FLAT PLATE: THE BLASIUS SOLUTION

Consider the incompressible, two-dimensional flow over a flat plate at 0° angle of attack, such as sketched in Figure 17.7. For such a flow, $\rho =$ constant, $\mu =$ constant, and $dp_e/dx = 0$ (because the inviscid flow over a flat plate at $\alpha = 0$ yields a constant pressure over the surface). Moreover, recall that the energy

equation is not needed to calculate the velocity field for an incompressible flow. Hence, the boundary-layer equations, Equations (17.28) to (17.31), reduce to

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{18.1}$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \nu\frac{\partial^2 u}{\partial y^2} \tag{18.2}$$

$$\frac{\partial p}{\partial y} = 0 \tag{18.3}$$

where ν is the *kinematic viscosity,* defined as $\nu \equiv \mu/\rho$.

We now embark on a procedure that is common to many boundary-layer solutions. Let us transform the independent variables (x, y) to (ξ, η), where

$$\xi = x \quad \text{and} \quad \eta = y\sqrt{\frac{V_\infty}{\nu x}} \tag{18.4}$$

Using the chain rule, we obtain the derivatives

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial}{\partial \eta}\frac{\partial \eta}{\partial x} \tag{18.5}$$

$$\frac{\partial}{\partial y} = \frac{\partial}{\partial \xi}\frac{\partial \xi}{\partial y} + \frac{\partial}{\partial \eta}\frac{\partial \eta}{\partial y} \tag{18.6}$$

However, from Equations (18.4) we have

$$\frac{\partial \xi}{\partial x} = 1 \qquad \frac{\partial \xi}{\partial y} = 0 \qquad \frac{\partial \eta}{\partial y} = \sqrt{\frac{V_\infty}{\nu x}} \tag{18.7}$$

(We do not have to explicitly obtain $\partial\eta/\partial x$ because these terms will eventually cancel from our equations.) Substituting Equations (18.7) into (18.5) and (18.6), we have

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial \xi} + \frac{\partial \eta}{\partial x}\frac{\partial}{\partial \eta} \tag{18.8}$$

$$\frac{\partial}{\partial y} = \sqrt{\frac{V_\infty}{\nu x}}\frac{\partial}{\partial \eta} \tag{18.9}$$

$$\frac{\partial^2}{\partial y^2} = \frac{V_\infty}{\nu x}\frac{\partial^2}{\partial \eta^2} \tag{18.10}$$

Also, let us define a stream function ψ such that

$$\psi = \sqrt{\nu x V_\infty}\, f(\eta) \tag{18.11}$$

where $f(\eta)$ is strictly a function of η only. This expression for ψ identically satisfies the continuity equation, Equation (18.1); therefore, it is a physically possible stream function. [Show yourself that ψ satisfies Equation (18.1); to do this, you will have to carry out many of the same manipulations described below.]

From the definition of the stream function, and using Equations (18.8), (18.9), and (18.11), we have

$$u = \frac{\partial \psi}{\partial y} = \sqrt{\frac{V_\infty}{\nu x}} \frac{\partial \psi}{\partial \eta} = V_\infty f'(\eta) \tag{18.12}$$

$$v = -\frac{\partial \psi}{\partial x} = -\left(\frac{\partial \psi}{\partial \xi} + \frac{\partial \eta}{\partial x} \frac{\partial \psi}{\partial \eta} \right) = -\frac{1}{2} \sqrt{\frac{\nu V_\infty}{x}} f - \sqrt{\nu x V_\infty} \frac{\partial \eta}{\partial x} f' \tag{18.13}$$

Equation (18.12) is of particular note. The function $f(\eta)$ defined in Equation (18.11) has the property that its derivative f' gives the x component of velocity as

$$f'(\eta) = \frac{u}{V_\infty}$$

Substitute Equations (18.8) to (18.10), (18.12), and (18.13) into the momentum equation, Equation (18.2). Writing each term explicitly so that you can see what is happening, we have

$$V_\infty f' \left(V_\infty \frac{\partial \eta}{\partial x} f'' \right) - \left(\frac{1}{2} \sqrt{\frac{\nu V_\infty}{x}} f + \sqrt{\nu x V_\infty} \frac{\partial \eta}{\partial x} f' \right) V_\infty \sqrt{\frac{V_\infty}{\nu x}} f'' = \nu V_\infty \frac{V_\infty}{\nu x} f'''$$

Simplifying, we obtain

$$V_\infty^2 \frac{\partial \eta}{\partial x} f' f'' - \frac{1}{2} \frac{V_\infty^2}{x} f f'' - V_\infty^2 \left(\frac{\partial \eta}{\partial x} \right) f' f'' = \frac{V_\infty^2}{x} f''' \tag{18.14}$$

The first and third terms cancel, and Equation (18.14) becomes

$$\boxed{2f''' + f f'' = 0} \tag{18.15}$$

Equation (18.15) is important; it is called *Blasius' equation,* after H. Blasius, who obtained it in his Ph.D. dissertation in 1908. Blasius was a student of Prandtl, and his flat-plate solution using Equation (18.15) was the first practical application of Prandtl's boundary-layer hypothesis since its announcement in 1904. Examine Equation (18.15) closely. Amazingly enough it is an *ordinary differential equation*. Look what has happened! Starting with the partial differential equations for a flat-plate boundary layer given by Equations (18.1) to (18.3), and transforming *both* the independent and dependent variables through Equations (18.4) and (18.11), we obtain an ordinary differential equation for $f(\eta)$. In the same breath, we can say that Equation (18.15) is also an equation for the velocity u because $u = V_\infty f'(\eta)$. Because Equation (18.15) is a single ordinary differential equation, it is simpler to solve than the original boundary-layer equations. However, it is still a nonlinear equation and must be solved numerically, subject to the transformed boundary conditions,

$$\text{At } \eta = 0: \qquad f = 0,\ f' = 0$$

$$\text{At } \eta \to \infty: \qquad f' = 1$$

[Note that at the wall where $\eta = 0$, $f' = 0$ because $u = 0$, and therefore $f = 0$ from Equation (18.13) evaluated at the wall.]

Equation (18.15) is a third-order, nonlinear, ordinary differential equation; it can be solved numerically by means of standard techniques, such as the Runge-Kutta method (such as that described in Reference 52). The integration begins at the wall and is carried out in small increments Δy in the direction of increasing y away from the wall. However, since Equation (18.15) is third order, three boundary conditions must be known at $\eta = 0$; from the above, only two are specified. A third boundary condition, namely, some value for $f''(0)$, must be *assumed;* Equation (18.15) is then integrated across the boundary layer to a large value of η. The value of f' at large eta is then examined. Does it match the boundary condition at the edge of the boundary layer, namely, is $f' = 1$ satisfied at the edge of the boundary layer? If not, assume a different value of $f''(0)$ and integrate again. Repeat this process until convergence is obtained. This numerical approach is called the "shooting technique"; it is a classical approach, and its basic philosophy and details are discussed at great length in Section 16.4. Its application to Equation (18.15) is more straightforward than the discussion in Section 16.4, because here we are dealing with an incompressible flow and only one equation, namely, the momentum equation as embodied in Equation (18.15).

The solution of Equation (18.15) is plotted in Figure 18.2 in the form of $f'(\eta) = u/V_\infty$ as a function of η. Note that this curve is the *velocity profile* and that it is a function of η only. Think about this for a moment. Consider two different x stations along the plate, as shown in Figure 18.3. In general, $u = u(x, y)$, and the velocity profiles in terms of $u = u(y)$ at given x stations will be *different*. Clearly, the variation of u normal to the wall will change as the flow progresses downstream. However, when plotted versus η, we see that the profile, $u = u(\eta)$, is

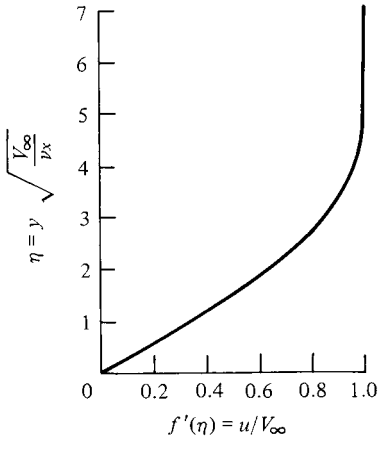


Figure 18.2 Incompressible velocity profile for a flat plate; solution of the Blasius equation.

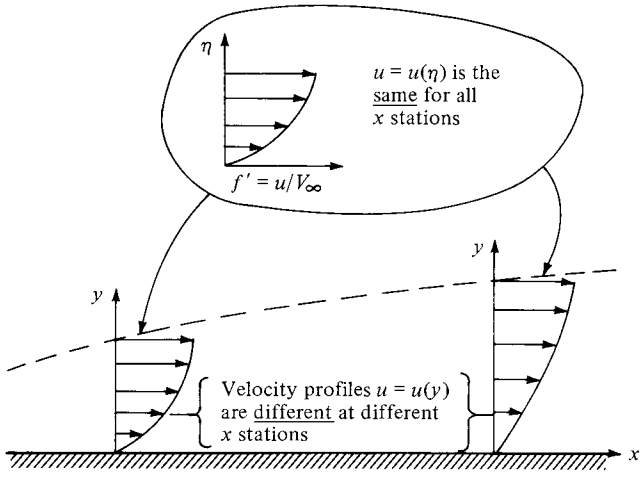


Figure 18.3 Velocity profiles in physical and transformed space, demonstrating the meaning of self-similar solutions.

the same for *all* x stations, as illustrated in Figure 18.3. This result is an example of a *self-similar solution*—solutions where the boundary-layer profiles, when plotted versus a similarity variable η are the same for all x stations. For such self-similar solutions, the governing boundary-layer equations reduce to one or more ordinary differential equations in terms of a transformed independent variable. Self-similar solutions occur only for certain special types of flows—the flow over a flat plate is one such example. In general, for the flow over an arbitrary body, the boundary-layer solutions are nonsimilar; the governing partial differential equations cannot be reduced to ordinary differential equations.

Numerical values of f, f', and f'' tabulated versus η can be found in Reference 42. Of particular interest is the value of f'' at the wall; $f''(0) = 0.332$. Consider the local skin-friction coefficient defined as $c_f = \tau_w / \frac{1}{2}\rho_\infty V_\infty^2$. From Equation (17.34), the shear stress at the wall is given by

$$\tau_w = \mu \left(\frac{\partial u}{\partial y} \right)_{y=0} \tag{18.16}$$

However, from Equations (18.9) and (18.11),

$$\frac{\partial u}{\partial y} = V_\infty \frac{\partial f'}{\partial y} = V_\infty \sqrt{\frac{V_\infty}{\nu x}} \frac{\partial f'}{\partial \eta} = V_\infty \sqrt{\frac{V_\infty}{\nu x}} f'' \tag{18.17}$$

Evaluating Equation (18.17) at the wall, where $y = \eta = 0$, we obtain

$$\left(\frac{\partial u}{\partial y} \right)_{y=0} = V_\infty \sqrt{\frac{V_\infty}{\nu x}} f''(0) \tag{18.18}$$

Combining Equations (18.16) and (18.18), we have

$$c_f = \frac{\tau_w}{\frac{1}{2}\rho_\infty V_\infty^2} = \frac{2\mu}{\rho_\infty V_\infty^2} V_\infty \sqrt{\frac{V_\infty}{\nu x}}\, f''(0) \tag{18.19}$$

$$= 2\sqrt{\frac{\mu}{\rho_\infty V_\infty x}}\, f''(0) = \frac{2f''(0)}{\sqrt{\mathrm{Re}_x}}$$

where Re_x is the local Reynolds number. Since $f''(0) = 0.332$ from the numerical solution of Equation (18.15), then Equation (18.19) yields

$$\boxed{c_f = \frac{0.664}{\sqrt{\mathrm{Re}_x}}} \tag{18.20}$$

which is a classic expression for the local skin-friction coefficient for the incompressible laminar flow over a flat plate—a result that stems directly from boundary-layer theory. Its validity has been amply verified by experiment. Note that $c_f \propto \mathrm{Re}_x^{-1/2} \propto x^{-1/2}$; that is, c_f decreases inversely proportional to the square root of distance from the leading edge. Examining the flat plate sketched in Figure 17.7, the total drag on the top surface of the entire plate is the integrated contribution of $\tau_w(x)$ from $x = 0$ to $x = c$. Letting C_f denote the skin friction drag coefficient, we obtain from Equation (1.16)

$$C_f = \frac{1}{c}\int_0^c c_f\, dx \tag{18.21}$$

Substituting Equation (18.20) into (18.21), we obtain

$$C_f = \frac{1}{c}(0.664)\sqrt{\frac{\mu}{\rho_\infty V_\infty}}\int_0^c x^{-1/2}\, dx = \frac{1.328}{c}\sqrt{\frac{\mu c}{\rho_\infty V_\infty}}$$

or

$$\boxed{C_f = \frac{1.328}{\sqrt{\mathrm{Re}_c}}} \tag{18.22}$$

where Re_c is the Reynolds number based on the total plate length c.

An examination of Figure 18.2 shows that $f' = 0.99$ at approximately $\eta = 5.0$. Hence, the boundary-layer thickness, which was defined earlier as that distance above the surface where $u = 0.99u_e$, is

$$\eta = y\sqrt{\frac{V_\infty}{\nu x}} = \delta\sqrt{\frac{V_\infty}{\nu x}} = 5.0$$

or

$$\boxed{\delta = \frac{5.0x}{\sqrt{\mathrm{Re}_x}}} \tag{18.23}$$

Note that the boundary-layer thickness is inversely proportional to the square root of the Reynolds number (based on the local distance x). Also, $\delta \propto x^{1/2}$; the

laminar boundary layer over a flat plate grows parabolically with distance from the leading edge.

The displacement thickness δ^*, defined by Equation (17.3), becomes for an incompressible flow

$$\delta^* = \int_0^{y_1} \left(1 - \frac{u}{u_e}\right) dy \tag{18.24}$$

In terms of the transformed variables f' and η given by Equations (18.4) and (18.12), the integral in Equation (18.24) can be written as

$$\delta^* = \sqrt{\frac{\nu x}{V_\infty}} \int_0^{\eta_1} [1 - f'(\eta)]\, d\eta = \sqrt{\frac{\nu x}{V_\infty}} [\eta_1 - f(\eta_1)] \tag{18.25}$$

where η_1 is an arbitrary point above the boundary layer. The numerical solution for $f(\eta)$ obtained from Equation (18.15) shows that, amazingly enough, $\eta_1 - f(\eta) = 1.72$ for all values of η above 5.0. Therefore, from Equation (18.25), we have

$$\delta^* = 1.72\sqrt{\frac{\nu x}{V_\infty}}$$

or

$$\boxed{\delta^* = \frac{1.72x}{\sqrt{\mathrm{Re}_x}}} \tag{18.26}$$

Note that, as in the case of the boundary-layer thickness itself, δ^* varies inversely with the square root of the Reynolds number, and $\delta^* \propto x^{1/2}$. Also, comparing Equations (18.23) and (18.26), we see that $\delta^* = 0.34\delta$; the displacement thickness is smaller than the boundary-layer thickness, confirming our earlier statement in Section 17.2.

The momentum thickness for an incompressible flow is, from Equation (17.10),

$$\theta = \int_0^{y_1} \frac{u}{u_e} \left(1 - \frac{u}{u_e}\right) dy$$

or in terms of our transformed variables,

$$\theta = \sqrt{\frac{\nu x}{V_\infty}} \int_0^{\eta_1} f'(1 - f')\, d\eta \tag{18.27}$$

Equation (18.27) can be integrated numerically from $\eta = 0$ to any arbitrary point $\eta_1 > 5.0$. The result gives

$$\theta = \sqrt{\frac{\eta x}{V_\infty}}(0.664)$$

or

$$\boxed{\theta = \frac{0.664x}{\sqrt{\mathrm{Re}_x}}} \tag{18.28}$$

Note that, as in the case of our previous thicknesses, θ varies inversely with the square root of the Reynolds number and that $\theta \propto x^{1/2}$. Also, $\theta = 0.39\delta^*$, and

$\theta = 0.13\delta$. Another property of momentum thickness can be demonstrated by evaluating θ at the trailing edge of the flat plate sketched in Figure 17.7. In this case, $x = c$, and from Equation (18.28), we obtain

$$\theta_{x=c} = \frac{0.664c}{\sqrt{\mathrm{Re}_c}} \tag{18.29}$$

Comparing Equations (18.22) and (18.29), we have

$$C_f = \frac{2\theta_{x=c}}{c} \tag{18.30}$$

Equation (18.30) demonstrates that the integrated skin-friction coefficient for the flat plate is directly proportional to the value of θ evaluated at the trailing edge.

18.3 COMPRESSIBLE FLOW OVER A FLAT PLATE

The properties of the incompressible, laminar, flat-plate boundary layer were developed in Section 18.2. These results hold at low Mach numbers where the density is essentially constant through the boundary layer. However, what happens to these properties at high Mach numbers where the density becomes a variable; that is, what are the compressibility effects? The purpose of the present section is to outline briefly the effects of compressibility on both the derivations and the final results for laminar flow over a flat plate. We do not intend to present much detail; rather, we examine some of the salient aspects which distinguish compressible from incompressible boundary layers.

The compressible boundary-layer equations were derived in Section 17.3, and were presented as Equations (17.28) to (17.31). For flow over a flat plate, where $dp_e/dx = 0$, these equations become

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0 \tag{18.31}$$

$$\rho u\frac{\partial u}{\partial x} + \rho v\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}\left(\mu\frac{\partial u}{\partial y}\right) \tag{18.32}$$

$$\frac{\partial p}{\partial y} = 0 \tag{18.33}$$

$$\rho u\frac{\partial h}{\partial x} + \rho v\frac{\partial h}{\partial y} = \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \mu\left(\frac{\partial u}{\partial y}\right)^2 \tag{18.34}$$

Compare these equations with those for the incompressible case given by Equations (18.1) to (18.3). Note that, for a compressible boundary layer, (1) the energy equation must be included, (2) the density is treated as a variable, and (3) in general, μ and k are functions of temperature and hence also must be treated as variables. As a result, the system of equations for the compressible case, Equations (18.31) to (18.34), is more complex than for the incompressible case, Equations (18.1) to (18.3).

It is sometimes convenient to deal with total enthalpy, $h_0 = h + V^2/2$, as the dependent variable in the energy equation, rather than the static enthalpy as given in Equation (18.34). Note that, consistent with the boundary-layer approximation, where v is small, $h_0 = h + V^2/2 = h + (u + v^2)/2 \approx h + u^2/2$. To obtain the energy equation in terms of h_0, multiply Equation (18.32) by u, and add to Equation (18.34), as follows. From Equation (18.32) multiplied by u,

$$\rho u \frac{\partial (u^2/2)}{\partial x} + \rho v \frac{\partial (u^2/2)}{\partial y} = u \frac{\partial}{\partial y} \left(\mu \frac{\partial u}{\partial y} \right) \tag{18.35}$$

Adding Equation (18.35) to (18.34), we obtain

$$\rho u \frac{\partial (h + u^2/2)}{\partial x} + \rho v \frac{\partial (h + u^2/2)}{\partial y} = \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \mu \left(\frac{\partial u}{\partial y} \right)^2 + u \frac{\partial}{\partial y} \left(\mu \frac{\partial u}{\partial y} \right) \tag{18.36}$$

Recall that for a calorically perfect gas, $dh = c_p \, dT$; hence,

$$\frac{\partial T}{\partial y} = \frac{1}{c_p} \frac{\partial h}{\partial y} = \frac{1}{c_p} \frac{\partial}{\partial y} \left(h_0 - \frac{u^2}{2} \right) \tag{18.37}$$

Substituting Equation (18.37) into (18.36), we obtain

$$\rho u \frac{\partial h_0}{\partial x} + \rho v \frac{\partial h_0}{\partial y} = \frac{\partial}{\partial y} \left[\frac{k}{c_p} \frac{\partial}{\partial y} \left(h_0 - \frac{u^2}{2} \right) \right] + \mu \left(\frac{\partial u}{\partial y} \right)^2 + u \frac{\partial}{\partial y} \left(\mu \frac{\partial u}{\partial y} \right) \tag{18.38}$$

Note that

$$\frac{k}{c_p} \frac{\partial}{\partial y} \left(h_0 - \frac{u^2}{2} \right) = \frac{\mu k}{\mu c_p} \frac{\partial}{\partial y} \left(h_0 - \frac{u^2}{2} \right) = \frac{\mu}{\text{Pr}} \left(\frac{\partial h_0}{\partial y} - u \frac{\partial u}{\partial y} \right) \tag{18.39}$$

and

$$\mu \left(\frac{\partial u}{\partial y} \right)^2 + u \frac{\partial}{\partial y} \left(\mu \frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial y} \left(\mu u \frac{\partial u}{\partial y} \right) \tag{18.40}$$

Substituting Equations (18.39) and (18.40) into (18.38), we obtain

$$\rho u \frac{\partial h_0}{\partial x} + \rho v \frac{\partial h_0}{\partial y} = \frac{\partial}{\partial y} \left[\frac{\mu}{\text{Pr}} \frac{\partial h_0}{\partial y} + \left(1 - \frac{1}{\text{Pr}} \right) \mu u \frac{\partial u}{\partial y} \right] \tag{18.41}$$

which is an alternate form of the boundary-layer energy equation. In this equation, Pr is the local Prandtl number, which, in general, is a function of T and hence varies throughout the boundary layer.

For the laminar, compressible flow over a flat plate, the system of governing equations can now be considered to be Equations (18.31) to (18.33) and (18.41). These are nonlinear partial differential equations. As in the incompressible case, let us seek a self-similar solution; however, the transformed independent variables

must be defined differently:

$$\xi = \rho_e \mu_e u_e x \qquad \xi = \xi(x)$$

$$\eta = \frac{u_e}{\sqrt{2\xi}} \int_0^y \rho \, dy \qquad \eta = \eta(x, y)$$

The dependent variables are transformed as follows:

$$f' = \frac{u}{u_e} \qquad \text{(which is consistent with defining stream function } \psi = \sqrt{2\xi} f\text{)}$$

$$g = \frac{h_0}{(h_0)_e}$$

The mechanics of the transformation using the chain rule are similar to that described in Section 18.2. Hence, without detailing the precise steps (which are left for your entertainment), Equations (18.32) and (18.41) transform into

$$\left(\frac{\rho \mu}{\rho_e \mu_e} f'' \right)' + f f' = 0 \tag{18.42}$$

and

$$\left(\frac{\rho \mu}{\rho_e \mu_e} \frac{1}{\Pr} g' \right)' + f g' + \frac{u_e^2}{(h_0)_e} \left[\left(1 - \frac{1}{\Pr} \right) \frac{\rho \mu}{\rho_e \mu_e} f' f'' \right]' = 0 \tag{18.43}$$

Examine Equations (18.42) and (18.43) closely. They are ordinary differential equations—recall that the primes denote differentiation with respect to η. Therefore, the compressible, laminar flow over a flat plate does lend itself to a self-similar solution, where $f' = f'(\eta)$ and $g = g(\eta)$. That is, the velocity and total enthalpy profiles plotted versus η are the same at any station. Furthermore, the product $\rho\mu$ is a variable and depends in part on temperature. Hence, Equation (18.42) is coupled to the energy equation, Equation (18.43), via $\rho\mu$. Of course, the energy equation is strongly coupled to Equation (18.42) via the appearance of f, f', and f'' in Equation (18.43). Hence, we are dealing with a *system* of coupled ordinary differential equations which must be solved simultaneously. The boundary conditions for these equations are

$$\text{At } \eta = 0: \qquad f = f' = 0 \qquad g = g_w$$

$$\text{At } \eta \to \infty: \qquad f' = 1 \qquad g = 1$$

Note that the coefficient $u_e^2/(h_0)_e$ appearing in Equation (18.43) is simply a function of the Mach number:

$$\frac{u_e^2}{(h_0)_e} = \frac{u_e^2}{h_e + u_e^2/2} = \frac{1}{h_e/u_e^2 + \frac{1}{2}} = \frac{1}{c_p T_e/u_e^2 + \frac{1}{2}} = \frac{1}{RT_e/(\gamma - 1)u_e^2 + \frac{1}{2}}$$

$$= \frac{1}{1/(\gamma - 1)M_e^2 + \frac{1}{2}} = \frac{2(\gamma - 1)M_e^2}{2 + (\gamma - 1)M_e^2}$$

Therefore, Equation (18.43) involves as a parameter the Mach number of the flow at the outer edge of the boundary layer, that is, for the flat-plate case, the

freestream Mach number. Hence, we can explicitly see that the compressible boundary-layer solutions will depend on the Mach number. Moreover, because of the appearance of the local Pr in Equation (18.43), the solutions will also depend on the freestream Prandtl number as a parameter. Finally, note from the boundary conditions that the value of g at the wall g_w is a given quantity. Note that at the wall where $u = 0$, $g_w = h_w/(h_0)_e = c_p T_w/(h_0)_e$. Hence, instead of referring to a given enthalpy at the wall g_w, we usually deal with a given wall temperature T_w. An alternative to a given value of T_w is the assumption of an *adiabatic wall,* that is, a case where there is no heat transfer to the wall. If $\dot{q}_w = k(\partial T/\partial y)_w = 0$, then $(\partial T/\partial y)_w = 0$. Hence, for an adiabatic wall, the boundary condition at the wall becomes simply $(\partial T/\partial y)_w = 0$.

In short, we see from the above discussion that a numerical self-similar solution can be obtained for the compressible, laminar flow over a flat plate. However, this solution depends on the Mach number, the Prandtl number, and the condition of the wall (whether it is adiabatic or a constant temperature wall with T_w given). Such numerical solutions have been carried out; see Reference 43 for details. A classic solution to Equations (18.42) and (18.43) is the shooting technique described in Section 16.4. The approach here is directly analogous to that used for the solution of compressible Couette flow discussed in Section 16.4. Since Equation (18.42) is third order, we need three boundary conditions at $\eta = 0$. We have only two, namely, $f = f' = 0$. Therefore, *assume* a value for $f''(0)$, and iterate until the boundary condition at the boundary-layer edge, $f' = 1$, is matched. Similarly, Equation (18.43) is a second-order equation. It requires two boundary conditions at the wall in order to integrate numerically across the boundary layer; we have only one, namely, $g(0) = g_w$. Thus, *assume* $g'(0)$, and integrate Equation (18.43). Iterate until the outer boundary condition is satisfied; that is, $g = 1$. Since Equation (18.42) is coupled to Equation (18.43), that is, since $\rho\mu$ in Equation (18.42) requires a knowledge of the enthalpy (or temperature) profile across the boundary layer, the entire process must be repeated again. This is directly analogous to the two minor iterations nested within the major iteration that was described in the discussion of the shooting method in Section 16.4. The approach here is virtually the same philosophy as described in Section 16.4, which should be reviewed at this stage. Therefore, no further details will be given here.

Typical solutions of Equations (18.42) and (18.43) for the velocity and temperature profiles through a compressible boundary layer on a flat plate are shown in Figures 18.4–18.7, obtained from van Driest (Reference 79). Figures 18.4 and 18.5 contain results for an insulated flat plate (zero-heat transfer) using Sutherland's law for μ, and assuming a constant Pr = 0.75. The velocity profiles are shown in Figure 18.4 for different Mach numbers ranging from 0 (incompressible flow) to the large hypersonic value of 20. Note that at a given x station at a given Re_x, the boundary-layer thickness increases markedly as M_e is increased to hypersonic values. This clearly demonstrates one of the most important aspects of compressible boundary layers, namely, that the boundary-layer thickness becomes large at large Mach numbers. Figure 18.5 illustrates the temperature profiles for the same case as Figure 18.4. Note the obvious physical trend that, as M_e increases to large

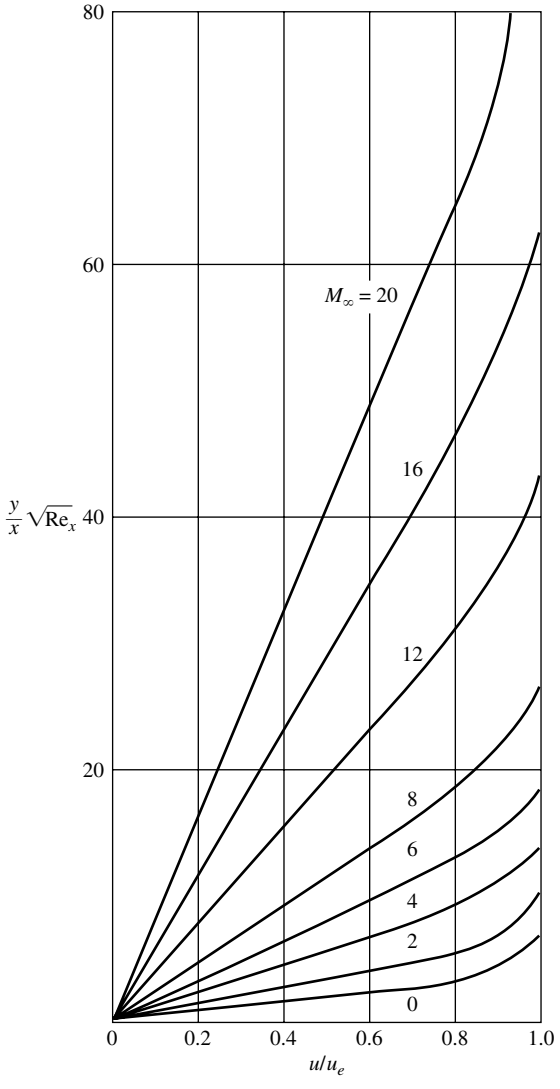


Figure 18.4 Velocity profiles in a compressible laminar boundary layer over an insulated flat plate (*Data Source: van Driest, Reference 79.*)

hypersonic values, the temperatures increase markedly. Also note in Figure 18.5 that at the wall ($y = 0$), $(\partial T/\partial y)_w = 0$, as it should be for an insulated surface ($q_w = 0$). Figures 18.6 and 18.7 also contain results by van Driest, but now for the case of heat transfer to the wall. Such a case is called a "cold wall" case, because $T_w < T_{aw}$. (The opposite case would be a "hot wall," where heat is transferred from the wall into the flow; in this case, $T_w > T_{aw}$.) For the results shown in

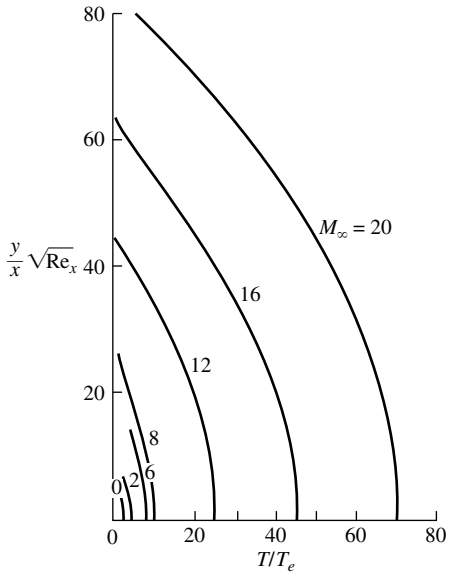


Figure 18.5 Temperature profiles in a compressible laminar boundary layer over an insulated flat plate. (*Data Source: van Driest, Reference 79.*)

Figures 18.6 and 18.7, $T_w/T_e = 0.25$ and Pr $= 0.75 =$ constant. Figure 18.6 shows velocity profiles for various different values of M_e, again demonstrating the rapid growth in boundary layer thickness with increasing M_e. In addition, the effect of a cold wall on the boundary layer thickness can be seen by comparing Figures 18.4 and 18.6. For example, consider the case of $M_e = 20$ in both figures. For the insulated wall at Mach 20 (Figure 18.4), the boundary layer thickness reaches out beyond a value of $(y/x)\sqrt{\text{Re}_x} = 60$, whereas for the cold wall at Mach 20 (Figure 18.6), the boundary-layer thickness is slightly above $(y/x)\sqrt{\text{Re}_w} = 30$. This illustrates the general fact that the effect of a cold wall is to reduce the boundary-layer thickness. This trend is easily explainable on a physical basis when we examine Figure 18.7, which illustrates the temperature profiles through the boundary layer for the cold-wall case. Comparing Figures 18.5 and 18.7, we note that, as expected, the temperature levels in the cold-wall case are considerably lower than in the insulated case. In turn, because the pressure is the same in both cases, we have from the equation of state $p = \rho RT$, that the density in the cold-wall case is much higher. If the density is higher, the mass flow within the boundary layer can be accommodated within a smaller boundary-layer thickness; hence, the effect of a cold wall is to thin the boundary layer. Also note in Figure 18.7 that, starting at the outer edge of the boundary layer and going toward the wall, the temperature first increases, reaches a peak somewhere within the boundary layer, and then decreases to its prescribed cold-wall value

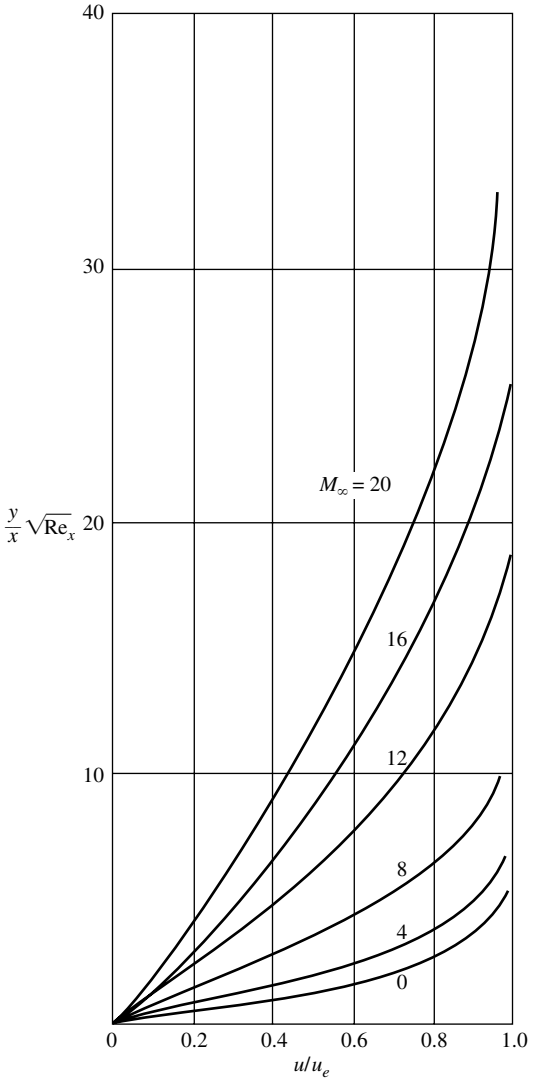


Figure 18.6 Velocity profiles in a laminar, compressible boundary layer over a cold flat plate. (*Data Source: van Driest Reference 79.*)

of T_w. The peak temperature inside the boundary layer is an indication of the amount of viscous dissipation occurring within the boundary layer. Figure 18.7 clearly demonstrates the rapidly growing effect of this viscous dissipation as M_e increases—yet another basic aspect of compressible boundary layers.

Carefully study the boundary-layer profiles shown in Figures 18.4–18.7. They are an example of the detailed results which emerge from a solution of

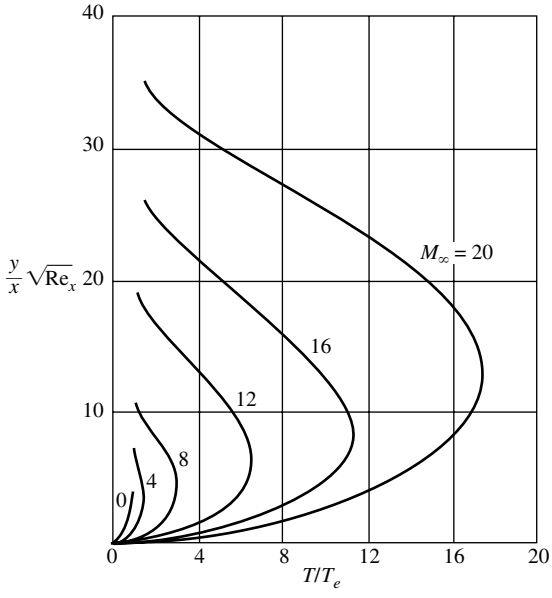


Figure 18.7 Temperature profiles in a laminar, compressible boundary layer over a cold flat plate. (*Data Source: van Driest, Reference 79.*)

Equations (18.42) and (18.43); indeed, these figures are graphical representations of Equations (18.43) and (18.42), with the results cast in the physical (x, y) space (rather than in terms of the transformed variable η). In turn, the surface values c_f and C_H can be obtained from the velocity and temperature gradients respectively at the wall as given by the velocity and temperature profiles evaluated at the wall. Recall from Equations (16.51) and (16.55) that c_f and C_H are defined as

$$c_f = \frac{\tau_w}{\frac{1}{2}\rho_e u_e^2} \tag{16.51}$$

and

$$C_H = \frac{\dot{q}_w}{\rho_e u_e (h_{aw} - h_e)} \tag{16.54}$$

where

$$\tau_w = \mu \left(\frac{\partial u}{\partial y}\right)_w \quad \text{and} \quad \dot{q}_w = -k \left(\frac{\partial T}{\partial y}\right)_w$$

and where $(\partial u/\partial y)_w$ and $(\partial T/\partial y)_w$ are the values obtained from the velocity and temperature profiles, respectively, evaluated at the wall. In turn, the overall flat plate skin friction drag coefficient C_f can be obtained by integrating c_f over the plate via Equation (18.21).

Return to Equation (18.22) for the friction drag coefficient for incompressible flow. The analogous compressible result can be written as

$$C_f = \frac{1.328}{\sqrt{\mathrm{Re}_c}} F\left(M_e, \mathrm{Pr}, \frac{T_w}{T_e}\right) \tag{18.44}$$

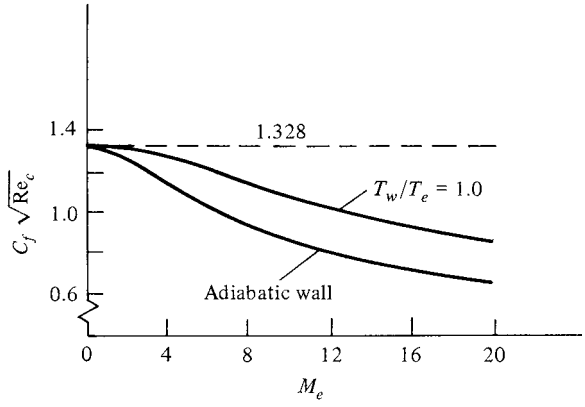


Figure 18.8 Friction drag coefficient for laminar, compressible flow over a flat plate, illustrating the effect of Mach number and wall temperature. Pr = 0.75. (*Calculations by E. R. van Driest, NACA Tech. Note 2597.*)

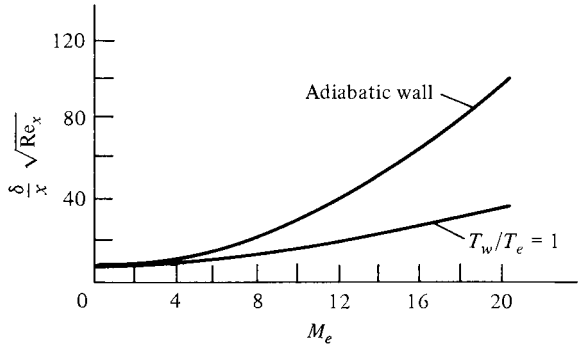


Figure 18.9 Boundary-layer thickness for laminar, compressible flow over a flat plate, illustrating the effect of Mach number and wall temperature. Pr = 0.75. (*Calculations by E. R. van Driest, NACA Tech. Note 2597.*)

In Equation (18.44), the function F is determined from the numerical solution. Sample results are given in Figure 18.8, which shows that the product $C_f\sqrt{\mathrm{Re}_c}$ decreases as M_e increases. Moreover, the adiabatic wall is warmer than the wall in the case of $T_w/T_e = 1.0$. Hence, Figure 18.8 demonstrates that a hot wall also reduces $C_f\sqrt{\mathrm{Re}_c}$.

Return to Equation (18.23) for the thickness of the incompressible flat-plate boundary layer. The analogous result for compressible flow is

$$\delta = \frac{5.0x}{\sqrt{\mathrm{Re}_x}} G\left(M_e, \mathrm{Pr}, \frac{T_w}{T_e}\right) \tag{18.45}$$

In Equation (18.45), the function G is obtained from the numerical solution. Sample results are given in Figure 18.9, which shows that the product $(\delta\sqrt{\mathrm{Re}_x}/x)$

increases as M_e increases. Everything else being equal, boundary layers are thicker at higher Mach numbers. This fact was stated earlier, as shown in Figures 18.4 and 18.6. Note also from Figure 18.9 that a hot wall thickens the boundary layer, as discussed earlier.

Recall our discussion of Couette flow in Chapter 16. There, we introduced the concept of the recovery factor r where

$$h_{aw} = h_e + r\frac{u_e^2}{2} \tag{18.46}$$

This is a general concept, and can be applied to the boundary-layer solutions here. If we assume a constant Prandtl number for the compressible flat-plate flow, the numerical solution shows that

$$r = \sqrt{\text{Pr}} \tag{18.47}$$

for the flat plate. Note that Equation (18.47) is analogous to the result given for Couette flow in that the recovery factor is a function of the Prandtl number only. However, for the flat plate, $r = \sqrt{\text{Pr}}$, whereas for Couette flow, $r = \text{Pr}$.

Aerodynamic heating for the flat plate can be treated via Reynolds analogy. The Stanton number and skin friction coefficients are defined respectively as

$$C_H = \frac{\dot{q}_w}{\rho_e u_e (h_{aw} - h_w)} \tag{18.48}$$

and

$$c_f = \frac{\tau_w}{\frac{1}{2}\rho_e u_e^2} \tag{18.49}$$

(See our discussion of these coefficients in Chapter 16.) Our results for Couette flow proved that a relation existed between C_H and c_f—namely, Reynolds analogy, given by Equation (16.59) for Couette flow. Moreover, in this relation, the ratio C_H/c_f was a function of the Prandtl number only. A directly analogous result holds for the compressible flat-plate flow. If we assume that the Prandtl number is constant, then for a flat plate, Reynolds analogy is, from the numerical solution,

$$\boxed{\frac{C_H}{c_f} = \frac{1}{2}\text{Pr}^{-2/3}} \tag{18.50}$$

In Equation (18.50), the local skin friction coefficient c_f which is given by Equation (18.20) for the incompressible flat-plate case, becomes the following form for the compressible flat-plate flow:

$$c_f = \frac{0.664}{\sqrt{\text{Re}}} F\left(M_e, \text{Pr}, \frac{T_w}{T_e}\right) \tag{18.51}$$

In Equation (18.51), F is the same function as appears in Equation (18.44), and its variation with M_e and T_w/T_e is the same as shown in Figure 18.8.

EXAMPLE 18.1

Consider a flat plate at zero angle of attack in an airflow at standard sea level conditions ($p_\infty = 1.01 \times 10^5$ N/m^2 and $T_\infty = 288$ K). The chord length of the plate (distance from the leading edge to the trailing edge) is 2 m. The planform area of the plate is 40 m^2. At standard sea level conditions, $\mu_\infty = 1.7894 \times 10^{-5}$ kg/(m)(s). Assume the wall temperature is the adiabatic wall temperature T_{aw}. Calculate the friction drag on the plate when the freestream velocity is (*a*) 100 m/s, (*b*) 1000 m/s.

■ Solution

(*a*) The freestream density is

$$\rho_\infty = \frac{p_\infty}{RT_\infty} = \frac{1.01 \times 10^5}{(287)(288)} = 1.22 \text{ kg/m}^3$$

The speed of sound is

$$a_\infty = \sqrt{\gamma RT_\infty} = \sqrt{(1.4)(287)(288)} = 340.2 \text{ m/s}$$

The Mach number is $M_\infty = 100/340.2 = 0.29$. Hence, M_∞ is low enough to assume incompressible flow, and we can use Equation (18.22),

$$C_f = \frac{1.328}{\sqrt{\text{Re}_c}}$$

Please note that for the flow over a flat plate at zero angle of attack, the freestream velocity and density, V_∞ and ρ_∞, are the same as the velocity and density at the outer edge of the boundary layer, u_e and ρ_e. Hence, these quantities can be used interchangeably. Thus,

$$\text{Re}_c = \frac{\rho_\infty V_\infty c}{\mu_\infty} = \frac{(1.22)(100)(2)}{1.7894 \times 10^{-5}} = 1.36 \times 10^7$$

Hence,

$$C_f = \frac{1.328}{\sqrt{\text{Re}_c}} = \frac{1.328}{\sqrt{1.36 \times 10^7}} = 3.60 \times 10^{-4}$$

The friction drag on one surface of the plate is given by

$$D_f = \tfrac{1}{2}\rho_\infty V_\infty^2 S C_f = \tfrac{1}{2}(1.22)(100)^2(40)(3.6 \times 10^{-4}) = 87.8 \text{ N}$$

The *total* drag due to friction is generated by the shear stress acting on *both* the top and bottom of the plate. Since D_f above is the friction drag on only one surface, we have

$$\text{Total friction drag} = D = 2D_f = 2(87.8) = \boxed{175.6 \text{ N}}$$

(*b*) For $V_\infty = 1000$ m/s, we have

$$M_\infty = \frac{V_\infty}{a_\infty} = \frac{1000}{340.2} = 2.94$$

Clearly, the flow is compressible, and we have to use Equation (18.44), or more directly, Figure 18.8. From Figure 18.8, we have for $M_\infty = M_e = 2.94$ and an adiabatic wall,

$$C_f \sqrt{\text{Re}_c} = 1.2$$

or
$$C_f = \frac{1.2}{\sqrt{\text{Re}_c}}$$

$$\text{Re}_c = \frac{\rho_\infty V_\infty c}{\mu_\infty} = \frac{(1.22)(1000)(2)}{1.7894 \times 10^{-5}} = 1.36 \times 10^8$$

Thus,
$$C_f = \frac{1.2}{\sqrt{1.36 \times 10^8}} = 1.03 \times 10^{-4}$$

The friction drag on one surface is

$$D_f = \tfrac{1}{2}\rho_\infty V_\infty^2 S C_f = \tfrac{1}{2}(1.22)(1000)^2(40)(1.03 \times 10^{-4}) = 2513 \text{ N}$$

Taking into account both the top and bottom surfaces,

$$\text{Total friction drag} = D = 2D_f = 2(2513) = \boxed{5026 \text{ N}}$$

18.3.1 A Comment on Drag Variation with Velocity

Beginning with Chapter 1, indeed beginning with the most elementary studies of fluid dynamics, the point is usually made that the aerodynamic force on a body immersed in a flowing fluid is proportional to the square of the flow velocity. For example, from Section 1.5,

$$L = \tfrac{1}{2}\rho_\infty V_\infty^2 S C_L$$

and
$$D = \tfrac{1}{2}\rho_\infty V_\infty^2 S C_D$$

As long as C_L and C_D are *independent* of velocity, then clearly $L \propto V_\infty^2$ and $D \propto V_\infty^2$. This is the case for an inviscid, incompressible flow, where C_L and C_D depend only on the shape and angle of attack of the body. However, from the dimensional analysis in Section 1.7, we also discovered that C_L and C_D in general are functions of both Reynolds number and Mach number,

$$C_L = f_1(\text{Re}, M_\infty)$$
$$C_D = f_2(\text{Re}, M_\infty)$$

Of course, for an inviscid, incompressible flow, Re and M_∞ are not players (indeed, for inviscid flow, $\text{Re} \to \infty$ and for incompressible flow, $M_\infty \to 0$). However for all other types of flow, Re and M_∞ *are* players, and the values of C_L and C_D depend not only on the shape and angle of attack of the body, but also on Re and M_∞. For this reason, in general the aerodynamic force is *not* exactly proportional to the square of the velocity. For example, examine the results from Example 18.1. In part (*a*), we calculated a value for drag to the 175.6 N when $V_\infty = 100$ m/s. If the drag were proportional to V_∞^2, then in part (*b*) where $V_\infty = 1000$ m/s, a factor of 10 larger, the drag would have been one hundred times larger, or 17,560 N. In contrast, our calculations in part (*b*) showed the drag to be considerably smaller, namely 5026 N. In other words, when V_∞ was increased by a factor of 10, the drag increased by only a factor of 28.6, *not* by

a factor of 100. The reason is obvious. The value of C_f *decreases* when the velocity is increased because: (1) the Reynolds number increases, which from Equation (18.22) causes C_f to decrease, and (2) the Mach number increases, which from Figure 18.8 causes C_f to decrease.

So be careful about thinking that aerodynamic force varies with the square of the velocity. For cases other than inviscid, incompressible flow, this is not true.

18.4 THE REFERENCE TEMPERATURE METHOD

In this section we discuss an approximate engineering method for predicting skin friction and heat transfer for laminar compressible flow. It is based on the simple idea of utilizing the formulas obtained from incompressible flow theory, wherein the thermodynamic and transport properties in these formulas are evaluated at some reference temperature indicative of the temperature somewhere inside the boundary layer. This idea was first advanced by Rubesin and Johnson in Reference 80 and was modified by Eckert (Reference 81) to include a reference enthalpy. In this fashion, in some sense the classical incompressible formulas were "corrected" for compressibility effects. Reference temperature (or reference enthalpy) methods have enjoyed frequent application in engineering-oriented analyses, because of their simplicity. For this reason, we briefly describe the approach here.

Consider the incompressible laminar flow over a flat plate, as discussed in Section 18.2. The local skin friction coefficient is given by Equation (18.20), repeated below:

$$c_f = \frac{0.664}{\sqrt{\text{Re}_x}} \tag{18.20}$$

For the *compressible* laminar flow over a flat plate, we write the analogous expression

$$\boxed{c_f^* = \frac{0.664}{\sqrt{\text{Re}_x^*}}} \tag{18.52}$$

except here c_f^* and Re_x^* are evaluated at a reference temperature T^*. That is,

$$\text{Re}_x^* = \frac{\rho^* u_e x}{\mu^*}$$

and

$$c_f^* = \frac{\tau_w}{\frac{1}{2}\rho^* u_e^2}$$

From Section 18.3, we know that for a compressible boundary layer c_f is a function of both M_e and T_w/T_e. Hence, the reference temperature T^* must be a function of M_e and T_w/T_e. This function is

$$\frac{T^*}{T_e} = 1 + 0.032M_e^2 + 0.58\left(\frac{T_w}{T_e} - 1\right) \tag{18.53}$$

For heat transfer for the compressible boundary layer, we write Reynolds analogy from Equation (18.50) repeated below.

$$\frac{C_H}{c_f} = \frac{1}{2}\,\mathrm{Pr}^{-2/3} \tag{18.50}$$

Inserting the incompressible formula for c_f, Equation (18.20), into Equation (18.50), we have

$$C_H = \frac{0.332}{\sqrt{\mathrm{Re}_x}}\,\mathrm{Pr}^{-2/3} \tag{18.54}$$

Evaluating Equation (18.54) at the reference temperature, we have

$$\boxed{C_H^* = \frac{0.332}{\sqrt{\mathrm{Re}_x^*}}(\mathrm{Pr}^*)^{-2/3}} \tag{18.55}$$

where $$C_H^* = \frac{\dot{q}_w}{\rho^* u_e^* (h_{aw} - h_w)}$$

EXAMPLE 18.2

Use the reference temperature method to calculate the friction drag on the same flat plate at the same flow conditions as described in Example 18.1*b*. Compare the reference temperature results with that obtained in Example 8.1*b*, which reflected the "exact" laminar boundary layer theory.

■ Solution

The reference temperature is calculated from Equation (18.53), where we need the ratio T_w/T_e. For the present case, the flat plate is at the adiabatic wall temperature, hence we need the ratio T_{aw}/T_e. To obtain this, we use the recovery factor, which for a flat plate laminar boundary layer is given by Equation (18.47):

$$r = \sqrt{\mathrm{Pr}} \qquad \text{where} \qquad \mathrm{Pr} = \frac{\mu c_p}{k}$$

For air at standard conditions, Pr = 0.71, and it is relatively constant with temperature up to about 800 K. Hence, we assume Pr = Pr* = 0.71, and

$$r = \sqrt{\mathrm{Pr}} = \sqrt{0.71} = 0.843$$

The recovery factor is defined by Equation (16.49) as

$$r = \frac{T_{aw} - T_e}{T_0 - T_e} \tag{16.49}$$

or $$T_{aw} = T_e + r(T_0 - T_e)$$

or $$\frac{T_{aw}}{T_e} = 1 + r\left(\frac{T_0}{T_e} - 1\right) \tag{18.56}$$

From Appendix A, for $M_e = 2.94$, $\frac{T_0}{T_e} = 2.74$. Hence, Equation (18.56) yields

$$\frac{T_{aw}}{T_e} = 1 + r\left(\frac{T_0}{T_e} - 1\right) = 1 + 0.843(2.74 - 1) = 2.467$$

From Equation (18.53),

$$\frac{T^*}{T_e} = 1 + 0.032M_e^2 + 0.58\left(\frac{T_{aw}}{T_e} - 1\right)$$

$$= 1 + 0.032(2.94)^2 + 0.58(2.467 - 1) = 2.1275$$

Thus, $$T^* = 2.1275T_e = 2.1275(288) = 612.7 \text{ K}$$

From the equation of state, the value of ρ^* that corresponds to T^* is

$$\rho^* = \frac{p^*}{RT^*} = \frac{1.01 \times 10^5}{(287)(612.7)} = 0.574 \text{ kg/m}^3$$

Also, the value of μ^* that corresponds to T^* is obtained from Sutherland's law, given by Equation (15.3)

$$\frac{\mu}{\mu_0} = \left(\frac{T}{T_0}\right)^{3/2} \frac{T_0 + 110}{T + 110} \tag{15.3}$$

Recall: In Equation (15.3), μ_0 is the reference viscosity coefficient at the reference temperature T_0. In Equation (15.3) T_0 denotes the *reference temperature,* not the total temperature. Here we have a case of the same notation for two different quantities, but the meaning of T_0 in Equation (15.3) is clear from its context. We will use the standard sea level conditions for the values of T_0 and μ_0, that is,

$$\mu_0 = 1.7894 \times 10^{-5} \text{ kg/(m)(s)} \quad \text{and} \quad T_0 = 288 \text{ K}$$

Hence, from Equation (15.3)

$$\frac{\mu^*}{\mu_0} = \left(\frac{T^*}{T_0}\right)^{3/2} \frac{T_0 + 110}{T^* + 110} = \left(\frac{612.7}{288}\right)^{3/2} \frac{288 + 110}{612.7 + 110} = 1.709$$

or $$\mu^* = 1.709\mu_0 = (1.709)(1.7894 \times 10^{-5}) = 3.058 \times 10^{-5} \text{ kg/(m)(s)}$$

Thus, $$\text{Re}_c^* = \frac{\rho^* u_e c}{\mu^*} = \frac{(0.574)(1000)(2)}{3.058 \times 10^{-5}} = 3.754 \times 10^7$$

From Equation (18.52) integrated over the entire chord of the plate, we have the same form as Equation (18.22), namely,

$$C_f^* = \frac{1.328}{\sqrt{\text{Re}_c^*}} \tag{18.57}$$

Thus, $$C_f^* = \frac{1.328}{\sqrt{\text{Re}_c^*}} = \frac{1.328}{\sqrt{3.754 \times 10^7}} = 2.167 \times 10^{-4}$$

Hence, the friction drag on one side of the plate is

$$D_f = \tfrac{1}{2}\rho^* V_\infty^2 S C_f^* = \tfrac{1}{2}(0.574)(1000)^2(40)(2.167 \times 10^{-4}) = 2488 \text{ N}$$

The total friction drag taking into account both the top and bottom surfaces of the plate is

$$D = 2(2488) = \boxed{4976 \text{ N}}$$

The result obtained from classical compressible boundary layer theory in Example 18.1*b* is $D = 5026$ N. The result from the reference temperature method used here is within one percent of the "exact" value found in Example 18.1*b*, a stunning example of the accuracy of the reference temperature method, at least for the case treated here.

18.4.1 Recent Advances: The Meador-Smart Reference Temperature Method

The reference temperature method discussed in Section 18.4 is a concept that dates back to the late 1940s, but it is still a work in progress. Very recently, Meador and Smart (William E. Meador and Michael K. Smart, "Reference Enthalpy Method Developed from Solutions of the Boundary-Layer Equations," *AIAA Journal,* vol. 43, no. 1, January 2005, pp. 135–139) published improved formulas for the calculation of the reference temperature, one for laminar flow and another for turbulent flow. This result for a laminar flow is

$$\frac{T^*}{T_e} = 0.45 + 0.55\frac{T_w}{T_e} + 0.16\, r \left(\frac{\gamma - 1}{2}\right) M_e^2$$

where r is the recovery factor for laminar flow, $r = \sqrt{\text{Pr}^*}$.

EXAMPLE 18.3

Repeat Example 18.2, using the Meador-Smart equation for the reference temperature.

■ Solution

Assuming the Prandtl number is reasonably constant,

$$r = \sqrt{\text{Pr}^*} = \sqrt{\text{Pr}} = \sqrt{0.71} = 0.843$$

Also, because the flat plate is at the adiabatic wall temperature, from Example 8.2,

$$\frac{T_w}{T_e} = \frac{T_{aw}}{T_e} = 2.467$$

So the Meador-Smart equation becomes

$$\frac{T^*}{T_e} = 0.45 + 0.55(2.467) + 0.16(0.843)(0.2)M_e^2$$

or

$$\frac{T^*}{T_e} = 1.807 + 0.027 M_e^2$$

For $M_e = 2.94$, we have

$$\frac{T^*}{T_e} = 1.807 + 0.027(2.94)^2 = 2.04$$

$$T^* = 2.04\, T_e = 2.04\,(288) = 587.5 \text{ K}$$

$$\rho^* = \frac{p^*}{RT^*} = \frac{1.01 \times 10^5}{(287)(587.5)} = 0.599 \text{ kg/m}^3$$

$$\frac{\mu^*}{\mu_0} = \left(\frac{T^*}{T_0}\right)^{3/2} \frac{T_0 + 110}{T^* + 110} = \left(\frac{587.5}{288}\right)^{3/2} \frac{288 + 110}{587.5 + 110} = 1.664$$

$$\mu^* = 1.664\,\mu_0 = 1.664(1.7894 \times 10^{-5}) = 2.978 \times 10^{-5}$$

$$\text{Re}_c^* = \frac{\rho^* u_e c}{\mu^*} = \frac{(0.500)(1000)(2)}{2.978 \times 10^{-5}} = 4.02 \times 10^7$$

$$C_f^* = \frac{1.328}{\sqrt{\text{Re}_c^*}} = \frac{1.328}{\sqrt{4.02 \times 10^7}} = 2.09 \times 10^4$$

$$D_f = \tfrac{1}{2}\rho^* V_\infty^2 S C_f^* = \tfrac{1}{2}(0.599)(1000)^2(40)(2.09 \times 10^{-4}) = 2504 \text{ N}$$

$$D = 2(2504) = \boxed{5008 \text{ N}}$$

The result from Example 18.2 is $D = 4976$ N, and the exact result from Example 18.1 is 5026 N. The Meador-Smith method is more accurate than Equation (18.53); it agrees to within 0.4 percent of the exact amount.

18.5 STAGNATION POINT AERODYNAMIC HEATING

Contrary to what you might think, even though the flow velocity is zero at a stagnation point, the boundary layer at the stagnation point can be defined and has a finite thickness. The flow conditions at the edge of the stagnation point boundary layer are given by the inviscid solution for a stagnation point; in particular, at the boundary-layer edge, the velocity is zero and the temperature is the total temperature, that is, $u_e = 0$ and $T_e = T_0$. This is shown in Figure 18.10. Moreover, along the vertical line in the η-direction shown in Figure 18.10, $u = 0$ at every point inside the boundary layer. However, the ratio $(u/u_e) = (0/0)$ is an indeterminant form that has a finite value at each point in the boundary layer. As in the case of the flat plate solutions discussed in Sections 18.2 and 18.3, we define a function $f(\eta)$ such that $(u/u_e) = f'(\eta)$, and f' has a definite profile through the boundary layer. Indeed, we can define the edge of the boundary layer as the point where $(u/u_e) = f'(\eta) = 0.99$. Finally, we note that the shear stress at the wall at the stagnation point (point A in Figure 18.10) is zero. This not only comes out of the solution of the boundary layer equations, but it is obvious by inspection. Along the wall above point A the shear stress acts upward, and below point A it acts downward. Hence right at point A the shear stress must go through zero.

If the above discussion sounds rather theoretical, the temperature profile through the stagnation point boundary layer is easier to visualize. The temperature at the outer edge is the total temperature T_0. The temperature at the wall at $\eta = 0$ is T_w. Hence, there is a temperature profile that exists in the normal direction through the stagnation point boundary layer. The heat transfer at the stagnation

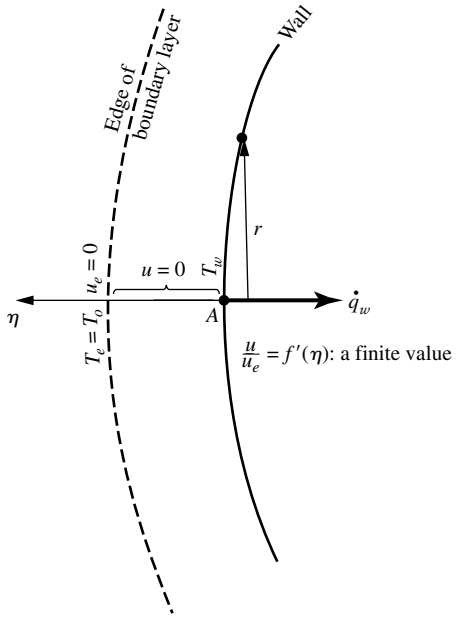


Figure 18.10 Schematic of the stagnation region boundary layer.

point is given by the temperature gradient at point A, namely,

$$\dot{q}_w = \left[-k \left(\frac{\partial T}{\partial y} \right)_w \right]_A \tag{18.58}$$

The practical purpose of a stagnation point boundary-layer solution is to calculate the heat transfer, $\dot{q}_w$.

The boundary-layer equations, Equation (17.28)–(17.31) applied at the stagnation point region are transformed using a version of the transformation described in Section 18.3, namely,

$$\xi = \int_0^x \rho_e \mu_e u_r \, dx \tag{18.59}$$

$$\eta = \frac{u_e}{\sqrt{2\xi}} \int_0^y \rho \, dy \tag{18.60}$$

$$f'(\eta) = \frac{u}{u_e} \tag{18.61}$$

$$g(\eta) = \frac{h}{h_e} \tag{18.62}$$

where h is the static enthalpy (since $u = 0$, the static and total enthalpies are the same). This leads to the stagnation point boundary layer equations given

below. For a detailed derivation of these equations, see, for example, Chapter 6 of Reference 55.

$$\boxed{(Cf'')' + ff'' = (f')^2 - g} \tag{18.63}$$

$$\boxed{\left(\frac{C}{\text{Pr}}g'\right)' + fg' = 0} \tag{18.64}$$

where $C = (\rho\mu/\rho_e\mu_e)$. Equations (18.63) and (18.64) are the governing equations for a compressible, stagnation-point boundary layer. Examining these equations, we see no ξ-dependency. Hence, the stagnation point boundary layer is a self-similar case.

Numerical solutions to Equations (18.63) and (18.64) can be obtained by the "shooting technique" as described earlier in the flat plate case. There is nothing to be gained in going through the details at this stage of our discussion. Instead, we simply state the result of solving Equations (18.63) and (18.64), correlated in the following expression obtained from Reference 82:

$$\textit{Cylinder:} \qquad \dot{q}_w = 0.57\,\text{Pr}^{-0.6}(\rho_e\mu_e)^{1/2}\sqrt{\frac{du_e}{dx}}(h_{aw} - h_w) \tag{18.65}$$

If we had considered an axisymmetric body, the original transformation given by Equations (18.59) and (18.60) would have been slightly modified as follows:

$$\xi = \int_0^x \rho_e u_e \mu_e r^2\,dx \tag{18.66}$$

and

$$\eta = \frac{u_e r}{\sqrt{2\xi}}\int_0^y \rho\,dy \tag{18.67}$$

where r is the vertical coordinate measured from the centerline, as shown in Figure 18.10. Equations (18.66) and (18.67) lead to equations for the axisymmetric stagnation point almost identical to Equation (18.63) and (18.64), namely,

$$(Cf'')' + ff'' = \tfrac{1}{2}[(f')^2 - g] \tag{18.68}$$

and

$$\left(\frac{C}{\text{Pr}}g'\right)' + fg' = 0 \tag{18.69}$$

where $C = (\rho\mu/\rho_e\mu_e)$. In turn, the resulting heat transfer expression is (Reference 82):

$$\textit{Sphere:} \qquad \dot{q}_w = 0.763\,\text{Pr}^{-0.65}(\rho_e\mu_e)\sqrt{\frac{du_e}{dx}}(h_{aw} - h_w) \tag{18.70}$$

Compare Equation (18.65) for the two-dimensional cylinder with Equation (18.70) for the axisymmetric sphere. The equations are the same except for the leading coefficient, which is higher for the sphere. Everything else being the same, this demonstrates that stagnation point heating to a sphere is larger

than to a two-dimensional cylinder. Why? The answer lies in a basic difference between two- and three-dimensional flows. In a two-dimensional flow, the gas has only two directions to move when it encounters a body—up or down. In contrast, in an axisymmetric flow, the gas has three directions to move—up, down, and sideways—and hence the flow is somewhat "relieved," that is, in comparing two- and three-dimensional flows over bodies with the same longitudinal section (such as a cylinder and a sphere), there is a well-known three-dimensional relieving effect for the three-dimensional flow. As a consequence of this relieving effect, the boundary-layer thickness δ at the stagnation point is smaller for the sphere than for the cylinder. In turn, the temperature gradient at the wall, $(\partial T/\partial y)_w$, which is of the order of (T_e/δ), is larger for the sphere. Since $\dot{q}_w = k(\partial T/\partial y)_w$, then $\dot{q}_w$ is larger for the sphere. This confirms the comparison between Equations (18.65) and (18.70).

The above results for aerodynamic heating to a stagnation point have a stunning impact on hypersonic vehicle design, namely, they impose the requirement for the vehicle to have a blunt, rather than a sharp, nose. To see this, consider the velocity gradient, du_e/dx, which appears in Equations (18.65) and (18.70). From Euler's equation applied at the edge of the boundary layer

$$dp_e = -\rho_e u_e \, du_e \tag{18.71}$$

we have

$$\frac{du_e}{dx} = -\frac{1}{\rho_e u_e}\frac{dp_e}{dx} \tag{18.72}$$

Assuming a newtonian pressure distribution over the surface, we have from Equation (14.4)

$$C_p = 2\sin^2\theta$$

where θ is defined as the angle between a tangent to the surface and the freestream direction. If we define ϕ as the angle between the normal to the surface and the freestream, then Equation (14.4) can be written as

$$C_p = 2\cos^2\phi \tag{18.73}$$

From the definition of C_p, Equation (18.73) becomes

$$\frac{p_e - p_\infty}{q_\infty} = 2\cos^2\phi$$

or

$$p_e = 2q_\infty \cos^2\phi + p_\infty \tag{18.74}$$

Differentiating Equation (18.74), we obtain

$$\frac{dp_e}{dx} = -4q_\infty \cos\phi \sin\phi \frac{d\phi}{dx} \tag{18.75}$$

Combining Equations (18.72) and (18.75), we have

$$\frac{du_e}{dx} = \frac{4q_\infty}{\rho_e u_e}\cos\phi \sin\phi \frac{d\phi}{dx} \tag{18.76}$$

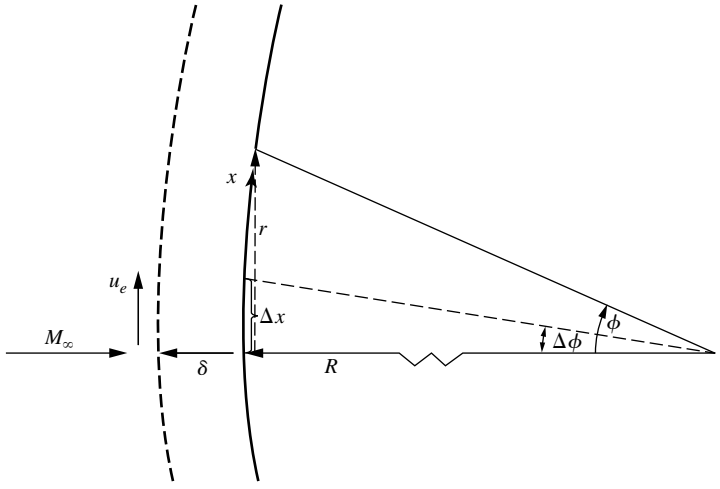


Figure 18.11 Geometry of the stagnation region.

Equation (18.76) is a general result which applies at all points along the body. Now consider the stagnation-point region, as sketched in Figure 18.11. In this region, let Δx be a small increment of surface distance above the stagnation point, corresponding to the small change in ϕ, $\Delta\phi$. The inviscid velocity variation in the stagnation region can be shown to be

$$u_e = \left(\frac{du_e}{dx}\right)_s \Delta x \tag{18.77}$$

Also, in the stagnation region ϕ is small, hence, from Figure 18.11,

$$\cos\phi \approx 1 \tag{18.78}$$

$$\sin\phi \approx \phi \approx \Delta\phi \approx \frac{\Delta x}{R} \tag{18.79}$$

$$\frac{d\phi}{dx} = \frac{1}{R} \tag{18.80}$$

where R is the local radius of curvature of the body at the stagnation point. Finally, at the stagnation point, Equation (18.73) becomes

$$C_p = 2 = \frac{p_e - p_\infty}{q_\infty}$$

or

$$q_\infty = \tfrac{1}{2}(p_e - p_\infty) \tag{18.81}$$

Substituting Equations (18.77)–(18.81) into (18.76), we have

$$\left(\frac{du_e}{dx}\right)^2 = \frac{2(p_e - p_\infty)}{\rho_e \Delta x}\left(\frac{\Delta x}{R}\right)\left(\frac{1}{R}\right)$$

or
$$\frac{du_e}{dx} = \frac{1}{R}\sqrt{\frac{2(p_e - p_\infty)}{\rho_e}} \tag{18.82}$$

Examine Equations (18.65) and (18.70) in light of Equation (18.82). We see that

$$\boxed{\dot{q}_w \propto \frac{1}{\sqrt{R}}} \tag{18.83}$$

This states that stagnation-point heating varies inversely with the square root of the nose radius; hence, to reduce the heating, increase the nose radius. This is the reason why the nose and leading edge regions of hypersonic vehicles are blunt; otherwise, the severe aerothermal conditions in the stagnation region would quickly melt a sharp leading edge.

Return to Section 1.1 and review our *qualitative* discussion contrasting the aerodynamic heating for slender and blunt reentry vehicles. There we argued on a qualitative basis that to minimize aerodynamic heating a blunt nose must be used. We have now quantitatively proven this fact with the derivation of Equation (18.83).

The fact that $\dot{q}_w$ is inversely proportional to $\sqrt{R}$ is experimentally verified in Figure 18.12, obtained from Reference 83. Here, various sets of experimental data for C_H at the stagnation point are plotted versus Reynolds number based on nose diameter; the abscissa is essentially proportional to R. This is a log-log plot, and the data exhibit a slope of -0.5, hence verifying that $\dot{q}_w \propto 1/\sqrt{R}$.

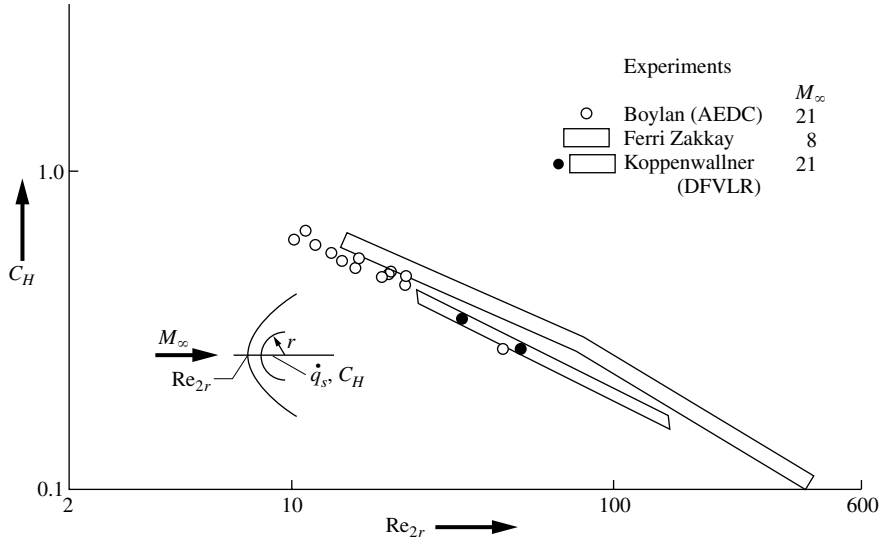


Figure 18.12 Stagnation point Stanton number versus Re based on nose radius. (*Source: Koppenwallner, Reference 83.*)

18.6 BOUNDARY LAYERS OVER ARBITRARY BODIES: FINITE-DIFFERENCE SOLUTION

"Exact" solutions of the boundary layer equations, Equations (17.28)–(17.31), for the flow over bodies of arbitrary shape did not occur until the advent of the high-speed digital computer and ultimately not until the beginnings of computational fluid dynamics. In this section we discuss a finite-difference technique for solving the general boundary layer equations; such finite-difference solutions represent the current state of the art in the analysis of boundary layers.

Let us set the perspective for our discussion. Equations (17.28)–(17.31) are the general boundary layer equations. For the special case of the flat plate, these equations reduced to Equations (18.42) and (18.43), and for the stagnation region they reduced to Equations (18.63) and (18.64). In both special cases, these equations in terms of the transformed dependent and independent variables led to self-similar solutions (flow variations only in the transformed η direction). For the general case of an arbitrary body, it is still useful to transform the full boundary-layer equations, Equations (17.28)–(17.31), via the transformation given by Equations (18.59)–(18.62). For a detailed derivation of these transformed equations, see Chapter 6 of Reference 55. The resulting form of the equation is:

x momentum:

$$\boxed{(Cf'')' + ff'' = \frac{2\xi}{u_e}\left[\left((f')^2 - \frac{\rho_e}{\rho}\right)\frac{du_e}{d\xi} + 2\xi\left(f'\frac{\partial f'}{\partial \xi} - \frac{\partial f}{\partial \xi}f''\right)\right]} \tag{18.84}$$

y momentum:

$$\boxed{\frac{\partial p}{\partial \eta} = 0} \tag{18.85}$$

Energy:

$$\boxed{\left(\frac{C}{\Pr}g'\right)' + fg' = 2\xi\left[f'\frac{\partial g}{\partial \xi} + \frac{f'g}{h_e}\frac{\partial h_e}{\partial \xi} - g'\frac{\partial f}{\partial \xi} + \frac{\rho_e u_e}{\rho h_e}f'\frac{du_e}{d\xi}\right] - C\frac{u_e^2}{h_e}(f'')^2} \tag{18.86}$$

where as before $C = \rho\mu/\rho_e\mu_e$, $f' = u/u_e$, and $g = h/h_e$. In Equations (18.84)–(18.86), the prime denotes the *partial* derivative with respect to η, that is, $f' \equiv \partial f/\partial \eta$. Equations (18.84)–(18.86) are simply the transformed versions of Equations (17.28)–(17.31), with no loss of authority.

Examine Equations (18.84)–(18.86); they are the transformed compressible boundary layer equations. They are still *partial* differential equations, where both f and g are functions of ξ and η. They contain no further approximations or assumptions beyond those associated with the original boundary-layer equations. However, they are certainly in a less recognizable, somewhat more

complicated-looking form than the original equations. However, do not be disturbed by this; in reality Equations (18.84)–(18.86) are in a form that proves to be practical and useful.

The above transformed boundary-layer equations must be solved subject to the following boundary conditions. The physical boundary conditions were given immediately following Equations (17.28)–(17.31); the corresponding transformed boundary conditions are:

At the wall: $\eta = 0 \quad f = f' = 0 \quad g = g_w$ (fixed wall temperature)

or $g' = 0$ (adiabatic wall)

At the boundary-layer edge: $\eta \to \infty \quad f' = 1 \quad g = 1$

In general, solutions of Equations (18.84), (18.85), and (18.86) along with the appropriate boundary conditions yield variations of velocity and enthalpy throughout the boundary layer, via $u = u_e f'(\xi, \eta)$ and $h = h_e g(\xi, \eta)$. The pressure throughout the boundary layer is known, because the known pressure distribution (or equivalently the known velocity distribution) at the edge of the boundary is given by $p_e = p_e(\xi)$, and this pressure is impressed without change through the boundary layer in the locally normal direction via Equation (18.85), which says that $p =$ constant in the normal direction at any ξ location. Finally, knowing h and p throughout the boundary layer, equilibrium thermodynamics provides the remaining variables through the appropriate equations of state, for example, $T = T(h, p)$, $\rho = \rho(h, p)$, etc.

18.6.1 Finite-Difference Method

Return for a moment to Section 2.17.2 where we introduced some ideas from computation fluid dynamics, and especially review the finite-difference expressions derived there. Recall that we can simulate the partial derivatives with forward, rearward, or central differences. We will use these concepts in the following discussion.

Also consider Figure 18.13, which shows a schematic of a finite-difference grid inside the boundary layer. The grid is shown in the physical x-y space, where it is curvilinear and unequally spaced. However, in the ξ-η space, where the calculations are made, the grid takes the form of a rectangular grid with uniform spacing $\Delta\xi$ and $\Delta\eta$. In Figure 18.13, the portion of the grid at four different ξ (or x) stations is shown, namely, at $(i - 2)$, $(i - 1)$, i, and $(i + 1)$.

Consider again the general, transformed boundary-layer equations given by Equations (18.84) and (18.86). Assume that we wish to calculate the boundary layer at station $(i + 1)$ in Figure 18.13. As discussed in Section 2.17.2, the general philosophy of finite-difference approaches is to evaluate the governing partial differential equations at a given grid point by replacing the derivatives by finite-difference quotients at that point. Consider, for example, the grid point (i, j) in Figure 18.13. At this point, replace the derivatives in Equations (18.84)

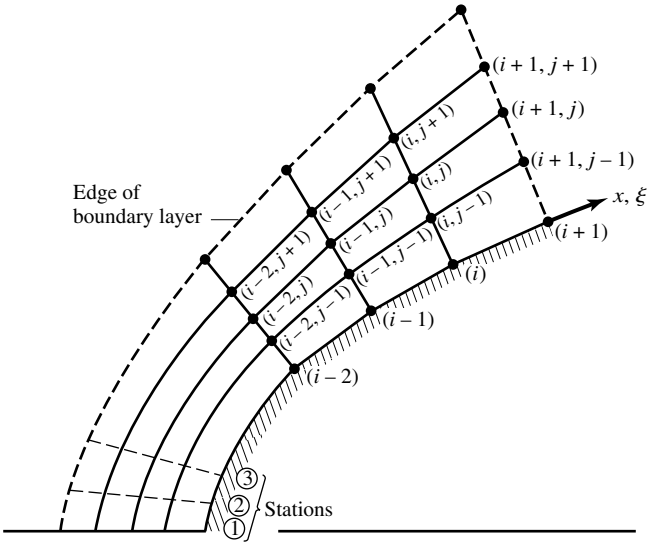


Figure 18.13 Schematic for finite-difference solution of the boundary layer.

and (18.86) by finite-difference expressions of the form:

$$\frac{\partial f}{\partial \xi} = \frac{f_{i+1,j} - f_{i,j}}{\Delta \xi} \tag{18.87}$$

$$\frac{\partial f}{\partial \eta} = \frac{\theta(f_{i+1,j+1} - f_{i+1,j-1})}{2\,\Delta\eta} + \frac{(1-\theta)(f_{i,j+1} - f_{i,j-1})}{2\,\Delta\eta} \tag{18.88}$$

$$\frac{\partial^2 f}{\partial \eta^2} = \frac{\theta(f_{i+1,j+1} - 2f_{i+1,j} + f_{i+1,j-1})}{(\Delta\eta)^2} + \frac{(1-\theta)(f_{i,j+1} - 2f_{i,j} + f_{i,j-1})}{(\Delta\eta)^2} \tag{18.89}$$

$$f = \theta f_{i+1,j} + (1-\theta) f_{i,j} \tag{18.90}$$

where θ is a parameter which adjusts Equations (18.87)–(18.90) to various finite-difference approaches (to be discussed below). Similar relations for the derivatives of g are employed. When Equations (18.87)–(18.90) are inserted into Equations (18.84) and (18.86), along with the analogous expressions for g, two algebraic equations are obtained. If $\theta = 0$, the only unknowns that appear are $f_{i+1,j}$ and $g_{i+1,j}$, which can be obtained directly from the two algebraic equations. This is an explicit approach. Using this approach, the boundary layer properties at grid point $(i + 1, j)$ are solved explicitly in terms of the known properties at points $(i, j + 1)$, (i, j) and $(i, j - 1)$. The boundary-layer solution is a downstream marching procedure; we are calculating the boundary layer profiles at station $(i + 1)$ only after the flow at the previous station (i) has been obtained.

When $0 < \theta \leq 1$, then $f_{i+1,j+1}$, $f_{i+1,j}$, $f_{i+1,j-1}$, $g_{i+1,j+1}$, $g_{i+1,j}$, and $g_{i+1,j-1}$ appear as unknowns in Equations (18.84) and (18.86). We have six unknowns and only two equations. Therefore, the finite-difference forms of Equations (18.84) and (18.86) must be evaluated at all the grid points through the boundary layer at station $(i+1)$ simultaneously, leading to an implicit formulation of the unknowns. In particular, if $\theta = \frac{1}{2}$, the scheme becomes the well-known Crank-Nicolson implicit procedure, and if $\theta = 1$, the scheme is called "fully implicit." These implicit schemes result in large systems of simultaneous algebraic equations, the coefficients of which constitute block tridiagonal matrices.

Already the reader can sense that implicit solutions are more elaborate than explicit solutions. Indeed, we remind ourselves that the subject of this book is the fundamentals of aerodynamics, and it is beyond our scope to go into great computational fluid dynamic detail. Therefore, we will not elaborate any further. Our purpose here is only to give the flavor of the finite-difference approach to boundary-layer solutions. For more information on explicit and implicit finite-difference methods, see the author's book *Computational Fluid Dynamics: The Basics with Applications* (Reference 64).

In summary, a finite-difference solution of a general, nonsimilar boundary-layer proceeds as follows:

1. The solution must be started from a given solution at the leading edge, or at a stagnation point (say station 1 in Figure 18.13). This can be obtained from appropriate self-similar solutions.

2. At station 2, the next downstream station, the finite-difference procedure reflected by Equations (18.87)–(18.90) yields a solution of the flowfield variables across the boundary layer.

3. Once the boundary-layer profiles of u and T are obtained, the skin friction and heat transfer at the wall are determined from

$$\tau = \left[\mu \left(\frac{\partial u}{\partial y}\right)\right]_w$$

and

$$\dot{q} = \left(k \frac{\partial T}{\partial y}\right)_w$$

Here, the velocity gradients can be obtained from the known profiles of u and T by using one-sided differences (see References 64), such as

$$\left(\frac{\partial u}{\partial y}\right)_w = \frac{-3u_1 + 4u_2 - u_3}{2\,\Delta y} \tag{18.91}$$

$$\left(\frac{\partial T}{\partial y}\right)_w = \frac{-3T_1 + 4T_2 - T_3}{2\,\Delta y} \tag{18.92}$$

In Equations (18.91) and (18.92), the subscripts 1, 2, and 3 denote the wall point and the next two adjacent grid points above the wall. Of course, due to the specified boundary conditions of no velocity slip and a fixed wall temperature, $u_1 = 0$ and $T_1 = T_w$ in Equations (18.91) and (18.92).

4. The above steps are repeated for the next downstream location, say station 3 in Figure 18.13. In this fashion, by repeating applications of these steps, the complete boundary layer is computed, marching downstream from a given initial solution.

An example of results obtained from such finite-difference boundary-layer solutions is given in Figures 18.14 and 18.15 obtained by Blottner (Reference 84). These are calculated for flow over an axisymmetric hyperboloid flying at 20,000 ft/s at an altitude of 100,000 ft, with a wall temperature of 1000 K. At these conditions, the boundary layer will involve dissociation, and such chemical

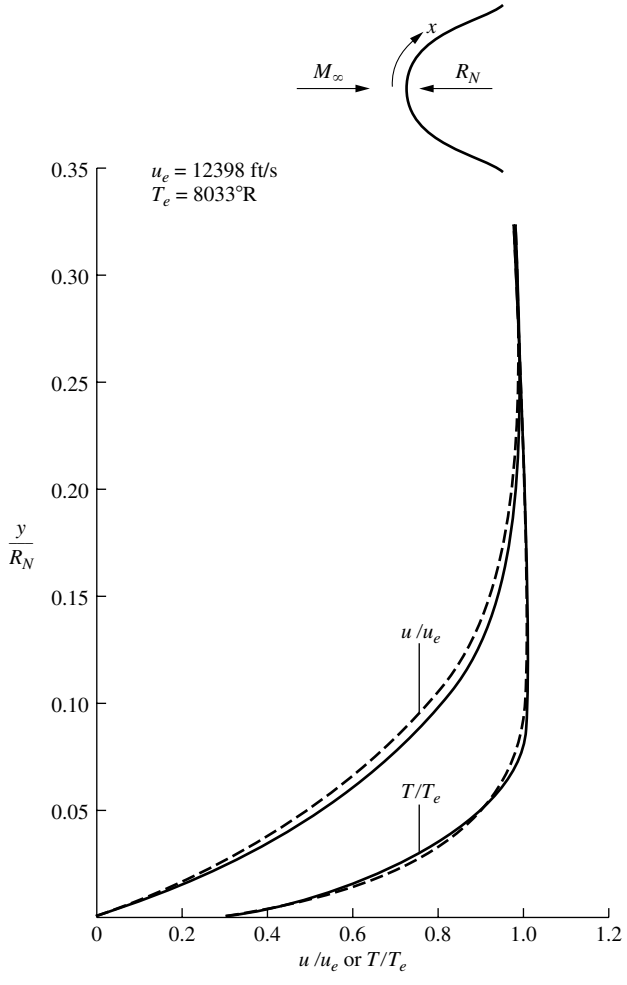


Figure 18.14 Velocity and temperature profiles across the boundary layer at $x/R_N = 50$ on an axisymmetric hyperboloid. (*Data Source: Blottner, Reference 84.*)

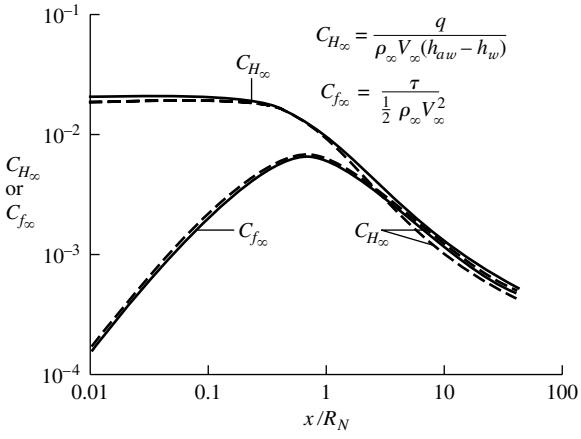


Figure 18.15 Stanton number and skin friction coefficient (based on freestream properties) along a hyperboloid. (*Data Source: Blottner, Reference 84.*)

reactions were included in the calculations of Reference 84. Chemically reacting boundary layers are not the purview of this book; however, some results of Reference 84 are presented here just to illustrate the finite-difference method. For example, Figure 18.14 gives the calculated velocity and temperature profiles as a station located at $x/R_N = 50$, where R_N is the nose radius. The local values of velocity and temperature at the boundary layer edge are also quoted in Figure 18.14. Considering the surface properties, the variations of C_H and c_f as functions of distance from the stagnation point are shown in Figure 18.15. Note the following physical trends illustrated in Figure 18.15.

1. The shear stress is zero at the stagnation point (as is always the case), then it increases around the nose, reaches a maximum, and decreases further downstream.
2. The values of C_H are relatively constant near the nose, and then decrease further downstream.
3. Reynolds analogy can be written as

$$C_H = \frac{c_f}{2s} \tag{18.93}$$

where s is called the "Reynolds analogy factor." For the flat plate case, we see from Equation (18.50) that $s = \text{Pr}^{2/3}$. However, clearly from the results of Figure 18.15 we see that s is a variable in the nose region because C_H is relatively constant while c_f is rapidly increasing. In contrast, for the downstream region, c_f and C_H are essentially equal, and we can state that Reynolds analogy becomes approximately $C_H/c_f = 1$. The point here is that Reynolds analogy is greatly affected by strong pressure gradients in the flow, and hence loses its usefulness as an engineering tool in such cases, at least when C_H and c_f are based on freestream quantities as shown in Figure 18.15.

18.7 SUMMARY

This brings to an end our discussion of laminar boundary layers. Return to the roadmap in Figure 18.1 and remind yourself of the territory we have covered. Some of the important results are summarized below.

For incompressible laminar flow over a flat plate, the boundary-layer equations reduce to the Blasius equation

$$2f''' + ff'' = 0 \tag{18.15}$$

where $f' = u/u_e$. This produces a self-similar solution where $f' = f'(\eta)$, independent of any particular x station along the surface. A numerical solution of Equation (17.48) yields numbers which lead to the following results.

Local skin friction coefficient:
$$c_f = \frac{\tau_w}{\frac{1}{2}\rho_\infty V_\infty^2} = \frac{0.664}{\sqrt{\mathrm{Re}_x}} \tag{18.20}$$

Integrated friction drag coefficient:
$$C_f = \frac{1.328}{\sqrt{\mathrm{Re}_c}} \tag{18.22}$$

Boundary-layer thickness:
$$\delta = \frac{5.0x}{\sqrt{\mathrm{Re}_x}} \tag{18.23}$$

Displacement thickness:
$$\delta^* = \frac{1.72x}{\sqrt{\mathrm{Re}_x}} \tag{18.26}$$

Momentum thickness:
$$\theta = \frac{0.664x}{\sqrt{\mathrm{Re}_x}} \tag{18.28}$$

Compressibility effects are such as to make boundary-layer solutions a function of Mach number, Prandtl number, and wall-to-freestream temperature ratio. Typical compressibility effects are shown in Figure 18.8. Generally, compressibility reduces C_f and increases δ.

The reference temperature method is an easy engineering calculation of skin friction and heat transfer to a flat plate taking into account compressibility effects, but using the incompressible equations for c_f and C_H. The reference temperature T^* is given by

$$\frac{T^*}{T_e} = 1 + 0.032M_e^2 + 0.58\left(\frac{T_w}{T_e} - 1\right) \tag{18.53}$$

(*continued*)

The local skin-friction coefficient is given by

$$c_f^* = \frac{0.644}{\sqrt{\mathrm{Re}_x^*}} \tag{18.52}$$

where $$c_f^* = \frac{\tau_w}{\frac{1}{2}\rho^* u_e^2}$$

and $$\mathrm{Re}_x^* = \frac{\rho^* u_e x}{\mu^*}$$

The Stanton number is given by

$$C_H^* = \frac{0.332}{\sqrt{\mathrm{Re}_x^*}} (\mathrm{Pr}^*)^{-2/3} \tag{18.55}$$

where $$C_H^* = \frac{\dot{q}_w}{\rho^* u_e (h_{aw} - h_w)}$$

18.8 PROBLEMS

Note: The homework problems for this chapter are deferred until the end of Chapter 19 so that both laminar and turbulent boundary layers can be dealt with together.

CHAPTER 19

Turbulent Boundary Layers

The one uncontroversial fact about turbulence is that it is the most complicated kind of fluid motion.

Peter Bradshaw
Imperial College of Science and Technology, London 1978

Turbulence was, and still is, one of the great unsolved mysteries of science, and it intrigued some of the best scientific minds of the day. Arnold Sommerfeld, the noted German theoretical physicist of the 1920s, once told me, for instance, that before he died he would like to understand two phenomena—quantum mechanics and turbulence. Sommerfeld died in 1924. I believe he was somewhat nearer to an understanding of the quantum, the discovery that led to modern physics, but no closer to the meaning of turbulence.

Theodore von Karman, 1967

PREVIEW BOX

Nature, when left to herself, always goes to the state of maximum disorder. This is particularly true for boundary-layer flows under real conditions. For most practical applications in aerodynamics, the flow in boundary layers is predominantly turbulent—nature going to the state of maximum disorder. Turbulent boundary layers can be bad news; for the same external flow conditions, the turbulent skin friction and aerodynamic heating is larger, frequently *much* larger, than laminar skin friction and aerodynamic heating. But turbulent boundary layers can also be good news because they remain attached to a surface for much larger distances downstream than a laminar boundary layer under the same external flow conditions. Hence, pressure drag due to flow separation is usually smaller for bodies with turbulent boundary layers compared to those with laminar boundary layers.

Turbulent boundary layers are the subject of this chapter. There are no stand-alone theoretical results for turbulent boundary layers—any analysis must incorporate empirical data in some form. So we will not beat around the bush. We go directly to empirically

based formulas that allow the estimation (and it is truly only an estimation) of turbulent boundary-layer thickness and skin friction. There are many different approaches to the calculation of turbulent flows, all requiring some input from experimental data. Whole books have been written on the subject of turbulent flow. The objective of this very short chapter is simply to give you some ability to compute turbulent boundary-layer results, albeit imprecisely. Hang on, and enjoy this short ride.

19.1 INTRODUCTION

The subject of turbulent flow is deep, extensively studied, but at the time of writing still imprecise. The basic nature of turbulence, and therefore our ability to predict its characteristics, is still an unsolved problem in classical physics. Many books have been written on turbulent flows, and many people have spent their professional lives working on the subject. As a result, it is presumptuous for us to try to carry out a thorough discussion of turbulent boundary layers in this chapter. Instead, the purpose of this chapter is simply to provide a contrast with our study of laminar boundary layers in Chapter 18. Here, we will only be able to provide a flavor of turbulent boundary layers, but this is all that is necessary in the present book. Turbulence is a subject that we leave for you to study more extensively as a subject on its own.

Before proceeding further, return to Section 15.2 and review the basic discussion of the nature of turbulence that is given there. In the present chapter, we will pick up where Section 15.2 leaves off.

Also, we note that no *pure theory* of turbulent flow exists. Every analysis of turbulent flow requires some type of *empirical data* in order to obtain a practical answer. As we examine the calculation of turbulent boundary layers in the following sections, the impact of this statement will become blatantly obvious. Finally, because this chapter is short, there is no need for a roadmap to act as a guide.

19.2 RESULTS FOR TURBULENT BOUNDARY LAYERS ON A FLAT PLATE

In this section, we discuss a few results for the turbulent boundary layer on a flat plate, both incompressible and compressible, simply to provide a basis of comparison with the laminar results described in the previous section. For considerably more detail on the subject of turbulent boundary layers, consult References 42 to 44.

For incompressible flow over a flat plate, the boundary-layer thickness is given approximately by

$$\boxed{\delta = \frac{0.37x}{\mathrm{Re}_x^{1/5}}} \tag{19.1}$$

Note from Equation (19.1) that the turbulent boundary-layer thickness varies approximately as $\mathrm{Re}_x^{-1/5}$ in contrast to $\mathrm{Re}_x^{-1/2}$ for a laminar boundary layer. Also, turbulent values of δ grow more rapidly with distance along the surface; $\delta \propto x^{4/5}$ for a turbulent flow in contrast to $\delta \propto x^{1/2}$ for a laminar flow. With regard to skin friction drag, for incompressible turbulent flow over a flat plate, we have

$$\boxed{C_f = \frac{0.074}{\mathrm{Re}_c^{1/5}}} \tag{19.2}$$

Note that for turbulent flow, C_f varies as $\mathrm{Re}_c^{-1/5}$ in comparison with the $\mathrm{Re}_c^{-1/2}$ variation for laminar flow. Hence, Equation (19.2) yields larger friction drag coefficients for turbulent flow in comparison with Equation (18.22) for laminar flow.

The effects of compressibility on Equation (19.2) are shown in Figure 19.1, where C_f is plotted versus Re_∞ with M_∞ as a parameter. The turbulent flow results are shown toward the right of Figure 19.1, at the higher values of Reynolds numbers where turbulent conditions are expected to occur, and laminar flow results are shown toward the left of the figure, at lower values of Reynolds numbers. This type of figure—friction drag coefficient for both laminar and turbulent flow as a function of Re on a log-log plot—is a classic picture, and it allows a ready contrast of the two types of flow. From this figure, we can see that, for the same

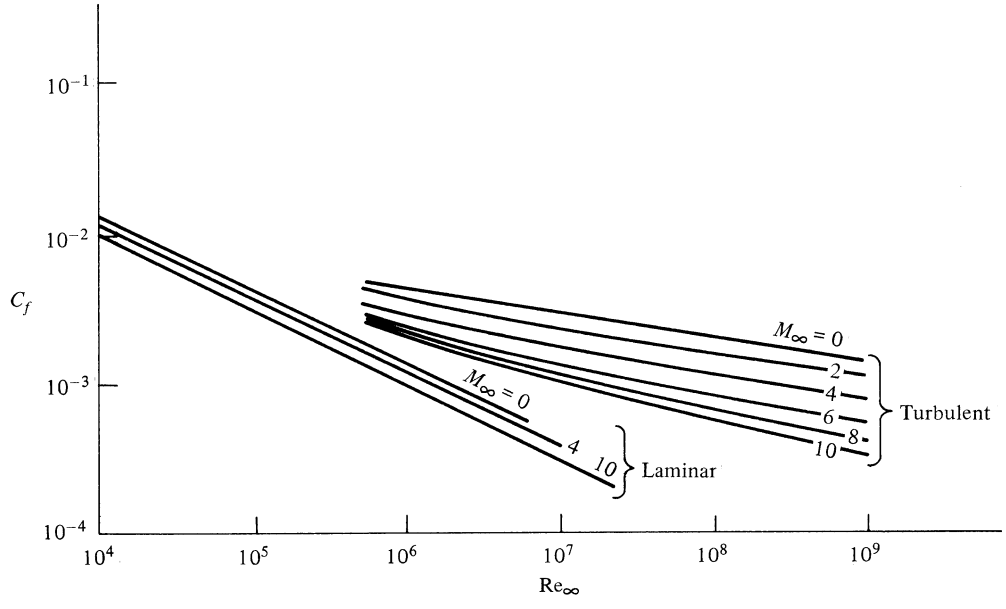


Figure 19.1 Turbulent friction drag coefficient for a flat plate as a function of Reynolds and Mach numbers. Adiabatic wall Pr = 0.75. For contrast, some laminar results are shown. (*Data are from the calculations of van Driest, Reference 47.*)

Re_∞, turbulent skin friction is higher than laminar; also, the slopes of the turbulent curves are smaller than the slopes of the laminar curves—a graphic comparison of the $\text{Re}^{-1/5}$ variation in contrast to the laminar $\text{Re}^{-1/2}$ variation. Note that the effect of increasing M_∞ is to reduce C_f at constant Re and that this effect is stronger on the turbulent flow results. Indeed, C_f for the turbulent results decreases by almost an order of magnitude (at the higher values of Re_∞) when M_∞ is increased from 0 to 10. For the laminar flow, the decrease in C_f as M_∞ is increased though the same Mach number range is far less pronounced.

19.2.1 Reference Temperature Method for Turbulent Flow

The reference temperature method discussed in Section 18.4 for laminar boundary layers can be applied to turbulent boundary layers as well. With the reference temperature T^* given by Equation (18.53), the incompressible turbulent flat plate result for C_f given by Equation (19.2) can be modified for compressible turbulent flow as

$$C_f^* = \frac{0.074}{(\text{Re}_c^*)^{1/5}} \tag{19.3}$$

where

$$C_f^* = \frac{D_f}{\frac{1}{2}\rho^* u_e^2 S} \tag{19.4}$$

EXAMPLE 19.1

Consider the same flat plate under the same external flow conditions given in Example 18.1. Calculate the friction drag on the plate assuming a *turbulent* boundary layer for a freestream velocity of (*a*) 100 m/s, and (*b*) 1000 m/s.

■ Solution

(*a*) From Example 18.1*a*, $\text{Re}_c = 1.36 \times 10^7$. Hence, from Equation (19.2)

$$C_f = \frac{0.074}{(\text{Re}_c)^{1/5}} = \frac{0.074}{(1.36 \times 10^7)^{1/5}} = \frac{0.074}{26.71} = 2.77 \times 10^{-3}$$

Also from Example 18.1, we have $\rho_\infty = 1.22$ kg/m^3 and $S = 40$ m^2. Hence, for one side of the plate,

$$D_f = \tfrac{1}{2}\rho_\infty V_\infty^2 S C_f = \tfrac{1}{2}(1.22)(100)^2(40)(2.77 \times 10^{-3}) = 675.9 \text{ N}$$

The total friction drag taking into account both sides of the plate is

$$D = 2D_f = 2(675.9) = \boxed{1352 \text{ N}}$$

Comparing this result for turbulent flow with the laminar result in Example 1.81*a*, we have

$$\frac{D_{\text{turbulent}}}{D_{\text{laminar}}} = \frac{1352}{175.6} = 7.7$$

Turbulent flow causes a factor of 7.7 *increase* in friction drag compared to the laminar flow. You can easily see why the understanding of, and prediction of, turbulent flow, especially the prediction of when the flow will transist from laminar to turbulent flow, is so important.

(*b*) For $V_\infty = 1000$ m/s, from Example 18.1*b*, $\text{Re}_c = 1.36 \times 10^8$, and $M_\infty = 2.94$. From Figure 19.1, we have

$$C_f = 1.34 \times 10^{-3}$$

Hence, $D_f = \frac{1}{2}\rho_\infty V_\infty^2 S C_f = \frac{1}{2}(1.22)(1000)^2(40)(1.34 \times 10^{-3}) = 32{,}700$ N.

The total friction drag is

$$D = 2(32{,}700) = \boxed{65{,}400 \text{ N}}$$

Again, comparing this result with that from Example 18.1*b*, we have

$$\frac{D_{\text{turbulent}}}{D_{\text{laminar}}} = \frac{65{,}400}{5026} = 13$$

Note that, at the higher Mach number of 2.92, turbulence increased the drag by a factor of 13, whereas for the incompressible case, the increase was 7.7, a smaller amount. The difference between the drag for laminar and turbulent flow is more pronounced at higher speeds.

EXAMPLE 19.2

Repeat Example 19.1*b*, except using the reference temperature method. Assume the plate has an adiabatic wall.

■ Solution

We draw on the results calculated in Example 18.2. The recovery factor for a turbulent flow is slightly different than that for a laminar flow. However, we will not account for that difference, and we will assume that the reference temperature for this case is the same as given in Example 18.2. Hence, from Example 18.2, we have

$$\text{Re}_c^* = 3.754 \times 10^7 \quad \text{and} \quad \rho^* = 0.574 \text{ kg/m}^3$$

From Equation (19.3) we have

$$C_f^* = \frac{0.074}{(\text{Re}_c^*)^{1/5}} = \frac{0.074}{(3.754 \times 10^7)^{1/5}} = 2.26 \times 10^{-3}$$

From Equation (19.4),

$$D_f = \frac{1}{2}\rho^* u_e^2 S C_f^* = \frac{1}{2}(0.574)(1000)^2(40)(2.26 \times 10^{-3}) = 25{,}945 \text{ N}$$

Hence, $\boxed{D = 2(25{,}945) = 51{,}890 \text{ N}}$

Comparing this answer with that obtained in Example 19.1*b*, we find a 20 percent discrepancy between the two methods of calculations. This is not surprising. It simply points out the great uncertainty in making calculations of turbulent skin friction.

19.2.2 The Meador-Smart Reference Temperature Method for Turbulent Flow

The method developed recently by Meador and Smart, discussed in Section 18.4.1, gives a reference temperature equation for turbulent flow slightly different than that for laminar flow. For a turbulent flow, their equation is

$$\frac{T^*}{T_e} = 0.5\left(1 + \frac{T_w}{T_e}\right) + 0.16\,r\left(\frac{\gamma - 1}{2}\right) M_e^2$$

They also give a *local* turbulent skin-friction coefficient for incompressible flow as

$$c_f = \frac{\tau_w}{\frac{1}{2}\rho_e u_e^2} = \frac{0.02296}{(\mathrm{Re}_x)^{0.139}}$$

When integrated over the entire plate of length c, this gives for the net skin-friction drag coefficient (prove it to yourself)

$$C_f = \frac{D_f}{\frac{1}{2}\rho_\infty V_\infty^2 S} = \frac{0.02667}{(\mathrm{Re}_c)^{0.139}}$$

EXAMPLE 19.3

Repeat Example 9.2 using the Meador-Smart reference temperature method.

■ **Solution**

From the above equation,

$$\frac{T^*}{T_e} = 0.5\left(1 + \frac{T_w}{T_e}\right) + 0.16r\left(\frac{\gamma - 1}{2}\right) M_e^2$$

For turbulent flow, the recovery factor is approximately

$$r = \mathrm{Pr}^{1/3} = (0.71)^{1/3} = 0.892$$

$$T_{aw} - T_e = r(T_0 - T_e)$$

or

$$\frac{T_{aw}}{T_e} = 1 + r\left(\frac{T_0}{T_e} - 1\right)$$

For $M_e = 2.94$,

$$\frac{T_0}{T_e} = 2.74$$

$$\frac{T_{aw}}{T_r} = 1 + 0.892(1.74) = 2.55$$

Since the flat plate has an adiabatic wall, $T_w = T_{aw}$. The Meador-Smith equation then becomes

$$\frac{T^*}{T_e} = 0.5\left(1 + \frac{T_w}{T_e}\right) + 0.16(0.892)(0.2)(2.94)^2 = 0.5(1 + 2.55) + 0.2467 = 2.02$$

$$T^* = 2.02\,T_e = 2.02\,(288) = 581.8\text{ K}$$

$$\rho^* = \frac{p}{RT^*} = \frac{1.01 \times 10^5}{(287)(581.8)} = 0.605\text{ kg/m}^3$$

From Sutherland's law (*note that T_0 in Sutherland's law is a reference temperature, not the total temperature*)

$$\frac{\mu^*}{\mu_0} = \left(\frac{T^*}{T_0}\right)^{3/2} \frac{T_0 + 110}{T^* + 110} = \left(\frac{581.8}{288}\right)^{3/2} \frac{398}{691.8} = 1.651$$

$$\mu^* = 1.651\,\mu_0 = 1.651(1.7894 \times 10^{-5}) = 2.05 \times 10^{-5}\text{ kg/m} \cdot \text{s}$$

$$\text{Re}_c^* = \frac{\rho^* u_e c}{\mu^*} = \frac{(0.605)(1000)(2)}{2.95 \times 10^{-5}} = 4.1 \times 10^7$$

From the Meador-Smith choice of the turbulent skin-friction coefficient equation,

$$C_f^* = \frac{0.02667}{(\text{Re}_c^*)^{0.139}} = \frac{0.02667}{(4.1 \times 10^7)^{0.139}} = 2.32 \times 10^{-3}$$

$$D_f = \tfrac{1}{2}\rho^* V_\infty^2 S C_f^* = \tfrac{1}{2}(0.605)(1000)^2(40)(2.32 \times 10^{-3}) = 28070\text{ N}$$

$$\text{Total drag} = D = 2D_f = 2(28070) = \boxed{56140\text{ N}}$$

Note: This result is more accurate than that obtained in Example 9.2; it shows a 14 percent discrepancy compared with the result obtained in Example 19.1*b*.

19.2.3 Prediction of Airfoil Drag

The flat plate results obtained in Chapter 18 for laminar flow, and in the present chapter for turbulent flow, can be used for engineering prediction of skin-friction drag on thin airfoils. Using results from Chapters 18 and 19, airfoil drag in low-speed incompressible flow is treated in Section 4.12, and supersonic airfoil drag is discussed in Section 12.4. If you have not read Sections 4.12 and 12.4, do so now. They give an important practical application of the boundary-layer results we have just covered. Indeed, Sections 4.12 and 12.4 provide a vital continuation of our discussion of viscous flow, and for all practical purposes they can be considered integral sections of Part 4 of this book, although they were inserted in the earlier chapters to provide some viscous flow reality to our otherwise inviscid flow presentations. The prediction of airfoil drag is one of the most important aspects of aerodynamics. Take it seriously, and make certain that you read, or have read, Sections 4.12 and 12.4.

19.3 TURBULENCE MODELING

The simple equations given in Section 19.2 for boundary-layer thickness and skin friction coefficient for a turbulent flow over a flat plate are simplified results that are heavily empirically based. Modern calculations of turbulent flows over

arbitrarily-shaped bodies involve the solution of the continuity, momentum, and energy equations along with some *model* of the turbulence. The calculations are carried out by means of computational fluid dynamic techniques. Here we will discuss only one model of tubulence, the Baldwin-Lomax turbulence model, which has become popular over the past two decades. *We emphasize that the following discussion is intended only to give you the flavor of what is meant by a turbulence model.*

19.3.1 The Baldwin-Lomax Model

In order to include the effects of turbulence in any analysis or computation, it is first necessary to have a model for the turbulence itself. Turbulence modeling is a state-of-the-art subject, and a recent survey of such modeling as applied to computations is given in Reference 85. Again, it is beyond the scope of the present book to give a detailed presentation of various turbulence models; the reader is referred to the literature for such matters. Instead, we choose to discuss only one such model here, because: (a) it is a typical example of an engineering-oriented turbulence model, (b) it is the model used in the majority of modern applications in turbulent, subsonic, supersonic, and hypersonic flows, and (c) we will discuss in the next chapter several applications which use this model. The model is called the Baldwin-Lomax turbulence model, first proposed in Reference 86. It is in the class of what is called an "eddy viscosity" model, where the effects of turbulence in the governing viscous flow equations (such as the boundary-layer equations or the Navier-Stokes equations) are included simply by adding an additional term to the transport coefficients. For example, in all our previous viscous flow equations, μ is replaced by $(\mu + \mu_T)$ and k by $(k + k_T)$ where μ_T and k_T are the eddy viscosity and eddy thermal conductivity, respectively—both due to turbulence. In these expressions, μ and k are denoted as the "molecular" viscosity and thermal conductivity, respectively. For example, the x momentum boundary-layer equation for turbulent flow is written as

$$\rho u \frac{\partial u}{\partial x} + \rho v \frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{\partial}{\partial y}\left[(\mu + \mu_T)\frac{\partial u}{\partial y}\right] \tag{19.5}$$

Moreover, the Baldwin-Lomax model is also in the class of "algebraic," or "zero-equation," models meaning that the formulation of the turbulence model utilizes just algebraic relations involving the flow properties. This is in contrast to one- and two-equation models which involve partial differential equations for the convection, creation, and dissipation of the turbulent kinetic energy and (frequently) the local vorticity. (See Reference 87 for a concise description of such one- and two-equation turbulence models.)

The Baldwin-Lomax turbulence model is described below. We give just a "cookbook" prescription for the model; the motivation and justification for the model are described at length in Reference 86. This, like all other turbulence models, is highly empirical. The final justification for its use is that it yields reasonable results across a wide range of Mach numbers, from subsonic to hypersonic. The

model assumes that the turbulent-boundary layer is split into two layers, an inner and an outer layer, with different expressions for μ_T in each layer:

$$\mu_T = \begin{cases} (\mu_T)_{\text{inner}} & y \le y_{\text{crossover}} \\ (\mu_T)_{\text{outer}} & y \ge y_{\text{crossover}} \end{cases} \tag{19.6}$$

where y is the local normal distance from the wall, and the crossover point from the inner to the outer layer is denoted by $y_{\text{crossover}}$. By definition, $y_{\text{crossover}}$ is that point in the turbulent boundary where $(\mu_T)_{\text{outer}}$ becomes less than $(\mu_T)_{\text{inner}}$. For the inner region:

$$(\mu_T)_{\text{inner}} = \rho l^2 |\omega| \tag{19.7}$$

where

$$l = ky\left[1 - \exp\left(\frac{-y^+}{A^+}\right)\right] \tag{19.8}$$

$$y^+ = \frac{\sqrt{\rho_w \tau_w}\, y}{\mu_w} \tag{19.9}$$

and k and A^+ are two dimensionless constants, specified later. In Equation (19.7), ω is the local vorticity, defined for a two dimensional flow as

$$\omega = \frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} \tag{19.10}$$

For the outer region:

$$(\mu_T)_{\text{outer}} = \rho K C_{\text{cp}} F_{\text{wake}} F_{\text{Kleb}} \tag{19.11}$$

where K and C_{cp} are two additional constants, and F_{wake} and F_{Kleb} are related to the function

$$f(y) = y|\omega|\left[1 - \exp\left(\frac{-y^+}{A^+}\right)\right] \tag{19.12}$$

Equation (19.12) will have a maximum value along a given normal distance y; this maximum value and the location where it occurs are denoted by $F_{\max}$ and $y_{\max}$, respectively. In Equation (19.11), F_{wake} is taken to be either $y_{\max} F_{\max}$ or $C_{\text{wk}} y_{\max} U_{\text{dif}}^2 / F_{\max}$, whichever is smaller, where C_{wk} is constant, and

$$U_{\text{dif}} = \sqrt{u^2 + v^2} \tag{19.13}$$

Also, in Equation (19.11), F_{Kleb} is the Klebanoff intermittency factor, given by

$$F_{\text{Kleb}}(y) = \left[1 + 5.5\left(C_{\text{Kleb}} \frac{y}{y_{\max}}\right)^6\right]^{-1} \tag{19.14}$$

The six dimensionless constants that appear in the above equations are: $A^+ = 26.0$, $C_{\text{cp}} = 1.6$, $C_{\text{Kleb}} = 0.3$, $C_{\text{wk}} = 0.25$, $k = 0.4$, and $K = 0.0168$. These constants are taken directly from Reference 86 with the understanding that, while they are not precisely the correct constants for most flows in general, they have been

used successfully for a number of different applications. Note that, unlike many algebraic eddy viscosity models that are based on a characteristic length, the Baldwin-Lomax model is based on the local vorticity ω. This is a distinct advantage for the analysis of flows without an obvious mixing length, such as separated flows. Note that, like all eddy-viscosity turbulent models, the value of μ_T obtained above is dependent on the flowfield properties themselves (for example ω and ρ); this is in contrast to the molecular viscosity μ, which is solely a property of the gas itself.

The molecular values of viscosity coefficient and thermal conductivity are related through the Prandtl number

$$k = \frac{\mu c_p}{\text{Pr}} \tag{19.15}$$

In lieu of developing a detailed turbulence model for the turbulent thermal conductivity k_T, the usual procedure is to define a "turbulent" Prandtl number as $\text{Pr}_T = \mu_T c_p / k_T$. Thus, analogous to Equation (19.15), we have

$$k_T = \frac{\mu_T c_p}{\text{Pr}_T} \tag{19.16}$$

where the usual assumption is that $\text{Pr}_T = 1$. Therefore, μ_T is obtained from a given eddy-viscosity model (such as the Baldwin-Lomax model), and the corresponding k_T is obtained from Equation (19.16).

Turbulence itself is a flowfield; it is *not* a simple property of the gas. This is why, as mentioned above, in an algebraic eddy viscosity model the values of μ_T and k_T *depend* on the solution of the flow field—they are not pure properties of the gas as are μ and k. This is clearly seen in the Baldwin-Lomax model via Equation (19.7), where μ_T is a function of the local vorticity in the flow, ω—a flow field variable which comes out as part of the solution for the particular case at hand.

19.4 FINAL COMMENTS

This chapter and the previous two have dealt with boundary layers, especially those on a flat plate. We end with the presentation of a photograph in Figure 19.2 showing the development of velocity profiles in the boundary layer over a flat plate. The fluid is water, which flows from left to right. The profiles are made visible by the hydrogen bubble technique, the same used for Figure 16.13. The Reynolds number is low (the freestream velocity is only 0.6 m/s); hence, the boundary-layer thickness is large. However, the thickness of the plate is only 0.5 mm, which means that the boundary layer shown here is on the order of 1 mm thick—still small on an absolute scale. In any event, if you need any further proof of the existence of boundary layers, Figure 19.2 is it.

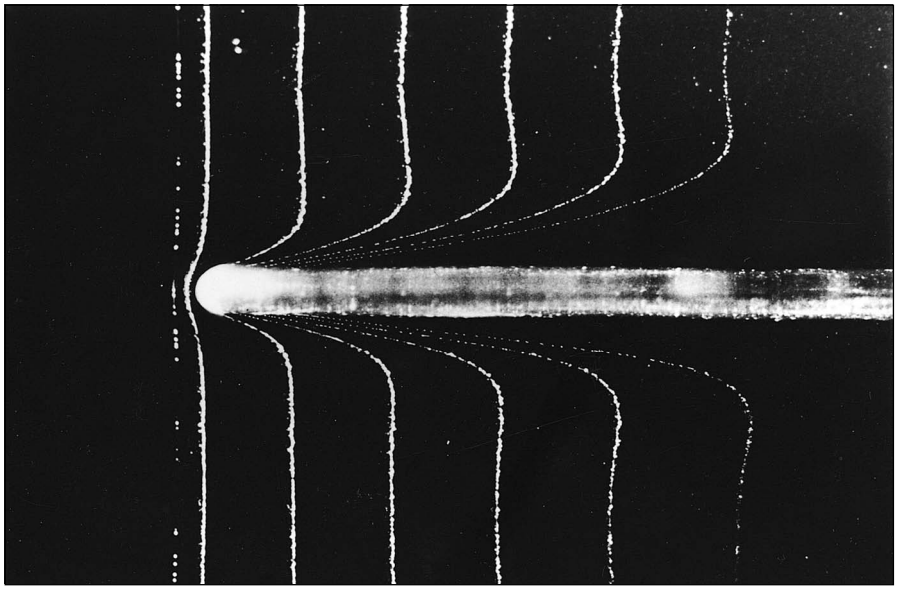


Figure 19.2 Photograph of velocity profiles for the laminar flow over a flat plate. Flow is from left to right. (*Courtesy of Yasuki Nakayama, Tokai University, Japan.*)

19.5 SUMMARY

Approximations for the turbulent, incompressible flow over a flat plate are

$$\delta = \frac{0.37x}{\mathrm{Re}_x^{1/5}} \tag{19.1}$$

$$C_f = \frac{0.074}{\mathrm{Re}_c^{1/5}} \tag{19.2}$$

To account for compressibility effects, the data shown in Figure 19.1 can be used, or alternatively the reference temperature method can be employed.

When the continuity, momentum, and energy equations are used to solve a turbulent flow, some type of turbulence model must be used. In the eddy viscosity concept, the viscosity coefficient and thermal conductivity in these equations must be the *sum* of the molecular and turbulent values.

19.6 PROBLEMS

Note: The standard sea level value of viscosity coefficient for air is $\mu = 1.7894 \times 10^{-5}$ kg/(m · s) $= 3.7373 \times 10^{-7}$ slug/(ft · s).

19.1 The wing on a Piper Cherokee general aviation aircraft is rectangular, with a span of 9.75 m and a chord of 1.6 m. The aircraft is flying at cruising speed (141 mi/h) at sea level. Assume that the skin friction drag on the wing can be approximated by the drag on a flat plate of the same dimensions. Calculate the skin friction drag:

a. If the flow were completely laminar (which is not the case in real life)

b. If the flow were completely turbulent (which is more realistic)

Compare the two results.

19.2 For the case in Problem 19.1, calculate the boundary-layer thickness at the trailing edge for

a. Completely laminar flow

b. Completely turbulent flow

19.3 For the case in Problem 19.1, calculate the skin friction drag accounting for transition. Assume the transition Reynolds number $= 5 \times 10^5$.

19.4 Consider Mach 4 flow at standard sea level conditions over a flat plate of chord 5 in. Assuming all laminar flow and adiabatic wall conditions, calculate the skin friction drag on the plate per unit span.

19.5 Repeat Problem 19.4 for the case of all turbulent flow.

19.6 Consider a compressible, laminar boundary layer over a flat plate. Assuming Pr $= 1$ and a calorically perfect gas, show that the profile of total temperature through the boundary layer is a function of the velocity profile via

$$T_0 = T_w + (T_{0,e} - T_w)\frac{u}{u_e}$$

where $T_w =$ wall temperature and $T_{0,e}$ and u_e are the total temperature and velocity, respectively, at the outer edge of the boundary layer. [*Hint:* Compare Equations (18.32) and (18.41).]

19.7 Consider a high-speed vehicle flying at a standard altitude of 35 km, where the ambient pressure and temperature are 583.59 N/m^2 and 246.1 K, respectively. The radius of the spherical nose of the vehicle is 2.54 cm. Assume the Prandtl number for air at these conditions is 0.72, that c_p is 1008 joules/(kg K), and that the viscosity coefficient is given by Sutherland's law. The wall temperature at the nose is 400 K. Assume the recovery factor at the nose is 1.0. Calculate the aerodynamic heat transfer to the stagnation point for flight velocities of (*a*) 1500 m/s, and (*b*) 4500 m/s. From these results, make a comment about how the heat transfer varies with flight velocity.

CHAPTER 20

Navier-Stokes Solutions: Some Examples

A numerical simulation of the flow over an airfoil using the Reynolds averaged Navier-Stokes equations can be conducted on today's supercomputers in less than a half hour for less than $1000 cost in computer time. If just one such simulation had been attempted 20 years ago on computers of that time (e.g., the IBM 704 class) and with algorithms then known, the cost in computer time would have amounted to roughly $10 million, and the results for that single flow would not be available until 10 years from now, since the computation would have taken about 30 years to complete.

Dean R. Chapman, NASA, 1977

PREVIEW BOX

This is a short chapter about a very extensive subject—the numerical solution of general viscous flows. This is the ultimate method for solving general flow fields including friction and thermal conduction throughout the whole flow field. This chapter is in the purview of computational fluid dynamics—a whole subject in itself. The purpose of this chapter is simply to round out our discussion of viscous flow and to bring some closure to the subject. So read on, and allow yourself to be rounded out.

In so doing, pay close attention to the section dealing with the accuracy of the prediction of skin-friction drag. You will see that no matter how hard we try, there is still room for improvement. This is a good thing. Aerodynamics in general is still an evolving intellectual subject, and there is plenty of room for you to make your own personal contributions to its improvement. I hope that by reading this book you are inspired to do so.

As we say in aerodynamics—onward and upward.

20.1 INTRODUCTION

This chapter is short. Its purpose is to discuss the third option for the solution of viscous flows as discussed in Section 15.7, namely, the exact numerical solution of the complete Navier-Stokes equations. This option is the purview of modern computational fluid dynamics—it is a state-of-the-art research activity which is currently in a rapid state of development. This subject now occupies volumes of modern literature; for a basic treatment, see the definitive text on computational fluid dynamics listed as Reference 54. We will only list a few sample calculations here.

20.2 THE APPROACH

Return to the complete Navier-Stokes equations, as derived in Chapter 15, and repeated and renumbered below for convenience:

Continuity:

$$\frac{\partial \rho}{\partial t} = -\left[\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z}\right] \tag{20.1}$$

x momentum:

$$\frac{\partial u}{\partial t} = -u\frac{\partial u}{\partial x} - v\frac{\partial u}{\partial y} - w\frac{\partial u}{\partial z} + \frac{1}{\rho}\left[-\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}\right] \tag{20.2}$$

y momentum:

$$\frac{\partial v}{\partial t} = -u\frac{\partial v}{\partial x} - v\frac{\partial v}{\partial y} - w\frac{\partial v}{\partial z} + \frac{1}{\rho}\left[-\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z}\right] \tag{20.3}$$

z momentum:

$$\frac{\partial w}{\partial t} = -u\frac{\partial w}{\partial x} - v\frac{\partial w}{\partial y} - w\frac{\partial w}{\partial z} + \frac{1}{\rho}\left[-\frac{\partial p}{\partial z} + \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z}\right] \tag{20.4}$$

Energy:

$$\begin{aligned}\frac{\partial(e + V^2/2)}{\partial t} = &-u\frac{\partial(e + V^2/2)}{\partial x} - v\frac{\partial(e + V^2/2)}{\partial y} - w\frac{\partial(e + V^2/2)}{\partial z} + \dot{q} \\ &+ \frac{1}{\rho}\left[\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right)\right. \\ &- \frac{\partial(pu)}{\partial x} - \frac{\partial(pv)}{\partial y} - \frac{\partial(pw)}{\partial z} + \frac{\partial(u\tau_{xx})}{\partial x} \\ &+ \frac{\partial(u\tau_{yx})}{\partial y} + \frac{\partial(u\tau_{zx})}{\partial z} + \frac{\partial(v\tau_{xy})}{\partial x} + \frac{\partial(v\tau_{yy})}{\partial y} + \frac{\partial(v\tau_{zy})}{\partial z} \\ &\left.+ \frac{\partial(w\tau_{xz})}{\partial x} + \frac{\partial(w\tau_{yz})}{\partial y} + \frac{\partial(w\tau_{zz})}{\partial z}\right]\end{aligned} \tag{20.5}$$

These equations have been written with the time derivatives on the left-hand side and all spatial derivatives on the right-hand side. This is the form suitable to a

time-dependent solution of the equations, as discussed in Chapters 13 and 16. Indeed, Equations (20.1) to (20.5) are partial differential equations that have a mathematically "elliptic" behavior; that is, on a physical basis they treat flow-field information and flow disturbances that can travel throughout the flow field, in both the upstream and downstream directions. The time-dependent technique is particularly suited to such a problem.

The time-dependent solution of Equations (20.1) to (20.5) can be carried out in direct parallel to the discussion in Section 16.4. It is important for you to return to that section and review our discussion of the time-dependent solution of compressible Couette flow using MacCormack's technique. We suggest doing this before reading further. The approach to the solution of Equations (20.1) to (20.5) for other problems is exactly the same. Therefore, we will not elaborate further here.

20.3 EXAMPLES OF SOME SOLUTIONS

In this section, we present samples of a few numerical solutions of the complete Navier-Stokes equations. Most of these solutions have the following in common:

1. They were obtained by means of a time-dependent solution using MacCormack's technique as described in Section 16.4.
2. They utilize the Baldwin-Lomax turbulence model (see Section 19.3.1 for a discussion of this model). Hence, turbulent flow is modeled in these calculations.
3. They require anywhere from thousands to close to a million grid points for their solution. Therefore, these are problems that must be solved on large-scale digital computers.

20.3.1 Flow over a Rearward-Facing Step

The supersonic viscous flow over a rearward-facing step was examined in Reference 46. Some results are shown in Figures 20.1 and 20.2. The flow is moving from left to right. In the velocity vector diagram in Figure 20.1, note the separated, recirculating flow region just downstream of the step. The calculation of such separated flows is the forte of solutions of the complete Navier-Stokes equations. In contrast, the boundary-layer equations discussed in Chapter 17 are not suited for the analysis of separated flows; boundary-layer calculations usually "blow up" in regions of separated flow. Figure 20.2 shows the temperature contours (lines of constant temperature) for the same flow in Figure 20.1.

20.3.2 Flow over an Airfoil

The viscous compressible flow over an airfoil was studied in Reference 56. For the treatment of this problem, a nonrectangular finite-difference grid is wrapped around the airfoil, as shown in Figure 20.3. Equations (20.1) to (20.5) have to be transformed into the new curvilinear coordinate system in Figure 20.3. The details

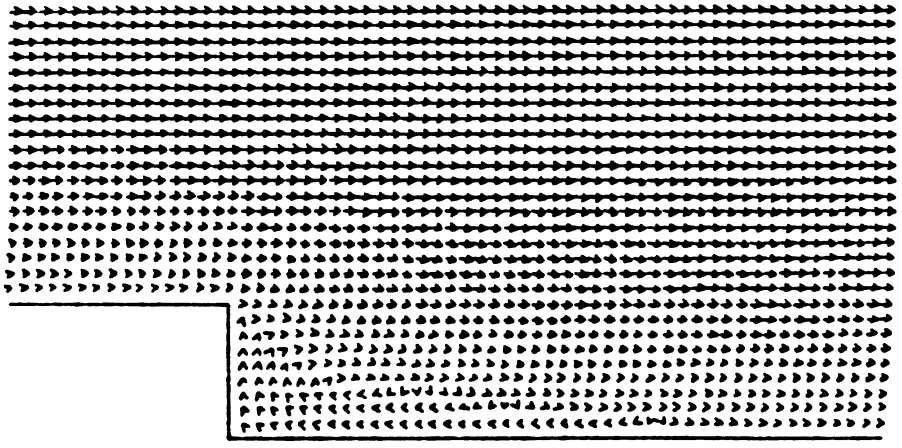


Figure 20.1 Velocity vector diagram for the flow over a rearward-facing step. $M = 2.19$, $T = 1005$ K, Re $= 70{,}000$ (based on step height) *(Reference 46)*. Note the recirculating flow region downstream of the step.

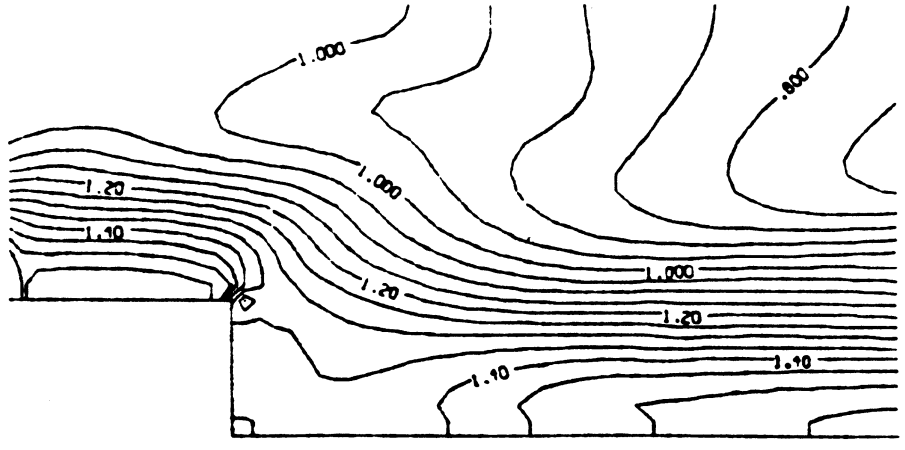


Figure 20.2 Temperature contours for the flow shown in Figure 20.1. The separated region just downstream of the step is a reasonably constant pressure, constant temperature region.

are beyond the scope of this book; see Reference 56 for a complete discussion. Some results for the streamline patterns are shown in Figure 20.4*a* and *b*. Here, the flow over a Wortmann airfoil at zero angle of attack is shown. The freestream Mach number is 0.5, and the Reynolds number based on chord is relatively low, Re $= 100{,}000$. The completely laminar flow over this airfoil is shown in Figure 20.4*a*. Because of the peculiar aerodynamic properties of some low Reynolds number flows over airfoils (see References 51 and 56), we note that the laminar flow separated over both the top and bottom surfaces of the airfoil. However, in Figure 20.4*b*, the turbulence model is turned on for the calculation; note that the flow is now completely attached. The differences in Figure 20.4*a* and *b* vividly

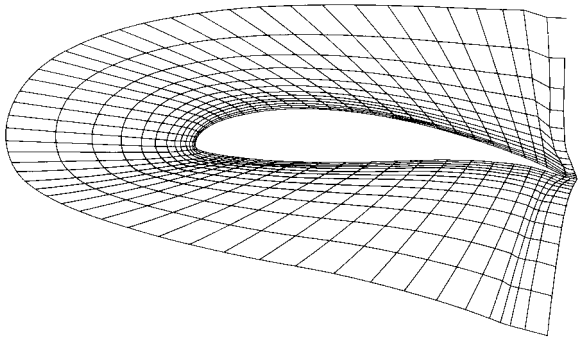
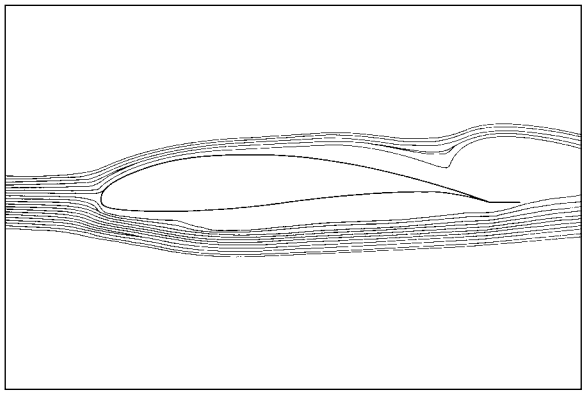
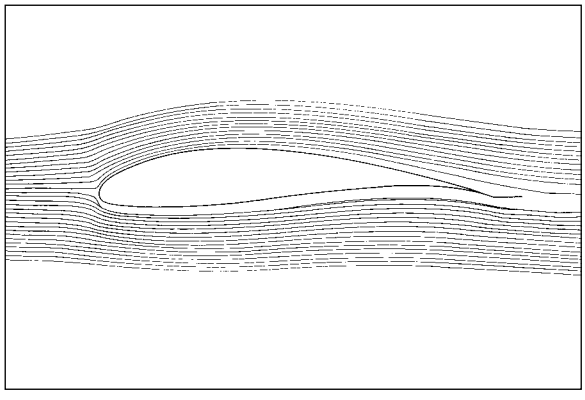


Figure 20.3 Curvilinear, boundary-fitted finite-difference grid for the solution of the flow over an airfoil. (*Data Source: Kothari and Anderson, Reference 56.*)



(*a*)



(*b*)

Figure 20.4 Streamlines for the low Reynolds flow over a Wortmann airfoil. Re = 100,000. (*a*) Laminar flow. (*b*) Turbulent flow. (*Data Source: Kothari and Anderson, Reference 56.*)

demonstrate the basic trend that turbulent flow resists flow separation much more strongly than laminar flow.

20.3.3 Flow over a Complete Airplane

We end this section by noting a history-making calculation. In Reference 57, a solution of the complete Navier-Stokes equations for the flow field over an entire airplane was reported—the first such calculation ever made. In this work, Shang and Scherr carried out a time-dependent solution using MacCormack's method—just as we have discussed it in Section 16.4. See Reference 57 for the details. Also, a lengthy description of this work can be found in Chapter 8 of Reference 55. Shang and Scherr applied their calculation to the hypersonic viscous flow over the X-24C hypersonic test vehicle. To illustrate the results, the surface streamline pattern is shown in Figure 20.5, as calculated in Reference 57. In reality, since

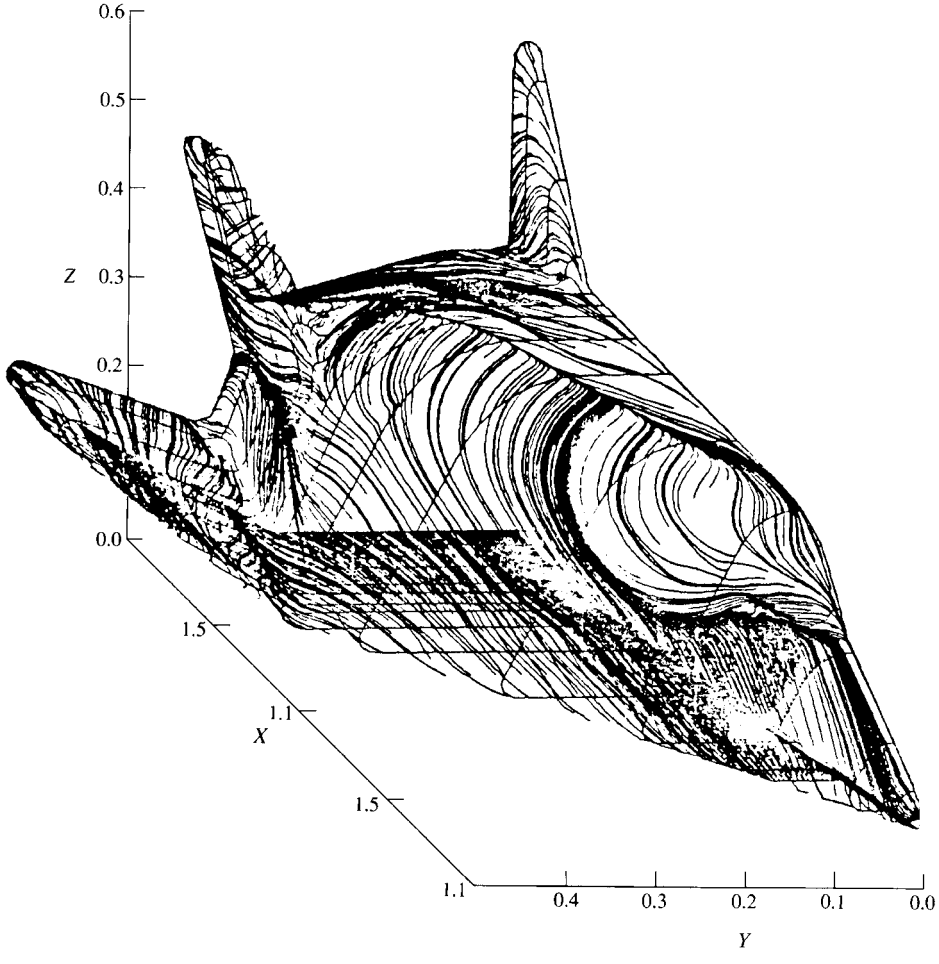


Figure 20.5 Surface shear stress lines on the X-24C. (*Data Source: Shang and Scherr, Reference 57.*)

the flow velocity is zero at the surface in a viscous flow (the no-slip condition), the lines shown in Figure 20.5 are the surface shear stress directions.

20.3.4 Shock-Wave/Boundary-Layer Interaction

The flow field that results when a shock wave impinges on a boundary layer can only be calculated in detail by means of a numerical solution of the complete Navier-Stokes equations. The qualitative physical aspects of a two-dimensional shock-wave/boundary-layer interaction are sketched in Figure 20.6. Here we see a boundary layer growing along a flat plate, where at some downstream location an incident shock wave impinges on the boundary layer. The large pressure rise across the shock wave acts as a severe adverse pressure gradient imposed on the boundary layer, thus causing the boundary layer to locally separate from the surface. Because the high pressure behind the shock feeds upstream through the subsonic portion of the boundary layer, the separation takes place ahead of the impingement point of the incident shock wave. In turn, the separated boundary layer induces a shock wave, identified here as the induced separation shock. The separated boundary layer subsequently turns back toward the plate, reattaching to the surface at the reattachment shock. Between the separation and reattachment shocks, expansion waves are generated where the boundary layer is turning back toward the surface. At the point of reattachment, the boundary layer has become relatively thin, the pressure is high, and consequently this becomes a region of high local aerodynamic heating. Further away from the plate, the separation and reattachment shocks merge to form the conventional "reflected shock wave" that is expected from the classical inviscid picture (see, for example, Figure 9.17). The scale and severity of the interaction picture shown in Figure 20.6 depends on whether the boundary layer is laminar or turbulent. Since laminar boundary layers

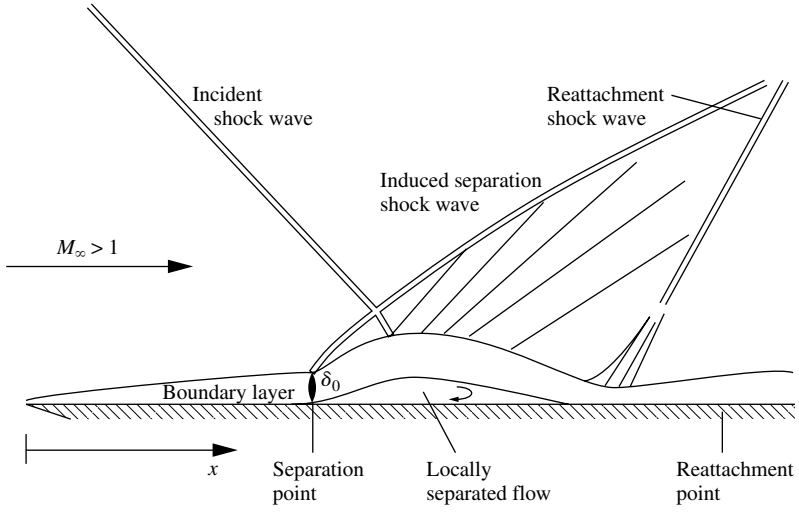


Figure 20.6 Schematic of the shock-wave/boundary-layer interaction.

separate more readily than turbulent boundary layers, the laminar interaction usually takes place more readily with more severe attendant consequences than the turbulent interaction. However, the general qualitative aspects of the interaction as sketched in Figure 20.6 are the same.

The first discussion in this book of the physical aspects of shock-wave/boundary-layer interactions can be found in Section 9.9, with a specific application to shock waves inside nozzles in Section 10.6. If you have not read Sections 9.9 and 10.6, now is the time to pause and read these two sections.

The fluid dynamic and mathematical details of the interaction region sketched in Figure 20.6 are complex, and the full prediction of this flow is still a state-of-the-art research problem. However, great strides have been made in recent years with the application of computational fluid dynamics to this problem, and solutions of the full Navier-Stokes equations for the flow sketched in Figure 20.6 have been obtained. For example, experimental and computational data for the two-dimensional interaction of a shock wave impinging on a turbulent flat plate boundary layer are given in Figure 20.7, obtained from Reference 86. In Figure 20.7*a*, the ratio of surface pressure to freestream total pressure is plotted versus distance along the surface (nondimensionalized by δ_0, the boundary-layer thickness ahead of the interaction). Here, x_0 is taken as the theoretical inviscid flow impingement point for the incident shock wave. The freestream Mach number is 3. The Reynolds number based on δ_0 is about 10^6. Note in Figure 20.7*a* that the surface pressure first increases at the front of the interaction region (ahead of the theoretical incident shock impingement point), reaches a plateau through the center of the separated region, and then increases again as the reattachment point is approached. The pressure variation shown in Figure 20.7*a* is typical of that for a two-dimensional shock-wave/boundary-layer interaction. The open circles correspond to experimental measurements of Reda and Murphy (Reference 88). The curve is obtained from a numerical solution of the Navier-Stokes equations as reported in Reference 86 and using the Baldwin-Lomax turbulence model discussed in Section 19.3.1. In Figure 20.7*b* the variation of surface shear stress plummets to zero, reverses its direction (negative values) in a rather complex variation, and then recovers to a positive value in the vicinity of the reattachment point. The two circles on the horizontal axis denote measured separation and reattachment points, and the curve is obtained from the calculations of Reference 86.

20.3.5 Flow over an Airfoil with a Protuberance

Here we show some very recent Navier-Stokes solutions carried out to study the aerodynamic effect of a small protuberance extending from the bottom surface of an airfoil. These calculations represent an example of the state-of-the-art of full Navier-Stokes solutions at the time of writing. The work was carried out by Beierle (Reference 89). The basic shape of the airfoil was an NACA 0015 section. The computational fluid dynamic solution of the Navier-Stokes equations was carried out using a time-marching finite volume code labeled OVERFLOW, developed by NASA (Reference 90). The flow was low speed, with a freestream Mach number

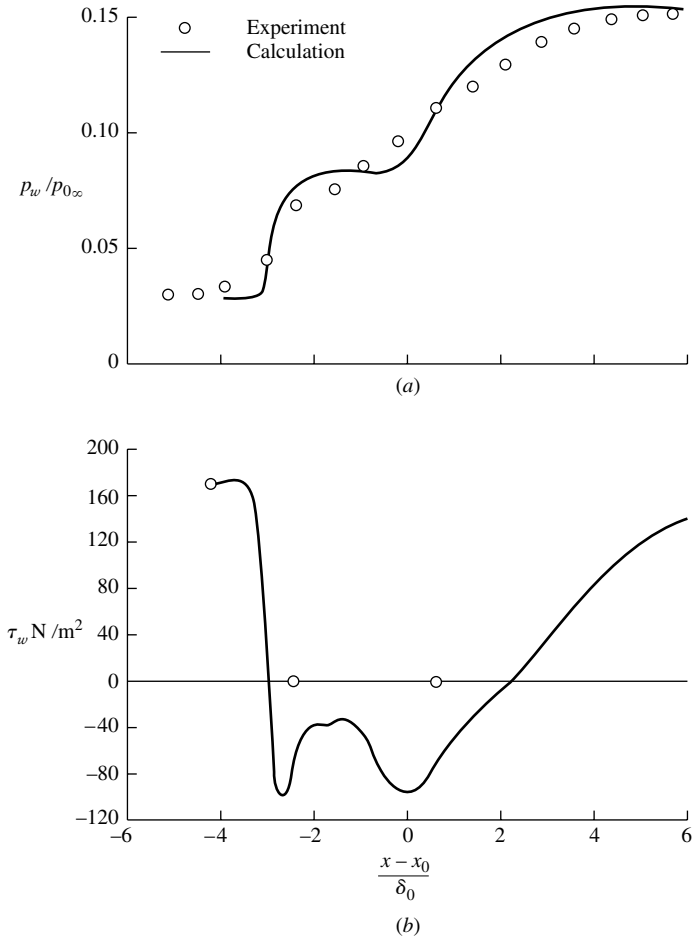


Figure 20.7 Effects of shock-wave/boundary-layer interaction on (*a*) pressure distribution, and (*b*) shear stress for Mach 3 turbulent flow over a flat plate.

of 0.15 and Reynolds number of 1.5×10^6. The fully turbulent flow field was simulated using a one-equation turbulence model.

Using a proper grid is vital to the integrity of any Navier-Stokes CFD solution. For the present case, Figures 20.8–20.11 show the grid used, progressing from the big picture of the whole grid (Figure 20.8) to the detail of the grid around the small protuberance on the bottom surface of the airfoil (Figure 20.11). The grid is an example of a chimera grid, a series of independent but overlapping grids that are generated about individual parts of the body and for specific flow regions.

Some results for the computed flow field are shown in Figures 20.12 and 20.13. In Figure 20.12, the local velocity vector field is shown; the flow separation

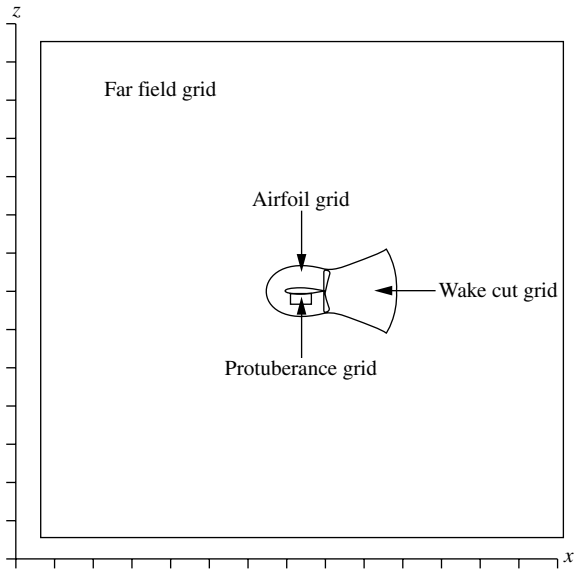


Figure 20.8 Individual grid boundary outlines used in the chimera grid scheme for calculating the flow over an airfoil with a protuberance.

and locally reversed flow can be seen downstream of the protuberance. In Figure 20.13, pressure contours are shown, illustrating how the small protuberance generates a substantially asymmetric flow over the otherwise symmetric airfoil.

Finally, results for a related flow are shown in Figure 20.14. Here, instead of a protuberance existing on the bottom surface, an array of small jets that are distributed over the bottom surface alternately blow and suck air into and out of the flow in such a manner that the net mass flow added is zero, so-called "zero-mass synthetic jets." The resulting series of large-scale vortices is shown in Figure 20.14—another example of a flow field that can only be solved in detail by means of a full Navier-Stokes solution. (See Hassan and JanakiRam, Reference 91, for details.)

20.4 THE ISSUE OF ACCURACY FOR THE PREDICTION OF SKIN FRICTION DRAG

The aerodynamic drag on a body is the sum of pressure drag and skin friction drag. For attached flows, the prediction of pressure drag is obtained from inviscid flow analyses such as those presented in Parts 2 and 3 of this book. For separated flows, various approximate theories for pressure drag have been advanced over the last century, but today the only viable and general method of the analysis of pressure drag for such flows is a complete numerical Navier-Stokes solution.

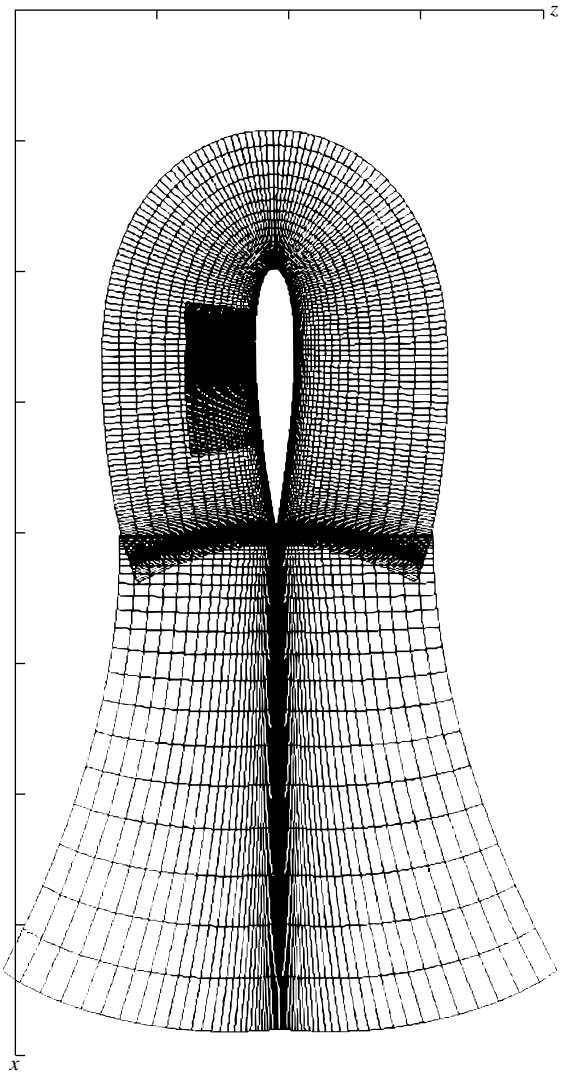


Figure 20.9 Zoom view of the airfoil, wake cut, and protuberance grids.

The prediction of skin friction on the surface of a body in an attached flow is nicely accomplished by means of a boundary-layer solution coupled with an inviscid flow analysis to define the flow conditions at the edge of the boundary layer. Such an approach is well-developed, and the calculations can be rapidly carried out on local computer workstations. Therefore, the use of boundary-layer solutions for skin friction and aerodynamic heating is the preferred engineering approach. However, as mentioned above, if regions of flow separation are present,

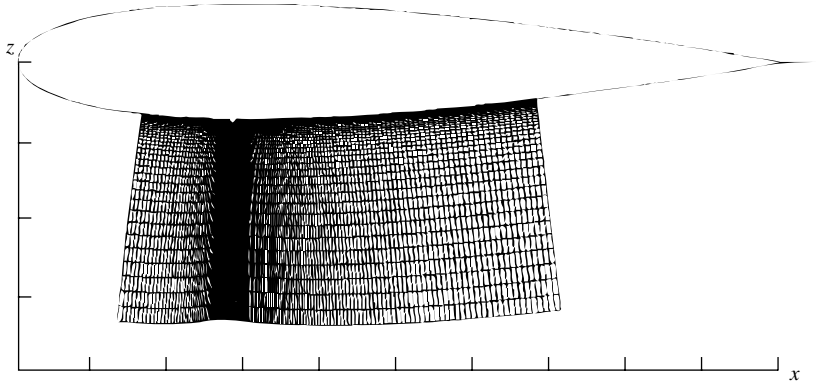


Figure 20.10 Zoom view of protuberance grid along the bottom surface of the airfoil.

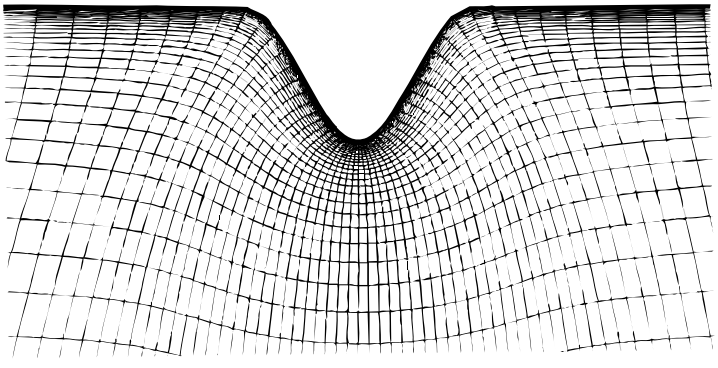


Figure 20.11 A detail of the grid in the vicinity of the protuberance.

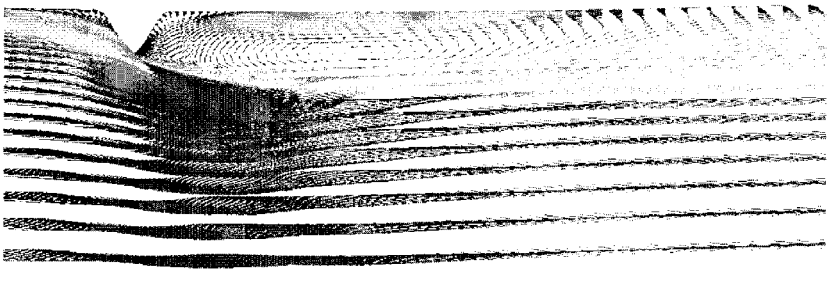


Figure 20.12 Computed velocity vector field around and downstream of the protuberance.

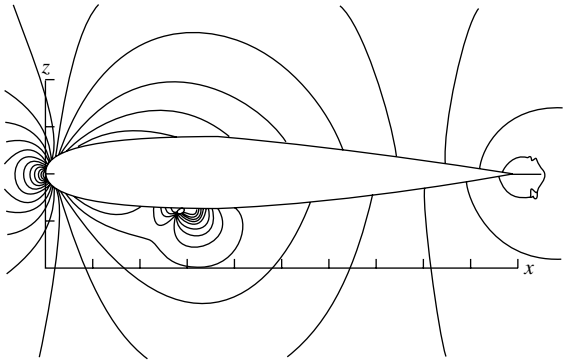


Figure 20.13 Computed pressure contours around the NACA 0012 airfoil with a protuberance.



Figure 20.14 Streamline pattern over the bottom of an NACA 0012 airfoil, over which a series of jets is distributed which are alternatively blowing or sucking mass into or out of the flow, with the overall net mass injected into the flow being zero—a zero-mass jet array. (*Data Source: Hassan and JanakiRam, Reference 91.*)

this approach cannot be used. In its place, a full Navier-Stokes solution can be used to obtain local skin friction and heat transfer, but these Navier-Stokes solutions are still not in the category of "quick engineering calculations."

This leads us to the question of the accuracy of CFD Navier-Stokes solutions for skin friction drag and heat transfer. There are three aspects that tend to diminish the accuracy of such solutions for the prediction of τ_w and $\dot{q}_w$ (or alternately, c_f and C_H):

1. The need to have a *very closely spaced grid in the vicinity of the wall* in order to obtain an accurate numerical value of $(\partial u/\partial y)_w$ and $(\partial T/\partial y)_w$, from which τ_w and $\dot{q}_w$ are obtained.
2. The uncertainty in the accuracy of turbulence models when a turbulent flow is being calculated.
3. The lack of ability of most turbulent models to predict transition from laminar to turbulent flow.

In spite of all the advances made in CFD to the present, and all the work that has gone into turbulence modeling, at the time of writing the ability of Navier-Stokes solutions to predict skin friction in a turbulent flow seems to be no better than about 20 percent accuracy, on the average. A study by Lombardi et al.

(Reference 92) has made this clear. They calculated the skin friction drag on an NACA 0012 airfoil at zero angle of attack in a low-speed flow using both a standard boundary-layer code and a state-of-the-art Navier-Stokes solver with three different state-of-the-art turbulence models. The results for friction drag from the boundary-layer code had been validated with experiment, and were considered the baseline for accuracy. The boundary-layer code also had a prediction for transition that was considered reliable. Some typical results reported in Reference 92 for the integrated friction drag coefficient C_f are as follows, where NS represents Navier-Stokes solver and with the turbulence model in parenthesis. The calculations were all for $\text{Re} = 3 \times 10^6$.

	$C_f \times 10^3$
NS (Standard $k - \varepsilon$)	7.486
NS (RNG $k - \varepsilon$)	6.272
NS (Reynolds stress)	6.792
Boundary Layer Solution	5.340

Clearly, the accuracy of the various Navier-Stokes calculations ranged from 18 percent to 40 percent.

More insight can be gained from the spatial distribution of the local skin-friction coefficient c_f along the surface of the airfoil, as shown in Figure 20.15. Again the three different Navier-Stokes calculations are compared with the results from the boundary layer code. All the Navier-Stokes calculations greatly overestimated the peak in c_f just downstream of the leading edge, and slightly underestimated c_f near the trailing edge.

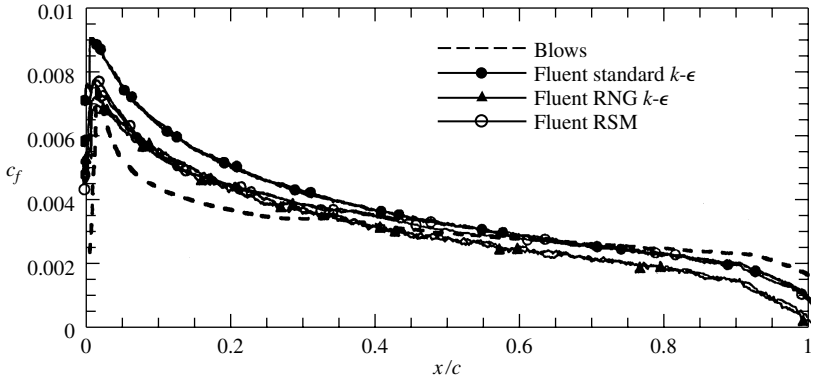


Figure 20.15 Distribution of the skin-friction coefficient over the surface of an NACA 0012 airfoil at zero angle of attack in low-speed flow. Comparison of three Navier-Stokes calculations using different turbulent models, and results obtained from a boundary-layer calculation. The boundary-layer results are given by the dashed curve labeled "Blows." (*Data Source: Lombardi, Salvetti, and Pinelli, Reference 92.*)

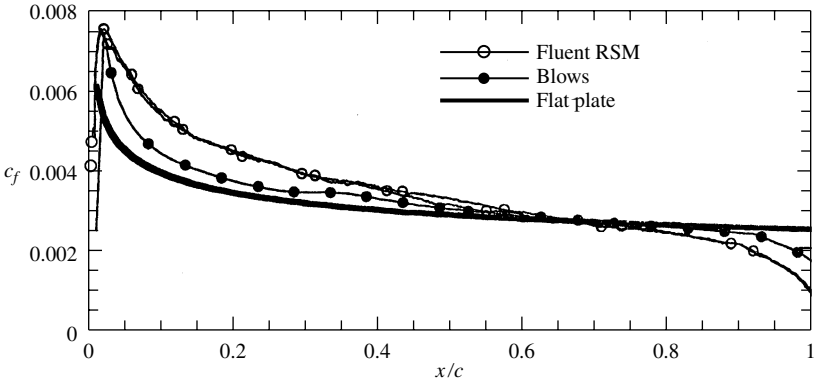


Figure 20.16 Computed skin-friction coefficient distributions over an NACA 0012 airfoil, comparing results from a Navier-Stokes solution and a boundary-layer solution. The heavy curve is for a flat plate, allowing a comparison with the skin friction distribution over a curved airfoil surface. (*Data Source: Lombardi et al., Reference 92.*)

For a completely different reason not having to do with our discussion of accuracy, but for purposes of showing and contrasting the physically different distribution of c_f along a flat plate compared with that along the surface of the airfoil, we show Figure 20.16. Here the heavy curve is the variation of c_f with distance from the leading edge for a flat plate; the monotonic decrease is expected from our previous discussions of flat plate boundary layers. In contrast, for the airfoil c_f rapidly increases from a value of zero at the stagnation point to a peak value shortly downstream of the leading edge. This rapid increase is due to the rapidly increasing velocity as the flow external to the boundary layer rapidly expands around the leading edge. Beyond the peak, c_f then monotonically decreases in the same qualitative manner as for a flat plate. It is simply interesting to note these different variations for c_f over an airfoil compared to that for a flat plate, especially since we devoted so much attention to flat plates in the previous chapters.

20.5 SUMMARY

With this, we end our discussion of viscous flow. The purpose of all of Part 4 has been to introduce you to the basic aspects of viscous flow. The subject is so vast that it demands a book in itself—many of which have been written (see, e.g., References 41 through 45). Here, we have presented only enough material to give you a flavor for some of the basic ideas and results. This is a subject of great importance in aerodynamics, and if you wish to expand your knowledge and expertise of aerodynamics in general, we encourage you to read further on the subject.

We are also out of our allotted space for this book. Therefore, we hope that you have enjoyed and benefited from our presentation of the fundamentals of

aerodynamics. However, before closing the cover, it might be useful to return once again to Figure 1.44, which is the block diagram categorizing the different general types of aerodynamic flows. Recall the curious, uninitiated thoughts you might have had when you first examined this figure during your study of Chapter 1, and compare these with the informed and mature thoughts that you now have—honed by the aerodynamic knowledge packed into the intervening pages. Hopefully, each block in Figure 1.44 has substantially more meaning for you now than when we first started. If this is true, then my efforts as an author have not gone in vain.

APPENDIX A

Isentropic Flow Properties

M	$\frac{p_0}{p}$	$\frac{\rho_0}{\rho}$	$\frac{T_0}{T}$	$\frac{A}{A^*}$
0.2000 − 01	0.1000 + 01	0.1000 + 01	0.1000 + 01	0.2894 + 02
0.4000 − 01	0.1001 + 01	0.1001 + 01	0.1000 + 01	0.1448 + 02
0.6000 − 01	0.1003 + 01	0.1002 + 01	0.1001 + 01	0.9666 + 01
0.8000 − 01	0.1004 + 01	0.1003 + 01	0.1001 + 01	0.7262 + 01
0.1000 + 00	0.1007 + 01	0.1005 + 01	0.1002 + 01	0.5822 + 01
0.1200 + 00	0.1010 + 01	0.1007 + 01	0.1003 + 01	0.4864 + 01
0.1400 + 00	0.1014 + 01	0.1010 + 01	0.1004 + 01	0.4182 + 01
0.1600 + 00	0.1018 + 01	0.1013 + 01	0.1005 + 01	0.3673 + 01
0.1800 + 00	0.1023 + 01	0.1016 + 01	0.1006 + 01	0.3278 + 01
0.2000 + 00	0.1028 + 01	0.1020 + 01	0.1008 + 01	0.2964 + 01
0.2200 + 00	0.1034 + 01	0.1024 + 01	0.1010 + 01	0.2708 + 01
0.2400 + 00	0.1041 + 01	0.1029 + 01	0.1012 + 01	0.2496 + 01
0.2600 + 00	0.1048 + 01	0.1034 + 01	0.1014 + 01	0.2317 + 01
0.2800 + 00	0.1056 + 01	0.1040 + 01	0.1016 + 01	0.2166 + 01
0.3000 + 00	0.1064 + 01	0.1046 + 01	0.1018 + 01	0.2035 + 01
0.3200 + 00	0.1074 + 01	0.1052 + 01	0.1020 + 01	0.1922 + 01
0.3400 + 00	0.1083 + 01	0.1059 + 01	0.1023 + 01	0.1823 + 01
0.3600 + 00	0.1094 + 01	0.1066 + 01	0.1026 + 01	0.1736 + 01
0.3800 + 00	0.1105 + 01	0.1074 + 01	0.1029 + 01	0.1659 + 01
0.4000 + 00	0.1117 + 01	0.1082 + 01	0.1032 + 01	0.1590 + 01
0.4200 + 00	0.1129 + 01	0.1091 + 01	0.1035 + 01	0.1529 + 01
0.4400 + 00	0.1142 + 01	0.1100 + 01	0.1039 + 01	0.1474 + 01
0.4600 + 00	0.1156 + 01	0.1109 + 01	0.1042 + 01	0.1425 + 01
0.4800 + 00	0.1171 + 01	0.1119 + 01	0.1046 + 01	0.1380 + 01
0.5000 + 00	0.1186 + 01	0.1130 + 01	0.1050 + 01	0.1340 + 01
0.5200 + 00	0.1202 + 01	0.1141 + 01	0.1054 + 01	0.1303 + 01
0.5400 + 00	0.1219 + 01	0.1152 + 01	0.1058 + 01	0.1270 + 01
0.5600 + 00	0.1237 + 01	0.1164 + 01	0.1063 + 01	0.1240 + 01
0.5800 + 00	0.1256 + 01	0.1177 + 01	0.1067 + 01	0.1213 + 01
0.6000 + 00	0.1276 + 01	0.1190 + 01	0.1072 + 01	0.1188 + 01

M	$\frac{p_0}{p}$	$\frac{\rho_0}{\rho}$	$\frac{T_0}{T}$	$\frac{A}{A^*}$
0.6200 + 00	0.1296 + 01	0.1203 + 01	0.1077 + 01	0.1166 + 01
0.6400 + 00	0.1317 + 01	0.1218 + 01	0.1082 + 01	0.1145 + 01
0.6600 + 00	0.1340 + 01	0.1232 + 01	0.1087 + 01	0.1127 + 01
0.6800 + 00	0.1363 + 01	0.1247 + 01	0.1092 + 01	0.1110 + 01
0.7000 + 00	0.1387 + 01	0.1263 + 01	0.1098 + 01	0.1094 + 01
0.7200 + 00	0.1412 + 01	0.1280 + 01	0.1104 + 01	0.1081 + 01
0.7400 + 00	0.1439 + 01	0.1297 + 01	0.1110 + 01	0.1068 + 01
0.7600 + 00	0.1466 + 01	0.1314 + 01	0.1116 + 01	0.1057 + 01
0.7800 + 00	0.1495 + 01	0.1333 + 01	0.1122 + 01	0.1047 + 01
0.8000 + 00	0.1524 + 01	0.1351 + 01	0.1128 + 01	0.1038 + 01
0.8200 + 00	0.1555 + 01	0.1371 + 01	0.1134 + 01	0.1030 + 01
0.8400 + 00	0.1587 + 01	0.1391 + 01	0.1141 + 01	0.1024 + 01
0.8600 + 00	0.1621 + 01	0.1412 + 01	0.1148 + 01	0.1018 + 01
0.8800 + 00	0.1655 + 01	0.1433 + 01	0.1155 + 01	0.1013 + 01
0.9000 + 00	0.1691 + 01	0.1456 + 01	0.1162 + 01	0.1009 + 01
0.9200 + 00	0.1729 + 01	0.1478 + 01	0.1169 + 01	0.1006 + 01
0.9400 + 00	0.1767 + 01	0.1502 + 01	0.1177 + 01	0.1003 + 01
0.9600 + 00	0.1808 + 01	0.1526 + 01	0.1184 + 01	0.1001 + 01
0.9800 + 00	0.1850 + 01	0.1552 + 01	0.1192 + 01	0.1000 + 01
0.1000 + 01	0.1893 + 01	0.1577 + 01	0.1200 + 01	0.1000 + 01
0.1020 + 01	0.1938 + 01	0.1604 + 01	0.1208 + 01	0.1000 + 01
0.1040 + 01	0.1985 + 01	0.1632 + 01	0.1216 + 01	0.1001 + 01
0.1060 + 01	0.2033 + 01	0.1660 + 01	0.1225 + 01	0.1003 + 01
0.1080 + 01	0.2083 + 01	0.1689 + 01	0.1233 + 01	0.1005 + 01
0.1100 + 01	0.2135 + 01	0.1719 + 01	0.1242 + 01	0.1008 + 01
0.1120 + 01	0.2189 + 01	0.1750 + 01	0.1251 + 01	0.1011 + 01
0.1140 + 01	0.2245 + 01	0.1782 + 01	0.1260 + 01	0.1015 + 01
0.1160 + 01	0.2303 + 01	0.1814 + 01	0.1269 + 01	0.1020 + 01
0.1180 + 01	0.2363 + 01	0.1848 + 01	0.1278 + 01	0.1025 + 01
0.1200 + 01	0.2425 + 01	0.1883 + 01	0.1288 + 01	0.1030 + 01
0.1220 + 01	0.2489 + 01	0.1918 + 01	0.1298 + 01	0.1037 + 01
0.1240 + 01	0.2556 + 01	0.1955 + 01	0.1308 + 01	0.1043 + 01
0.1260 + 01	0.2625 + 01	0.1992 + 01	0.1318 + 01	0.1050 + 01
0.1280 + 01	0.2697 + 01	0.2031 + 01	0.1328 + 01	0.1058 + 01
0.1300 + 01	0.2771 + 01	0.2071 + 01	0.1338 + 01	0.1066 + 01
0.1320 + 01	0.2847 + 01	0.2112 + 01	0.1348 + 01	0.1075 + 01
0.1340 + 01	0.2927 + 01	0.2153 + 01	0.1359 + 01	0.1084 + 01
0.1360 + 01	0.3009 + 01	0.2197 + 01	0.1370 + 01	0.1094 + 01
0.1380 + 01	0.3094 + 01	0.2241 + 01	0.1381 + 01	0.1104 + 01
0.1400 + 01	0.3182 + 01	0.2286 + 01	0.1392 + 01	0.1115 + 01
0.1420 + 01	0.3273 + 01	0.2333 + 01	0.1403 + 01	0.1126 + 01
0.1440 + 01	0.3368 + 01	0.2381 + 01	0.1415 + 01	0.1138 + 01
0.1460 + 01	0.3465 + 01	0.2430 + 01	0.1426 + 01	0.1150 + 01
0.1480 + 01	0.3566 + 01	0.2480 + 01	0.1438 + 01	0.1163 + 01
0.1500 + 01	0.3671 + 01	0.2532 + 01	0.1450 + 01	0.1176 + 01
0.1520 + 01	0.3779 + 01	0.2585 + 01	0.1462 + 01	0.1190 + 01
0.1540 + 01	0.3891 + 01	0.2639 + 01	0.1474 + 01	0.1204 + 01
0.1560 + 01	0.4007 + 01	0.2695 + 01	0.1487 + 01	0.1219 + 01
0.1580 + 01	0.4127 + 01	0.2752 + 01	0.1499 + 01	0.1234 + 01
0.1600 + 01	0.4250 + 01	0.2811 + 01	0.1512 + 01	0.1250 + 01

M	$\frac{p_0}{p}$	$\frac{\rho_0}{\rho}$	$\frac{T_0}{T}$	$\frac{A}{A^*}$
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0.1640 + 01	0.4511 + 01	0.2933 + 01	0.1538 + 01	0.1284 + 01
0.1660 + 01	0.4648 + 01	0.2996 + 01	0.1551 + 01	0.1301 + 01
0.1680 + 01	0.4790 + 01	0.3061 + 01	0.1564 + 01	0.1319 + 01
0.1700 + 01	0.4936 + 01	0.3128 + 01	0.1578 + 01	0.1338 + 01
0.1720 + 01	0.5087 + 01	0.3196 + 01	0.1592 + 01	0.1357 + 01
0.1740 + 01	0.5244 + 01	0.3266 + 01	0.1606 + 01	0.1376 + 01
0.1760 + 01	0.5406 + 01	0.3338 + 01	0.1620 + 01	0.1397 + 01
0.1780 + 01	0.5573 + 01	0.3411 + 01	0.1634 + 01	0.1418 + 01
0.1800 + 01	0.5746 + 01	0.3487 + 01	0.1648 + 01	0.1439 + 01
0.1820 + 01	0.5924 + 01	0.3564 + 01	0.1662 + 01	0.1461 + 01
0.1840 + 01	0.6109 + 01	0.3643 + 01	0.1677 + 01	0.1484 + 01
0.1860 + 01	0.6300 + 01	0.3723 + 01	0.1692 + 01	0.1507 + 01
0.1880 + 01	0.6497 + 01	0.3806 + 01	0.1707 + 01	0.1531 + 01
0.1900 + 01	0.6701 + 01	0.3891 + 01	0.1722 + 01	0.1555 + 01
0.1920 + 01	0.6911 + 01	0.3978 + 01	0.1737 + 01	0.1580 + 01
0.1940 + 01	0.7128 + 01	0.4067 + 01	0.1753 + 01	0.1606 + 01
0.1960 + 01	0.7353 + 01	0.4158 + 01	0.1768 + 01	0.1633 + 01
0.1980 + 01	0.7585 + 01	0.4251 + 01	0.1784 + 01	0.1660 + 01
0.2000 + 01	0.7824 + 01	0.4347 + 01	0.1800 + 01	0.1687 + 01
0.2050 + 01	0.8458 + 01	0.4596 + 01	0.1840 + 01	0.1760 + 01
0.2100 + 01	0.9145 + 01	0.4859 + 01	0.1882 + 01	0.1837 + 01
0.2150 + 01	0.9888 + 01	0.5138 + 01	0.1924 + 01	0.1919 + 01
0.2200 + 01	0.1069 + 02	0.5433 + 01	0.1968 + 01	0.2005 + 01
0.2250 + 01	0.1156 + 02	0.5746 + 01	0.2012 + 01	0.2096 + 01
0.2300 + 01	0.1250 + 02	0.6076 + 01	0.2058 + 01	0.2193 + 01
0.2350 + 01	0.1352 + 02	0.6425 + 01	0.2104 + 01	0.2295 + 01
0.2400 + 01	0.1462 + 02	0.6794 + 01	0.2152 + 01	0.2403 + 01
0.2450 + 01	0.1581 + 02	0.7183 + 01	0.2200 + 01	0.2517 + 01
0.2500 + 01	0.1709 + 02	0.7594 + 01	0.2250 + 01	0.2637 + 01
0.2550 + 01	0.1847 + 02	0.8027 + 01	0.2300 + 01	0.2763 + 01
0.2600 + 01	0.1995 + 02	0.8484 + 01	0.2352 + 01	0.2896 + 01
0.2650 + 01	0.2156 + 02	0.8965 + 01	0.2404 + 01	0.3036 + 01
0.2700 + 01	0.2328 + 02	0.9472 + 01	0.2458 + 01	0.3183 + 01
0.2750 + 01	0.2514 + 02	0.1001 + 02	0.2512 + 01	0.3338 + 01
0.2800 + 01	0.2714 + 02	0.1057 + 02	0.2568 + 01	0.3500 + 01
0.2850 + 01	0.2929 + 02	0.1116 + 02	0.2624 + 01	0.3671 + 01
0.2900 + 01	0.3159 + 02	0.1178 + 02	0.2682 + 01	0.3850 + 01
0.2950 + 01	0.3407 + 02	0.1243 + 02	0.2740 + 01	0.4038 + 01
0.3000 + 01	0.3673 + 02	0.1312 + 02	0.2800 + 01	0.4235 + 01
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0.3100 + 01	0.4265 + 02	0.1459 + 02	0.2922 + 01	0.4657 + 01
0.3150 + 01	0.4593 + 02	0.1539 + 02	0.2984 + 01	0.4884 + 01
0.3200 + 01	0.4944 + 02	0.1622 + 02	0.3048 + 01	0.5121 + 01
0.3250 + 01	0.5320 + 02	0.1709 + 02	0.3112 + 01	0.5369 + 01
0.3300 + 01	0.5722 + 02	0.1800 + 02	0.3178 + 01	0.5629 + 01
0.3350 + 01	0.6152 + 02	0.1896 + 02	0.3244 + 01	0.5900 + 01
0.3400 + 01	0.6612 + 02	0.1996 + 02	0.3312 + 01	0.6184 + 01
0.3450 + 01	0.7103 + 02	0.2101 + 02	0.3380 + 01	0.6480 + 01
0.3500 + 01	0.7627 + 02	0.2211 + 02	0.3450 + 01	0.6790 + 01

M	$\frac{p_0}{p}$	$\frac{\rho_0}{\rho}$	$\frac{T_0}{T}$	$\frac{A}{A^*}$
0.3550 + 01	0.8187 + 02	0.2325 + 02	0.3520 + 01	0.7113 + 01
0.3600 + 01	0.8784 + 02	0.2445 + 02	0.3592 + 01	0.7450 + 01
0.3650 + 01	0.9420 + 02	0.2571 + 02	0.3664 + 01	0.7802 + 01
0.3700 + 01	0.1010 + 03	0.2701 + 02	0.3738 + 01	0.8169 + 01
0.3750 + 01	0.1082 + 03	0.2838 + 02	0.3812 + 01	0.8552 + 01
0.3800 + 01	0.1159 + 03	0.2981 + 02	0.3888 + 01	0.8951 + 01
0.3850 + 01	0.1241 + 03	0.3129 + 02	0.3964 + 01	0.9366 + 01
0.3900 + 01	0.1328 + 03	0.3285 + 02	0.4042 + 01	0.9799 + 01
0.3950 + 01	0.1420 + 03	0.3446 + 02	0.4120 + 01	0.1025 + 02
0.4000 + 01	0.1518 + 03	0.3615 + 02	0.4200 + 01	0.1072 + 02
0.4050 + 01	0.1623 + 03	0.3791 + 02	0.4280 + 01	0.1121 + 02
0.4100 + 01	0.1733 + 03	0.3974 + 02	0.4362 + 01	0.1171 + 02
0.4150 + 01	0.1851 + 03	0.4164 + 02	0.4444 + 01	0.1224 + 02
0.4200 + 01	0.1975 + 03	0.4363 + 02	0.4528 + 01	0.1279 + 02
0.4250 + 01	0.2108 + 03	0.4569 + 02	0.4612 + 01	0.1336 + 02
0.4300 + 01	0.2247 + 03	0.4784 + 02	0.4698 + 01	0.1395 + 02
0.4350 + 01	0.2396 + 03	0.5007 + 02	0.4784 + 01	0.1457 + 02
0.4400 + 01	0.2553 + 03	0.5239 + 02	0.4872 + 01	0.1521 + 02
0.4450 + 01	0.2719 + 03	0.5480 + 02	0.4960 + 01	0.1587 + 02
0.4500 + 01	0.2894 + 03	0.5731 + 02	0.5050 + 01	0.1656 + 02
0.4550 + 01	0.3080 + 03	0.5991 + 02	0.5140 + 01	0.1728 + 02
0.4600 + 01	0.3276 + 03	0.6261 + 02	0.5232 + 01	0.1802 + 02
0.4650 + 01	0.3483 + 03	0.6542 + 02	0.5324 + 01	0.1879 + 02
0.4700 + 01	0.3702 + 03	0.6833 + 02	0.5418 + 01	0.1958 + 02
0.4750 + 01	0.3933 + 03	0.7135 + 02	0.5512 + 01	0.2041 + 02
0.4800 + 01	0.4177 + 03	0.7448 + 02	0.5608 + 01	0.2126 + 02
0.4850 + 01	0.4434 + 03	0.7772 + 02	0.5704 + 01	0.2215 + 02
0.4900 + 01	0.4705 + 03	0.8109 + 02	0.5802 + 01	0.2307 + 02
0.4950 + 01	0.4990 + 03	0.8457 + 02	0.5900 + 01	0.2402 + 02
0.5000 + 01	0.5291 + 03	0.8818 + 02	0.6000 + 01	0.2500 + 02
0.5100 + 01	0.5941 + 03	0.9579 + 02	0.6202 + 01	0.2707 + 02
0.5200 + 01	0.6661 + 03	0.1039 + 03	0.6408 + 01	0.2928 + 02
0.5300 + 01	0.7457 + 03	0.1127 + 03	0.6618 + 01	0.3165 + 02
0.5400 + 01	0.8335 + 03	0.1220 + 03	0.6832 + 01	0.3417 + 02
0.5500 + 01	0.9304 + 03	0.1320 + 03	0.7050 + 01	0.3687 + 02
0.5600 + 01	0.1037 + 04	0.1426 + 03	0.7272 + 01	0.3974 + 02
0.5700 + 01	0.1154 + 04	0.1539 + 03	0.7498 + 01	0.4280 + 02
0.5800 + 01	0.1283 + 04	0.1660 + 03	0.7728 + 01	0.4605 + 02
0.5900 + 01	0.1424 + 04	0.1789 + 03	0.7962 + 01	0.4951 + 02
0.6000 + 01	0.1579 + 04	0.1925 + 03	0.8200 + 01	0.5318 + 02
0.6100 + 01	0.1748 + 04	0.2071 + 03	0.8442 + 01	0.5708 + 02
0.6200 + 01	0.1933 + 04	0.2225 + 03	0.8688 + 01	0.6121 + 02
0.6300 + 01	0.2135 + 04	0.2388 + 03	0.8938 + 01	0.6559 + 02
0.6400 + 01	0.2355 + 04	0.2562 + 03	0.9192 + 01	0.7023 + 02
0.6500 + 01	0.2594 + 04	0.2745 + 03	0.9450 + 01	0.7513 + 02
0.6600 + 01	0.2855 + 04	0.2939 + 03	0.9712 + 01	0.8032 + 02
0.6700 + 01	0.3138 + 04	0.3145 + 03	0.9978 + 01	0.8580 + 02
0.6800 + 01	0.3445 + 04	0.3362 + 03	0.1025 + 02	0.9159 + 02
0.6900 + 01	0.3779 + 04	0.3591 + 03	0.1052 + 02	0.9770 + 02
0.7000 + 01	0.4140 + 04	0.3833 + 03	0.1080 + 02	0.1041 + 03

M	$\frac{p_0}{p}$	$\frac{\rho_0}{\rho}$	$\frac{T_0}{T}$	$\frac{A}{A^*}$
0.7100 + 01	0.4531 + 04	0.4088 + 03	0.1108 + 02	0.1109 + 03
0.7200 + 01	0.4953 + 04	0.4357 + 03	0.1137 + 02	0.1181 + 03
0.7300 + 01	0.5410 + 04	0.4640 + 03	0.1166 + 02	0.1256 + 03
0.7400 + 01	0.5903 + 04	0.4939 + 03	0.1195 + 02	0.1335 + 03
0.7500 + 01	0.6434 + 04	0.5252 + 03	0.1225 + 02	0.1418 + 03
0.7600 + 01	0.7006 + 04	0.5582 + 03	0.1255 + 02	0.1506 + 03
0.7700 + 01	0.7623 + 04	0.5928 + 03	0.1286 + 02	0.1598 + 03
0.7800 + 01	0.8285 + 04	0.6292 + 03	0.1317 + 02	0.1694 + 03
0.7900 + 01	0.8998 + 04	0.6674 + 03	0.1348 + 02	0.1795 + 03
0.8000 + 01	0.9763 + 04	0.7075 + 03	0.1380 + 02	0.1901 + 03
0.9000 + 01	0.2110 + 05	0.1227 + 04	0.1720 + 02	0.3272 + 03
0.1000 + 02	0.4244 + 05	0.2021 + 04	0.2100 + 02	0.5359 + 03
0.1100 + 02	0.8033 + 05	0.3188 + 04	0.2520 + 02	0.8419 + 03
0.1200 + 02	0.1445 + 06	0.4848 + 04	0.2980 + 02	0.1276 + 04
0.1300 + 02	0.2486 + 06	0.7144 + 04	0.3480 + 02	0.1876 + 04
0.1400 + 02	0.4119 + 06	0.1025 + 05	0.4020 + 02	0.2685 + 04
0.1500 + 02	0.6602 + 06	0.1435 + 05	0.4600 + 02	0.3755 + 04
0.1600 + 02	0.1028 + 07	0.1969 + 05	0.5229 + 02	0.5145 + 04
0.1700 + 02	0.1559 + 07	0.2651 + 05	0.5880 + 02	0.6921 + 04
0.1800 + 02	0.2311 + 07	0.3512 + 05	0.6580 + 02	0.9159 + 04
0.1900 + 02	0.3356 + 07	0.4584 + 05	0.7320 + 02	0.1195 + 05
0.2000 + 02	0.4783 + 07	0.5905 + 05	0.8100 + 02	0.1538 + 05
0.2200 + 02	0.9251 + 07	0.9459 + 05	0.9780 + 02	0.2461 + 05
0.2400 + 02	0.1691 + 08	0.1456 + 06	0.1162 + 03	0.3783 + 05
0.2600 + 02	0.2949 + 08	0.2165 + 06	0.1362 + 03	0.5624 + 05
0.2800 + 02	0.4936 + 08	0.3128 + 06	0.1578 + 03	0.8121 + 05
0.3000 + 02	0.7978 + 08	0.4408 + 06	0.1810 + 03	0.1144 + 06
0.3200 + 02	0.1250 + 09	0.6076 + 06	0.2058 + 03	0.1576 + 06
0.3400 + 02	0.1908 + 09	0.8216 + 06	0.2322 + 03	0.2131 + 06
0.3600 + 02	0.2842 + 09	0.1092 + 07	0.2602 + 03	0.2832 + 06
0.3800 + 02	0.4143 + 09	0.1430 + 07	0.2898 + 03	0.3707 + 06
0.4000 + 02	0.5926 + 09	0.1846 + 07	0.3210 + 03	0.4785 + 06
0.4200 + 02	0.8330 + 09	0.2354 + 07	0.3538 + 03	0.6102 + 06
0.4400 + 02	0.1153 + 10	0.2969 + 07	0.3882 + 03	0.7694 + 06
0.4600 + 02	0.1572 + 10	0.3706 + 07	0.4242 + 03	0.9603 + 06
0.4800 + 02	0.2116 + 10	0.4583 + 07	0.4618 + 03	0.1187 + 07
0.5000 + 02	0.2815 + 10	0.5618 + 07	0.5010 + 03	0.1455 + 07

APPENDIX B

Normal Shock Properties

M	$\frac{p_2}{p_1}$	$\frac{\rho_2}{\rho_1}$	$\frac{T_2}{T_1}$	$\frac{p_{0_2}}{p_{0_1}}$	$\frac{p_{0_2}}{p_1}$	M_2
0.1000 + 01	0.1000 + 01	0.1000 + 01	0.1000 + 01	0.1000 + 01	0.1893 + 01	0.1000 + 01
0.1020 + 01	0.1047 + 01	0.1033 + 01	0.1013 + 01	0.1000 + 01	0.1938 + 01	0.9805 + 00
0.1040 + 01	0.1095 + 01	0.1067 + 01	0.1026 + 01	0.9999 + 00	0.1984 + 01	0.9620 + 00
0.1060 + 01	0.1144 + 01	0.1101 + 01	0.1039 + 01	0.9998 + 00	0.2032 + 01	0.9444 + 00
0.1080 + 01	0.1194 + 01	0.1135 + 01	0.1052 + 01	0.9994 + 01	0.2082 + 01	0.9277 + 00
0.1100 + 01	0.1245 + 01	0.1169 + 01	0.1065 + 01	0.9989 + 00	0.2133 + 01	0.9118 + 00
0.1120 + 01	0.1297 + 01	0.1203 + 01	0.1078 + 01	0.9982 + 00	0.2185 + 01	0.8966 + 00
0.1140 + 01	0.1350 + 01	0.1238 + 01	0.1090 + 01	0.9973 + 00	0.2239 + 01	0.8820 + 00
0.1160 + 01	0.1403 + 01	0.1272 + 01	0.1103 + 01	0.9961 + 00	0.2294 + 01	0.8682 + 00
0.1180 + 01	0.1458 + 01	0.1307 + 01	0.1115 + 01	0.9946 + 00	0.2350 + 01	0.8549 + 00
0.1200 + 01	0.1513 + 01	0.1342 + 01	0.1128 + 01	0.9928 + 00	0.2408 + 01	0.8422 + 00
0.1220 + 01	0.1570 + 01	0.1376 + 01	0.1141 + 01	0.9907 + 00	0.2466 + 01	0.8300 + 00
0.1240 + 01	0.1627 + 01	0.1411 + 01	0.1153 + 01	0.9884 + 00	0.2526 + 01	0.8183 + 00
0.1260 + 01	0.1686 + 01	0.1446 + 01	0.1166 + 01	0.9857 + 00	0.2588 + 01	0.8071 + 00
0.1280 + 01	0.1745 + 01	0.1481 + 01	0.1178 + 01	0.9827 + 00	0.2650 + 01	0.7963 + 00
0.1300 + 01	0.1805 + 01	0.1516 + 01	0.1191 + 01	0.9794 + 00	0.2714 + 01	0.7860 + 00
0.1320 + 01	0.1866 + 01	0.1551 + 01	0.1204 + 01	0.9758 + 00	0.2778 + 01	0.7760 + 00
0.1340 + 01	0.1928 + 01	0.1585 + 01	0.1216 + 01	0.9718 + 00	0.2844 + 01	0.7664 + 00
0.1360 + 01	0.1991 + 01	0.1620 + 01	0.1229 + 01	0.9676 + 00	0.2912 + 01	0.7572 + 00
0.1380 + 01	0.2055 + 01	0.1655 + 01	0.1242 + 01	0.9630 + 00	0.2980 + 01	0.7483 + 00
0.1400 + 01	0.2120 + 01	0.1690 + 01	0.1255 + 01	0.9582 + 00	0.3049 + 01	0.7397 + 00
0.1420 + 01	0.2186 + 01	0.1724 + 01	0.1268 + 01	0.9531 + 00	0.3120 + 01	0.7314 + 00
0.1440 + 01	0.2253 + 01	0.1759 + 01	0.1281 + 01	0.9476 + 00	0.3191 + 01	0.7235 + 00
0.1460 + 01	0.2320 + 01	0.1793 + 01	0.1294 + 01	0.9420 + 00	0.3264 + 01	0.7157 + 00
0.1480 + 01	0.2389 + 01	0.1828 + 01	0.1307 + 01	0.9360 + 00	0.3338 + 01	0.7083 + 00
0.1500 + 01	0.2458 + 01	0.1862 + 01	0.1320 + 01	0.9298 + 00	0.3413 + 01	0.7011 + 00
0.1520 + 01	0.2529 + 01	0.1896 + 01	0.1334 + 01	0.9233 + 00	0.3489 + 01	0.6941 + 00
0.1540 + 01	0.2600 + 01	0.1930 + 01	0.1347 + 01	0.9166 + 00	0.3567 + 01	0.6874 + 00
0.1560 + 01	0.2673 + 01	0.1964 + 01	0.1361 + 01	0.9097 + 00	0.3645 + 01	0.6809 + 00
0.1580 + 01	0.2746 + 01	0.1998 + 01	0.1374 + 01	0.9026 + 00	0.3724 + 01	0.6746 + 00

M	$\frac{p_2}{p_1}$	$\frac{\rho_2}{\rho_1}$	$\frac{T_2}{T_1}$	$\frac{p_{0_2}}{p_{0_1}}$	$\frac{p_{0_2}}{p_1}$	M_2
0.1600 + 01	0.2820 + 01	0.2032 + 01	0.1388 + 01	0.8952 + 00	0.3805 + 01	0.6684 + 00
0.1620 + 01	0.2895 + 01	0.2065 + 01	0.1402 + 01	0.8877 + 00	0.3887 + 01	0.6625 + 00
0.1640 + 01	0.2971 + 01	0.2099 + 01	0.1416 + 01	0.8799 + 00	0.3969 + 01	0.6568 + 00
0.1660 + 01	0.3048 + 01	0.2132 + 01	0.1430 + 01	0.8720 + 00	0.4053 + 01	0.6512 + 00
0.1680 + 01	0.3126 + 01	0.2165 + 01	0.1444 + 01	0.8639 + 00	0.4138 + 01	0.6458 + 00
0.1700 + 01	0.3205 + 01	0.2198 + 01	0.1458 + 01	0.8557 + 00	0.4224 + 01	0.6405 + 00
0.1720 + 01	0.3285 + 01	0.2230 + 01	0.1473 + 01	0.8474 + 00	0.4311 + 01	0.6355 + 00
0.1740 + 01	0.3366 + 01	0.2263 + 01	0.1487 + 01	0.8389 + 00	0.4399 + 01	0.6305 + 00
0.1760 + 01	0.3447 + 01	0.2295 + 01	0.1502 + 01	0.8302 + 00	0.4488 + 01	0.6257 + 00
0.1780 + 01	0.3530 + 01	0.2327 + 01	0.1517 + 01	0.8215 + 00	0.4578 + 01	0.6210 + 00
0.1800 + 01	0.3613 + 01	0.2359 + 01	0.1532 + 01	0.8127 + 00	0.4670 + 01	0.6165 + 00
0.1820 + 01	0.3698 + 01	0.2391 + 01	0.1547 + 01	0.8038 + 00	0.4762 + 01	0.6121 + 00
0.1840 + 01	0.3783 + 01	0.2422 + 01	0.1562 + 01	0.7948 + 00	0.4855 + 01	0.6078 + 00
0.1860 + 01	0.3870 + 01	0.2454 + 01	0.1577 + 01	0.7857 + 00	0.4950 + 01	0.6036 + 00
0.1880 + 01	0.3957 + 01	0.2485 + 01	0.1592 + 01	0.7765 + 00	0.5045 + 01	0.5996 + 00
0.1900 + 01	0.4045 + 01	0.2516 + 01	0.1608 + 01	0.7674 + 00	0.5142 + 01	0.5956 + 00
0.1920 + 01	0.4134 + 01	0.2546 + 01	0.1624 + 01	0.7581 + 00	0.5239 + 01	0.5918 + 00
0.1940 + 01	0.4224 + 01	0.2577 + 01	0.1639 + 01	0.7488 + 00	0.5338 + 01	0.5880 + 00
0.1960 + 01	0.4315 + 01	0.2607 + 01	0.1655 + 01	0.7395 + 00	0.5438 + 01	0.5844 + 00
0.1980 + 01	0.4407 + 01	0.2637 + 01	0.1671 + 01	0.7302 + 00	0.5539 + 01	0.5808 + 00
0.2000 + 01	0.4500 + 01	0.2667 + 01	0.1687 + 01	0.7209 + 00	0.5640 + 01	0.5774 + 00
0.2050 + 01	0.4736 + 01	0.2740 + 01	0.1729 + 01	0.6975 + 00	0.5900 + 01	0.5691 + 00
0.2100 + 01	0.4978 + 01	0.2812 + 01	0.1770 + 01	0.6742 + 00	0.6165 + 01	0.5613 + 00
0.2150 + 01	0.5226 + 01	0.2882 + 01	0.1813 + 01	0.6511 + 00	0.6438 + 01	0.5540 + 00
0.2200 + 01	0.5480 + 01	0.2951 + 01	0.1857 + 01	0.6281 + 00	0.6716 + 01	0.5471 + 00
0.2250 + 01	0.5740 + 01	0.3019 + 01	0.1901 + 01	0.6055 + 00	0.7002 + 01	0.5406 + 00
0.2300 + 01	0.6005 + 01	0.3085 + 01	0.1947 + 01	0.5833 + 00	0.7294 + 01	0.5344 + 00
0.2350 + 01	0.6276 + 01	0.3149 + 01	0.1993 + 01	0.5615 + 00	0.7592 + 01	0.5286 + 00
0.2400 + 01	0.6553 + 01	0.3212 + 01	0.2040 + 01	0.5401 + 00	0.7897 + 01	0.5231 + 00
0.2450 + 01	0.6836 + 01	0.3273 + 01	0.2088 + 01	0.5193 + 00	0.8208 + 01	0.5179 + 00
0.2500 + 01	0.7125 + 01	0.3333 + 01	0.2137 + 01	0.4990 + 00	0.8526 + 01	0.5130 + 00
0.2550 + 01	0.7420 + 01	0.3392 + 01	0.2187 + 01	0.4793 + 00	0.8850 + 01	0.5083 + 00
0.2600 + 01	0.7720 + 01	0.3449 + 01	0.2238 + 01	0.4601 + 00	0.9181 + 01	0.5039 + 00
0.2650 + 01	0.8026 + 01	0.3505 + 01	0.2290 + 01	0.4416 + 00	0.9519 + 01	0.4996 + 00
0.2700 + 01	0.8338 + 01	0.3559 + 01	0.2343 + 01	0.4236 + 00	0.9862 + 01	0.4956 + 00
0.2750 + 01	0.8656 + 01	0.3612 + 01	0.2397 + 01	0.4062 + 00	0.1021 + 02	0.4918 + 00
0.2800 + 01	0.8980 + 01	0.3664 + 01	0.2451 + 01	0.3895 + 00	0.1057 + 02	0.4882 + 00
0.2850 + 01	0.9310 + 01	0.3714 + 01	0.2507 + 01	0.3733 + 00	0.1093 + 02	0.4847 + 00
0.2900 + 01	0.9645 + 01	0.3763 + 01	0.2563 + 01	0.3577 + 00	0.1130 + 02	0.4814 + 00
0.2950 + 01	0.9986 + 01	0.3811 + 01	0.2621 + 01	0.3428 + 00	0.1168 + 02	0.4782 + 00
0.3000 + 01	0.1033 + 02	0.3857 + 01	0.2679 + 01	0.3283 + 00	0.1206 + 02	0.4752 + 00
0.3050 + 01	0.1069 + 02	0.3902 + 01	0.2738 + 01	0.3145 + 00	0.1245 + 02	0.4723 + 00
0.3100 + 01	0.1104 + 02	0.3947 + 01	0.2799 + 01	0.3012 + 00	0.1285 + 02	0.4695 + 00
0.3150 + 01	0.1141 + 02	0.3990 + 01	0.2860 + 01	0.2885 + 00	0.1325 + 02	0.4669 + 00
0.3200 + 01	0.1178 + 02	0.4031 + 01	0.2922 + 01	0.2762 + 00	0.1366 + 02	0.4643 + 00
0.3250 + 01	0.1216 + 02	0.4072 + 01	0.2985 + 01	0.2645 + 00	0.1407 + 02	0.4619 + 00
0.3300 + 01	0.1254 + 02	0.4112 + 01	0.3049 + 01	0.2533 + 00	0.1449 + 02	0.4596 + 00
0.3350 + 01	0.1293 + 02	0.4151 + 01	0.3114 + 01	0.2425 + 00	0.1492 + 02	0.4573 + 00
0.3400 + 01	0.1332 + 02	0.4188 + 01	0.3180 + 01	0.2322 + 00	0.1535 + 02	0.4552 + 00
0.3450 + 01	0.1372 + 02	0.4225 + 01	0.3247 + 01	0.2224 + 00	0.1579 + 02	0.4531 + 00

M	$\frac{p_2}{p_1}$	$\frac{\rho_2}{\rho_1}$	$\frac{T_2}{T_1}$	$\frac{p_{0_2}}{p_{0_1}}$	$\frac{p_{0_2}}{p_1}$	M_2
0.3500 + 01	0.1412 + 02	0.4261 + 01	0.3315 + 01	0.2129 + 00	0.1624 + 02	0.4512 + 00
0.3550 + 01	0.1454 + 02	0.4296 + 01	0.3384 + 01	0.2039 + 00	0.1670 + 02	0.4492 + 00
0.3600 + 01	0.1495 + 02	0.4330 + 01	0.3454 + 01	0.1953 + 00	0.1716 + 02	0.4474 + 00
0.3650 + 01	0.1538 + 02	0.4363 + 01	0.3525 + 01	0.1871 + 00	0.1762 + 02	0.4456 + 00
0.3700 + 01	0.1580 + 02	0.4395 + 01	0.3596 + 01	0.1792 + 00	0.1810 + 02	0.4439 + 00
0.3750 + 01	0.1624 + 02	0.4426 + 01	0.3669 + 01	0.1717 + 00	0.1857 + 02	0.4423 + 00
0.3800 + 01	0.1668 + 02	0.4457 + 01	0.3743 + 01	0.1645 + 00	0.1906 + 02	0.4407 + 00
0.3850 + 01	0.1713 + 02	0.4487 + 01	0.3817 + 01	0.1576 + 00	0.1955 + 02	0.4392 + 00
0.3900 + 01	0.1758 + 02	0.4516 + 01	0.3893 + 01	0.1510 + 00	0.2005 + 02	0.4377 + 00
0.3950 + 01	0.1804 + 02	0.4544 + 01	0.3969 + 01	0.1448 + 00	0.2056 + 02	0.4363 + 00
0.4000 + 01	0.1850 + 02	0.4571 + 01	0.4047 + 01	0.1388 + 00	0.2107 + 02	0.4350 + 00
0.4050 + 01	0.1897 + 02	0.4598 + 01	0.4125 + 01	0.1330 + 00	0.2159 + 02	0.4336 + 00
0.4100 + 01	0.1944 + 02	0.4624 + 01	0.4205 + 01	0.1276 + 00	0.2211 + 02	0.4324 + 00
0.4150 + 01	0.1993 + 02	0.4650 + 01	0.4285 + 01	0.1223 + 00	0.2264 + 02	0.4311 + 00
0.4200 + 01	0.2041 + 02	0.4675 + 01	0.4367 + 01	0.1173 + 00	0.2318 + 02	0.4299 + 00
0.4250 + 01	0.2091 + 02	0.4699 + 01	0.4449 + 01	0.1126 + 00	0.2372 + 02	0.4288 + 00
0.4300 + 01	0.2140 + 02	0.4723 + 01	0.4532 + 01	0.1080 + 00	0.2427 + 02	0.4277 + 00
0.4350 + 01	0.2191 + 02	0.4746 + 01	0.4616 + 01	0.1036 + 00	0.2483 + 02	0.4266 + 00
0.4400 + 01	0.2242 + 02	0.4768 + 01	0.4702 + 01	0.9948 − 01	0.2539 + 02	0.4255 + 00
0.4450 + 01	0.2294 + 02	0.4790 + 01	0.4788 + 01	0.9550 − 01	0.2596 + 02	0.4245 + 00
0.4500 + 01	0.2346 + 02	0.4812 + 01	0.4875 + 01	0.9170 − 01	0.2654 + 02	0.4236 + 00
0.4550 + 01	0.2399 + 02	0.4833 + 01	0.4963 + 01	0.8806 − 01	0.2712 + 02	0.4226 + 00
0.4600 + 01	0.2452 + 02	0.4853 + 01	0.5052 + 01	0.8459 − 01	0.2771 + 02	0.4217 + 00
0.4650 + 01	0.2506 + 02	0.4873 + 01	0.5142 + 01	0.8126 − 01	0.2831 + 02	0.4208 + 00
0.4700 + 01	0.2560 + 02	0.4893 + 01	0.5233 + 01	0.7809 − 01	0.2891 + 02	0.4199 + 00
0.4750 + 01	0.2616 + 02	0.4912 + 01	0.5325 + 01	0.7505 − 01	0.2952 + 02	0.4191 + 00
0.4800 + 01	0.2671 + 02	0.4930 + 01	0.5418 + 01	0.7214 − 01	0.3013 + 02	0.4183 + 00
0.4850 + 01	0.2728 + 02	0.4948 + 01	0.5512 + 01	0.6936 − 01	0.3075 + 02	0.4175 + 00
0.4900 + 01	0.2784 + 02	0.4966 + 01	0.5607 + 01	0.6670 − 01	0.3138 + 02	0.4167 + 00
0.4950 + 01	0.2842 + 02	0.4983 + 01	0.5703 + 01	0.6415 − 01	0.3201 + 02	0.4160 + 00
0.5000 + 01	0.2900 + 02	0.5000 + 01	0.5800 + 01	0.6172 − 01	0.3265 + 02	0.4152 + 00
0.5100 + 01	0.3018 + 02	0.5033 + 01	0.5997 + 01	0.5715 − 01	0.3395 + 02	0.4138 + 00
0.5200 + 01	0.3138 + 02	0.5064 + 01	0.6197 + 01	0.5297 − 01	0.3528 + 02	0.4125 + 00
0.5300 + 01	0.3260 + 02	0.5093 + 01	0.6401 + 01	0.4913 − 01	0.3663 + 02	0.4113 + 00
0.5400 + 01	0.3385 + 02	0.5122 + 01	0.6610 + 01	0.4560 − 01	0.3801 + 02	0.4101 + 00
0.5500 + 01	0.3512 + 02	0.5149 + 01	0.6822 + 01	0.4236 − 01	0.3941 + 02	0.4090 + 00
0.5600 + 01	0.3642 + 02	0.5175 + 01	0.7038 + 01	0.3938 − 01	0.4084 + 02	0.4079 + 00
0.5700 + 01	0.3774 + 02	0.5200 + 01	0.7258 + 01	0.3664 − 01	0.4230 + 02	0.4069 + 00
0.5800 + 01	0.3908 + 02	0.5224 + 01	0.7481 + 01	0.3412 − 01	0.4378 + 02	0.4059 + 00
0.5900 + 01	0.4044 + 02	0.5246 + 01	0.7709 + 01	0.3180 − 01	0.4528 + 02	0.4050 + 00
0.6000 + 01	0.4183 + 02	0.5268 + 01	0.7941 + 01	0.2965 − 01	0.4682 + 02	0.4042 + 00
0.6100 + 01	0.4324 + 02	0.5289 + 01	0.8176 + 01	0.2767 − 01	0.4837 + 02	0.4033 + 00
0.6200 + 01	0.4468 + 02	0.5309 + 01	0.8415 + 01	0.2584 − 01	0.4996 + 02	0.4025 + 00
0.6300 + 01	0.4614 + 02	0.5329 + 01	0.8658 + 01	0.2416 − 01	0.5157 + 02	0.4018 + 00
0.6400 + 01	0.4762 + 02	0.5347 + 01	0.8905 + 01	0.2259 − 01	0.5320 + 02	0.4011 + 00
0.6500 + 01	0.4912 + 02	0.5365 + 01	0.9156 + 01	0.2115 − 01	0.5486 + 02	0.4004 + 00
0.6600 + 01	0.5065 + 02	0.5382 + 01	0.9411 + 01	0.1981 − 01	0.5655 + 02	0.3997 + 00
0.6700 + 01	0.5220 + 02	0.5399 + 01	0.9670 + 01	0.1857 − 01	0.5826 + 02	0.3991 + 00
0.6800 + 01	0.5378 + 02	0.5415 + 01	0.9933 + 01	0.1741 − 01	0.6000 + 02	0.3985 + 00
0.6900 + 01	0.5538 + 02	0.5430 + 01	0.1020 + 02	0.1635 − 01	0.6176 + 02	0.3979 + 00

M	$\frac{p_2}{p_1}$	$\frac{\rho_2}{\rho_1}$	$\frac{T_2}{T_1}$	$\frac{p_{0_2}}{p_{0_1}}$	$\frac{p_{0_2}}{p_1}$	M_2
0.7000 + 01	0.5700 + 02	0.5444 + 01	0.1047 + 02	0.1535 − 01	0.6355 + 02	0.3974 + 00
0.7100 + 01	0.5864 + 02	0.5459 + 01	0.1074 + 02	0.1443 − 01	0.6537 + 02	0.3968 + 00
0.7200 + 01	0.6031 + 02	0.5472 + 01	0.1102 + 02	0.1357 − 01	0.6721 + 02	0.3963 + 00
0.7300 + 01	0.6200 + 02	0.5485 + 01	0.1130 + 02	0.1277 − 01	0.6908 + 02	0.3958 + 00
0.7400 + 01	0.6372 + 02	0.5498 + 01	0.1159 + 02	0.1202 − 01	0.7097 + 02	0.3954 + 00
0.7500 + 01	0.6546 + 02	0.5510 + 01	0.1188 + 02	0.1133 − 01	0.7289 + 02	0.3949 + 00
0.7600 + 01	0.6722 + 02	0.5522 + 01	0.1217 + 02	0.1068 − 01	0.7483 + 02	0.3945 + 00
0.7700 + 01	0.6900 + 02	0.5533 + 01	0.1247 + 02	0.1008 − 01	0.7680 + 02	0.3941 + 00
0.7800 + 01	0.7081 + 02	0.5544 + 01	0.1277 + 02	0.9510 − 02	0.7880 + 02	0.3937 + 00
0.7900 + 01	0.7264 + 02	0.5555 + 01	0.1308 + 02	0.8982 − 02	0.8082 + 02	0.3933 + 00
0.8000 + 01	0.7450 + 02	0.5565 + 01	0.1339 + 02	0.8488 − 02	0.8287 + 02	0.3929 + 00
0.9000 + 01	0.9433 + 02	0.5651 + 01	0.1669 + 02	0.4964 − 02	0.1048 + 03	0.3898 + 00
0.1000 + 02	0.1165 + 03	0.5714 + 01	0.2039 + 02	0.3045 − 02	0.1292 + 03	0.3876 + 00
0.1100 + 02	0.1410 + 03	0.5762 + 01	0.2447 + 02	0.1945 − 02	0.1563 + 03	0.3859 + 00
0.1200 + 02	0.1678 + 03	0.5799 + 01	0.2894 + 02	0.1287 − 02	0.1859 + 03	0.3847 + 00
0.1300 + 02	0.1970 + 03	0.5828 + 01	0.3380 + 02	0.8771 − 03	0.2181 + 03	0.3837 + 00
0.1400 + 02	0.2285 + 03	0.5851 + 01	0.3905 + 02	0.6138 − 03	0.2528 + 03	0.3829 + 00
0.1500 + 02	0.2623 + 03	0.5870 + 01	0.4469 + 02	0.4395 − 03	0.2902 + 03	0.3823 + 00
0.1600 + 02	0.2985 + 03	0.5885 + 01	0.5072 + 02	0.3212 − 03	0.3301 + 03	0.3817 + 00
0.1700 + 02	0.3370 + 03	0.5898 + 01	0.5714 + 02	0.2390 − 03	0.3726 + 03	0.3813 + 00
0.1800 + 02	0.3778 + 03	0.5909 + 01	0.6394 + 02	0.1807 − 03	0.4176 + 03	0.3810 + 00
0.1900 + 02	0.4210 + 03	0.5918 + 01	0.7114 + 02	0.1386 − 03	0.4653 + 03	0.3806 + 00
0.2000 + 02	0.4665 + 03	0.5926 + 01	0.7872 + 02	0.1078 − 03	0.5155 + 03	0.3804 + 00
0.2200 + 02	0.5645 + 03	0.5939 + 01	0.9506 + 02	0.6741 − 04	0.6236 + 03	0.3800 + 00
0.2400 + 02	0.6718 + 03	0.5948 + 01	0.1129 + 03	0.4388 − 04	0.7421 + 03	0.3796 + 00
0.2600 + 02	0.7885 + 03	0.5956 + 01	0.1324 + 03	0.2953 − 04	0.8709 + 03	0.3794 + 00
0.2800 + 02	0.9145 + 03	0.5962 + 01	0.1534 + 03	0.2046 − 04	0.1010 + 04	0.3792 + 00
0.3000 + 02	0.1050 + 04	0.5967 + 01	0.1759 + 03	0.1453 − 04	0.1159 + 04	0.3790 + 00
0.3200 + 02	0.1194 + 04	0.5971 + 01	0.2001 + 03	0.1055 − 04	0.1319 + 04	0.3789 + 00
0.3400 + 02	0.1348 + 04	0.5974 + 01	0.2257 + 03	0.7804 − 05	0.1489 + 04	0.3788 + 00
0.3600 + 02	0.1512 + 04	0.5977 + 01	0.2529 + 03	0.5874 − 05	0.1669 + 04	0.3787 + 00
0.3800 + 02	0.1684 + 04	0.5979 + 01	0.2817 + 03	0.4488 − 05	0.1860 + 04	0.3786 + 00
0.4000 + 02	0.1866 + 04	0.5981 + 01	0.3121 + 03	0.3477 − 05	0.2061 + 04	0.3786 + 00
0.4200 + 02	0.2058 + 04	0.5983 + 01	0.3439 + 03	0.2727 − 05	0.2272 + 04	0.3785 + 00
0.4400 + 02	0.2258 + 04	0.5985 + 01	0.3774 + 03	0.2163 − 05	0.2493 + 04	0.3785 + 00
0.4600 + 02	0.2468 + 04	0.5986 + 01	0.4124 + 03	0.1733 − 05	0.2725 + 04	0.3784 + 00
0.4800 + 02	0.2688 + 04	0.5987 + 01	0.4489 + 03	0.1402 − 05	0.2967 + 04	0.3784 + 00
0.5000 + 02	0.2916 + 04	0.5988 + 01	0.4871 + 03	0.1144 − 05	0.3219 + 04	0.3784 + 00

APPENDIX C

Prandtl-Meyer Function and Mach Angle

M	ν	μ	M	ν	μ
0.1000 + 01	0.0000	0.9000 + 02	0.1600 + 01	0.1486 + 02	0.3868 + 02
0.1020 + 01	0.1257 + 00	0.7864 + 02	0.1620 + 01	0.1545 + 02	0.3812 + 02
0.1040 + 01	0.3510 + 00	0.7406 + 02	0.1640 + 01	0.1604 + 02	0.3757 + 02
0.1060 + 01	0.6367 + 00	0.7063 + 02	0.1660 + 01	0.1663 + 02	0.3704 + 02
0.1080 + 01	0.9680 + 00	0.6781 + 02	0.1680 + 01	0.1722 + 02	0.3653 + 02
0.1100 + 01	0.1336 + 01	0.6538 + 02	0.1700 + 01	0.1781 + 02	0.3603 + 02
0.1120 + 01	0.1735 + 01	0.6323 + 02	0.1720 + 01	0.1840 + 02	0.3555 + 02
0.1140 + 01	0.2160 + 01	0.6131 + 02	0.1740 + 01	0.1898 + 02	0.3508 + 02
0.1160 + 01	0.2607 + 01	0.5955 + 02	0.1760 + 01	0.1956 + 02	0.3462 + 02
0.1180 + 01	0.3074 + 01	0.5794 + 02	0.1780 + 01	0.2015 + 02	0.3418 + 02
0.1200 + 01	0.3558 + 01	0.5644 + 02	0.1800 + 01	0.2073 + 02	0.3375 + 02
0.1220 + 01	0.4057 + 01	0.5505 + 02	0.1820 + 01	0.2130 + 02	0.3333 + 02
0.1240 + 01	0.4569 + 01	0.5375 + 02	0.1840 + 01	0.2188 + 02	0.3292 + 02
0.1260 + 01	0.5093 + 01	0.5253 + 02	0.1860 + 01	0.2245 + 02	0.3252 + 02
0.1280 + 01	0.5627 + 01	0.5138 + 02	0.1880 + 01	0.2302 + 02	0.3213 + 02
0.1300 + 01	0.6170 + 01	0.5028 + 02	0.1900 + 01	0.2359 + 02	0.3176 + 02
0.1320 + 01	0.6721 + 01	0.4925 + 02	0.1920 + 01	0.2415 + 02	0.3139 + 02
0.1340 + 01	0.7279 + 01	0.4827 + 02	0.1940 + 01	0.2471 + 02	0.3103 + 02
0.1360 + 01	0.7844 + 01	0.4733 + 02	0.1960 + 01	0.2527 + 02	0.3068 + 02
0.1380 + 01	0.8413 + 01	0.4644 + 02	0.1980 + 01	0.2583 + 02	0.3033 + 02
0.1400 + 01	0.8987 + 01	0.4558 + 02	0.2000 + 01	0.2638 + 02	0.3000 + 02
0.1420 + 01	0.9565 + 01	0.4477 + 02	0.2050 + 01	0.2775 + 02	0.2920 + 02
0.1440 + 01	0.1015 + 02	0.4398 + 02	0.2100 + 01	0.2910 + 02	0.2844 + 02
0.1460 + 01	0.1073 + 02	0.4323 + 02	0.2150 + 01	0.3043 + 02	0.2772 + 02
0.1480 + 01	0.1132 + 02	0.4251 + 02	0.2200 + 01	0.3173 + 02	0.2704 + 02
0.1500 + 01	0.1191 + 02	0.4181 + 02	0.2250 + 01	0.3302 + 02	0.2639 + 02
0.1520 + 01	0.1249 + 02	0.4114 + 02	0.2300 + 01	0.3428 + 02	0.2577 + 02
0.1540 + 01	0.1309 + 02	0.4049 + 02	0.2350 + 01	0.3553 + 02	0.2518 + 02
0.1560 + 01	0.1368 + 02	0.3987 + 02	0.2400 + 01	0.3675 + 02	0.2462 + 02
0.1580 + 01	0.1427 + 02	0.3927 + 02	0.2450 + 01	0.3795 + 02	0.2409 + 02

M	ν	μ	M	ν	μ
0.2500 + 01	0.3912 + 02	0.2358 + 02	0.5000 + 01	0.7692 + 02	0.1154 + 02
0.2550 + 01	0.4028 + 02	0.2309 + 02	0.5100 + 01	0.7784 + 02	0.1131 + 02
0.2600 + 01	0.4141 + 02	0.2262 + 02	0.5200 + 01	0.7873 + 02	0.1109 + 02
0.2650 + 01	0.4253 + 02	0.2217 + 02	0.5300 + 01	0.7960 + 02	0.1088 + 02
0.2700 + 01	0.4362 + 02	0.2174 + 02	0.5400 + 01	0.8043 + 02	0.1067 + 02
0.2750 + 01	0.4469 + 02	0.2132 + 02	0.5500 + 01	0.8124 + 02	0.1048 + 02
0.2800 + 01	0.4575 + 02	0.2092 + 02	0.5600 + 01	0.8203 + 02	0.1029 + 02
0.2850 + 01	0.4678 + 02	0.2054 + 02	0.5700 + 01	0.8280 + 02	0.1010 + 02
0.2900 + 01	0.4779 + 02	0.2017 + 02	0.5800 + 01	0.8354 + 02	0.9928 + 01
0.2950 + 01	0.4878 + 02	0.1981 + 02	0.5900 + 01	0.8426 + 02	0.9758 + 01
0.3000 + 01	0.4976 + 02	0.1947 + 02	0.6000 + 01	0.8496 + 02	0.9594 + 01
0.3050 + 01	0.5071 + 02	0.1914 + 02	0.6100 + 01	0.8563 + 02	0.9435 + 01
0.3100 + 01	0.5165 + 02	0.1882 + 02	0.6200 + 01	0.8629 + 02	0.9282 + 01
0.3150 + 01	0.5257 + 02	0.1851 + 02	0.6300 + 01	0.8694 + 02	0.9133 + 01
0.3200 + 01	0.5347 + 02	0.1821 + 02	0.6400 + 01	0.8756 + 02	0.8989 + 01
0.3250 + 01	0.5435 + 02	0.1792 + 02	0.6500 + 01	0.8817 + 02	0.8850 + 01
0.3300 + 01	0.5522 + 02	0.1764 + 02	0.6600 + 01	0.8876 + 02	0.8715 + 01
0.3350 + 01	0.5607 + 02	0.1737 + 02	0.6700 + 01	0.8933 + 02	0.8584 + 01
0.3400 + 01	0.5691 + 02	0.1710 + 02	0.6800 + 01	0.8989 + 02	0.8457 + 01
0.3450 + 01	0.5773 + 02	0.1685 + 02	0.6900 + 01	0.9044 + 02	0.8333 + 01
0.3500 + 01	0.5853 + 02	0.1660 + 02	0.7000 + 01	0.9097 + 02	0.8213 + 01
0.3550 + 01	0.5932 + 02	0.1636 + 02	0.7100 + 01	0.9149 + 02	0.8097 + 01
0.3600 + 01	0.6009 + 02	0.1613 + 02	0.7200 + 01	0.9200 + 02	0.7984 + 01
0.3650 + 01	0.6085 + 02	0.1590 + 02	0.7300 + 01	0.9249 + 02	0.7873 + 01
0.3700 + 01	0.6160 + 02	0.1568 + 02	0.7400 + 01	0.9297 + 02	0.7766 + 01
0.3750 + 01	0.6233 + 02	0.1547 + 02	0.7500 + 01	0.9344 + 02	0.7662 + 01
0.3800 + 01	0.6304 + 02	0.1526 + 02	0.7600 + 01	0.9390 + 02	0.7561 + 01
0.3850 + 01	0.6375 + 02	0.1505 + 02	0.7700 + 01	0.9434 + 02	0.7462 + 01
0.3900 + 01	0.6444 + 02	0.1486 + 02	0.7800 + 01	0.9478 + 02	0.7366 + 01
0.3950 + 01	0.6512 + 02	0.1466 + 02	0.7900 + 01	0.9521 + 02	0.7272 + 01
0.4000 + 01	0.6578 + 02	0.1448 + 02	0.8000 + 01	0.9562 + 02	0.7181 + 01
0.4050 + 01	0.6644 + 02	0.1429 + 02	0.9000 + 01	0.9932 + 02	0.6379 + 01
0.4100 + 01	0.6708 + 02	0.1412 + 02	0.1000 + 02	0.1023 + 03	0.5739 + 01
0.4150 + 01	0.6771 + 02	0.1394 + 02	0.1100 + 02	0.1048 + 03	0.5216 + 01
0.4200 + 01	0.6833 + 02	0.1377 + 02	0.1200 + 02	0.1069 + 03	0.4780 + 01
0.4250 + 01	0.6894 + 02	0.1361 + 02	0.1300 + 02	0.1087 + 03	0.4412 + 01
0.4300 + 01	0.6954 + 02	0.1345 + 02	0.1400 + 02	0.1102 + 03	0.4096 + 01
0.4350 + 01	0.7013 + 02	0.1329 + 02	0.1500 + 02	0.1115 + 03	0.3823 + 01
0.4400 + 01	0.7071 + 02	0.1314 + 02	0.1600 + 02	0.1127 + 03	0.3583 + 01
0.4450 + 01	0.7127 + 02	0.1299 + 02	0.1700 + 02	0.1137 + 03	0.3372 + 01
0.4500 + 01	0.7183 + 02	0.1284 + 02	0.1800 + 02	0.1146 + 03	0.3185 + 01
0.4550 + 01	0.7238 + 02	0.1270 + 02	0.1900 + 02	0.1155 + 03	0.3017 + 01
0.4600 + 01	0.7292 + 02	0.1256 + 02	0.2000 + 02	0.1162 + 03	0.2866 + 01
0.4650 + 01	0.7345 + 02	0.1242 + 02	0.2200 + 02	0.1175 + 03	0.2605 + 01
0.4700 + 01	0.7397 + 02	0.1228 + 02	0.2400 + 02	0.1186 + 03	0.2388 + 01
0.4750 + 01	0.7448 + 02	0.1215 + 02	0.2600 + 02	0.1195 + 03	0.2204 + 01
0.4800 + 01	0.7499 + 02	0.1202 + 02	0.2800 + 02	0.1202 + 03	0.2047 + 01
0.4850 + 01	0.7548 + 02	0.1190 + 02	0.3000 + 02	0.1209 + 03	0.1910 + 01
0.4900 + 01	0.7597 + 02	0.1178 + 02	0.3200 + 02	0.1215 + 03	0.1791 + 01
0.4950 + 01	0.7645 + 02	0.1166 + 02	0.3400 + 02	0.1220 + 03	0.1685 + 01

M	ν	μ	M	ν	μ
$0.3600 + 02$	$0.1225 + 03$	$0.1592 + 01$	$0.4400 + 02$	$0.1239 + 03$	$0.1302 + 01$
$0.3800 + 02$	$0.1229 + 03$	$0.1508 + 01$	$0.4600 + 02$	$0.1242 + 03$	$0.1246 + 01$
$0.4000 + 02$	$0.1233 + 03$	$0.1433 + 01$	$0.4800 + 02$	$0.1245 + 03$	$0.1194 + 01$
$0.4200 + 02$	$0.1236 + 03$	$0.1364 + 01$	$0.5000 + 02$	$0.1247 + 03$	$0.1146 + 01$

APPENDIX D

Standard Atmosphere, SI Units

D.1 NOTE ABOUT THE STANDARD ATMOSPHERE TABLES IN APPENDICES D AND E

The following Standard Atmosphere Tables are compiled from mean experimental data for the temperature variation with altitude, combined with the laws of physics to compute the corresponding variation of pressure and density. The laws of physics used are the hydrostatic equation (Equation 1.52) and the equation of state (Equation 7.1). The construction of the Standard Atmosphere Tables is discussed in detail in Chapter 3 of Reference 2, which you should read to learn the whole story. In Appendices D and E, the temperature, pressure, and density are tabulated versus altitude. Two columns for altitude are given, the first for the geometric altitude, h_G, and the second for the geopotential altitude, h. The geometric altitude is the actual height above standard sea level, and the geopotential altitude is a related altitude based on the assumption of a constant value of the acceleration of gravity used for the calculations. See Reference 2 for an explanation of the difference. In this book, whenever a reference is made to a certain value of the standard altitude, it means the value of h_G, the first column in the tables.

Appendix D gives the Standard Atmosphere in SI units, and Appendix E gives the Standard Atmosphere in English Engineering units. There exist tables of the standard atmosphere compiled by various organizations over the years. The values tabulated in Appendices D and E are taken from the 1959 ARDC model atmosphere compiled by the U.S. Air Force.

Altitude				
h_G, m	h, m	Temperature T, K	Pressure p, N/m^2	Density ρ, kg/m^3
−5,000	−5,004	320.69	1.7761 + 5	1.9296 + 0
−4,900	−4,904	320.03	1.7587	1.9145
−4,800	−4,804	319.38	1.7400	1.8980
−4,700	−4,703	318.73	1.7215	1.8816
−4,600	−4,603	318.08	1.7031	1.8653
−4,500	−4,503	317.43	1.6848	1.8491
−4,400	−4,403	316.78	1.6667	1.8330
−4,300	−4,303	316.13	1.6488	1.8171
−4,200	−4,203	315.48	1.6311	1.8012
−4,100	−4,103	314.83	1.6134	1.7854
−4,000	−4,003	314.18	1.5960 + 5	1.7698 + 0
−3,900	−3,902	313.53	1.5787	1.7542
−3,800	−3,802	312.87	1.5615	1.7388
−3,700	−3,702	212.22	1.5445	1.7234
−3,600	−3,602	311.57	1.5277	1.7082
−3,500	−3,502	310.92	1.5110	1.6931
−3,400	−3,402	310.27	1.4945	1.6780
−3,300	−3,302	309.62	1.4781	1.6631
−3,200	−3,202	308.97	1.4618	1.6483
−3,100	−3,102	308.32	1.4457	1.6336
−3,000	−3,001	307.67	1.4297 + 5	1.6189 + 0
−2,900	−2,901	307.02	1.4139	1.6044
−2,800	−2,801	306.37	1.3982	1.5900
−2,700	−2,701	305.72	1.3827	1.5757
−2,600	−2,601	305.07	1.3673	1.5615
−2,500	−2,501	304.42	1.3521	1.5473
−2,400	−2,401	303.77	1.3369	1.5333
−2,300	−2,301	303.12	1.3220	1.5194
−2,200	−2,201	302.46	1.3071	1.5056
−2,100	−2,101	301.81	1.2924	1.4918
−2,000	−2,001	301.16	1.2778 + 5	1.4782 + 0
−1,900	−1,901	300.51	1.2634	1.4646
−1,800	−1,801	299.86	1.2491	1.4512
−1,700	−1,701	299.21	1.2349	1.4379
−1,600	−1,600	298.56	1.2209	1.4246
−1,500	−1,500	297.91	1.2070	1.4114
−1,400	−1,400	297.26	1.1932	1.3984
−1,300	−1,300	296.61	1.1795	1.3854
−1,200	−1,200	295.96	1.1660	1.3725
−1,100	−1,100	295.31	1.1526	1.3597
−1,000	−1,000	294.66	1.1393 + 5	1.3470 + 0
−900	−900	294.01	1.1262	1.3344
−800	−800	293.36	1.1131	1.3219
−700	−700	292.71	1.1002	1.3095
−600	−600	292.06	1.0874	1.2972
−500	−500	291.41	1.0748	1.2849
−400	−400	290.76	1.0622	1.2728
−300	−300	290.11	1.0498	1.2607
−200	−200	289.46	1.0375	1.2487
−100	−100	288.81	1.0253	1.2368

Altitude				
h_G, m	h, m	Temperature T, K	Pressure p, N/m^2	Density ρ, kg/m^3
0	0	288.16	1.01325 + 5	1.2250 + 0
100	100	287.51	1.0013	1.2133
200	200	286.86	9.8945 + 4	1.2071
300	300	286.21	9.7773	1.1901
400	400	285.56	9.6611	1.1787
500	500	284.91	9.5461	1.1673
600	600	284.26	9.4322	1.1560
700	700	283.61	9.3194	1.1448
800	800	282.96	9.2077	1.1337
900	900	282.31	9.0971	1.1226
1,000	1,000	281.66	8.9876 + 4	1.1117 + 0
1,100	1,100	281.01	8.8792	1.1008
1,200	1,200	280.36	8.7718	1.0900
1,300	1,300	279.71	8.6655	1.0793
1,400	1,400	279.06	8.5602	1.0687
1,500	1,500	278.41	8.4560	1.0581
1,600	1,600	277.76	8.3527	1.0476
1,700	1,700	277.11	8.2506	1.0373
1,800	1,799	276.46	8.1494	1.0269
1,900	1,899	275.81	8.0493	1.0167
2,000	1,999	275.16	7.9501 + 4	1.0066 + 0
2,100	2,099	274.51	7.8520	9.9649 − 1
2,200	2,199	273.86	7.7548	9.8649
2,300	2,299	273.22	7.6586	9.7657
2,400	2,399	272.57	7.5634	9.6673
2,500	2,499	271.92	7.4692	9.5696
2,600	2,599	271.27	7.3759	9.4727
2,700	2,699	270.62	7.2835	9.3765
2,800	2,799	269.97	7.1921	9.2811
2,900	2,899	269.32	7.1016	9.1865
3,000	2,999	268.67	7.0121 + 4	9.0926 − 1
3,100	3,098	268.02	6.9235	8.9994
3,200	3,198	267.37	6.8357	8.9070
3,300	3,298	266.72	6.7489	8.8153
3,400	3,398	266.07	6.6630	8.7243
3,500	3,498	265.42	6.5780	8.6341
3,600	3,598	264.77	6.4939	8.5445
3,700	3,698	264.12	6.4106	8.4557
3,800	3,798	263.47	6.3282	8.3676
3,900	3,898	262.83	6.2467	8.2802
4,000	3,997	262.18	6.1660 + 4	8.1935 − 1
4,100	4,097	261.53	6.0862	8.1075
4,200	4,197	260.88	6.0072	8.0222
4,300	4,297	260.23	5.9290	7.9376
4,400	4,397	259.58	5.8517	7.8536
4,500	4,497	258.93	5.7752	7.7704
4,600	4,597	258.28	5.6995	7.6878
4,700	4,697	257.63	5.6247	7.6059
4,800	4,796	256.98	5.5506	7.5247
4,900	4,896	256.33	5.4773	7.4442

Altitude				
h_G, m	h, m	Temperature T, K	Pressure p, N/m^2	Density ρ, kg/m^3
5,000	4,996	255.69	$5.4048 + 4$	$7.3643 - 1$
5,100	5,096	255.04	5.3331	7.2851
5,200	5,196	254.39	5.2621	7.2065
5,400	5,395	253.09	5.1226	7.0513
5,500	5,495	252.44	5.0539	6.9747
5,600	5,595	251.79	4.9860	6.8987
5,700	5,695	251.14	4.9188	6.8234
5,800	5,795	250.49	4.8524	6.7486
5,900	5,895	249.85	4.7867	6.6746
6,000	5,994	249.20	$4.7217 + 4$	$6.6011 - 1$
6,100	6,094	248.55	4.6575	6.5283
6,200	6,194	247.90	4.5939	6.4561
6,300	6,294	247.25	4.5311	6.3845
6,400	6,394	246.60	4.4690	6.3135
6,500	6,493	245.95	4.4075	6.2431
6,600	6,593	245.30	4.3468	6.1733
6,700	6,693	244.66	4.2867	6.1041
6,800	6,793	244.01	4.2273	6.0356
6,900	6,893	243.36	4.1686	5.9676
7,000	6,992	242.71	$4.1105 + 4$	$5.9002 - 1$
7,100	7,092	242.06	4.0531	5.8334
7,200	7,192	241.41	3.9963	5.7671
7,300	7,292	240.76	3.9402	5.7015
7,400	7,391	240.12	3.8848	5.6364
7,500	7,491	239.47	3.8299	5.5719
7,600	7,591	238.82	3.7757	5.5080
7,700	7,691	238.17	3.7222	5.4446
7,800	7,790	237.52	3.6692	5.3818
7,900	7,890	236.87	3.6169	5.3195
8,000	7,990	236.23	$3.5651 + 4$	$5.2578 - 1$
8,100	8,090	235.58	3.5140	5.1967
8,200	8,189	234.93	3.4635	5.1361
8,300	8,289	234.28	3.4135	5.0760
8,400	8,389	233.63	3.3642	5.0165
8,500	8,489	232.98	3.3154	4.9575
8,600	8,588	232.34	3.2672	4.8991
8,700	8,688	231.69	3.2196	4.8412
8,800	8,788	231.04	3.1725	4.7838
8,900	8,888	230.39	3.1260	4.7269
9,000	8,987	229.74	$3.0800 + 4$	$4.6706 - 1$
9,100	9,087	229.09	3.0346	4.6148
9,200	9,187	228.45	2.9898	4.5595
9,300	9,286	227.80	2.9455	4.5047
9,400	9,386	227.15	2.9017	4.4504
9,500	9,486	226.50	2.8584	4.3966
9,600	9,586	225.85	2.8157	4.3433
9,700	9,685	225.21	2.7735	4.2905
9,800	9,785	224.56	2.7318	4.2382
9,900	9,885	223.91	2.6906	4.1864

Altitude				
h_G, m	h, m	Temperature T, K	Pressure p, N/m^2	Density ρ, kg/m^3
10,000	9,984	223.26	2.6500 + 4	4.1351− 1
10,100	10,084	222.61	2.6098	4.0842
10,200	10,184	221.97	2.5701	4.0339
10,300	10,283	221.32	2.5309	3.9840
10,400	10,383	220.67	2.4922	3.9346
10,500	10,483	220.02	2.4540	3.8857
10,600	10,582	219.37	2.4163	3.8372
10,700	10,682	218.73	2.3790	3.7892
10,800	10,782	218.08	2.3422	3.7417
10,900	10,881	217.43	2.3059	3.6946
11,000	10,981	216.78	2.2700 + 4	3.6480 − 1
11,100	11,081	216.66	2.2346	3.5932
11,200	11,180	216.66	2.1997	3.5371
11,300	11,280	216.66	2.1654	3.4820
11,400	11,380	216.66	2.1317	3.4277
11,500	11,479	216.66	2.0985	3.3743
11,600	11,579	216.66	2.0657	3.3217
11,700	11,679	216.66	2.0335	3.2699
11,800	11,778	216.66	2.0018	3.2189
11,900	11,878	216.66	1.9706	3.1687
12,000	11,977	216.66	1.9399 + 4	3.1194 − 1
12,100	12,077	216.66	1.9097	3.0707
12,200	12,177	216.66	1.8799	3.0229
12,300	12,276	216.66	1.8506	2.9758
12,400	12,376	216.66	1.8218	2.9294
12,500	12,475	216.66	1.7934	2.8837
12,600	12,575	216.66	1.7654	2.8388
12,700	12,675	216.66	1.7379	2.7945
12,800	12,774	216.66	1.7108	2.7510
12,900	12,874	216.66	1.6842	2.7081
13,000	12,973	216.66	1.6579 + 4	2.6659 − 1
13,100	13,073	216.66	1.6321	2.6244
13,200	13,173	216.66	1.6067	2.5835
13,300	13,272	216.66	1.5816	2.5433
13,400	13,372	216.66	1.5570	2.5036
13,500	13,471	216.66	1.5327	2.4646
13,600	13,571	216.66	1.5089	2.4262
13,700	13,671	216.66	1.4854	2.3884
13,800	13,770	216.66	1.4622	2.3512
13,900	13,870	216.66	1.4394	2.3146
14,000	13,969	216.66	1.4170 + 4	2.2785 − 1
14,100	14,069	216.66	1.3950	2.2430
14,200	14,168	216.66	1.3732	2.2081
14,300	14,268	216.66	1.3518	2.1737
14,400	14,367	216.66	1.3308	2.1399
14,500	14,467	216.66	1.3101	2.1065
14,600	14,567	216.66	1.2896	2.0737
14,700	14,666	216.66	1.2696	2.0414
14,800	14,766	216.66	1.2498	2.0096
14,900	14,865	216.66	1.2303	1.9783

Altitude				
h_G, m	h, m	Temperature T, K	Pressure p, N/m^2	Density ρ, kg/m^3
15,000	14,965	216.66	$1.2112 + 4$	$1.9475 - 1$
15,100	15,064	216.66	1.1923	1.9172
15,200	15,164	216.66	1.1737	1.8874
15,300	15,263	216.66	1.1555	1.8580
15,400	15,363	216.66	1.1375	1.8290
15,500	15,462	216.66	1.1198	1.8006
15,600	15,562	216.66	1.1023	1.7725
15,700	15,661	216.66	1.0852	1.7449
15,800	15,761	216.66	1.0683	1.7178
15,900	15,860	216.66	1.0516	1.6910
16,000	15,960	216.66	$1.0353 + 4$	$1.6647 - 1$
16,100	16,059	216.66	1.0192	1.6388
16,200	16,159	216.66	1.0033	1.6133
16,300	16,258	216.66	$9.8767 + 3$	1.5882
16,400	16,358	216.66	9.7230	1.5634
16,500	16,457	216.66	9.5717	1.5391
16,600	16,557	216.66	9.4227	1.5151
16,700	16,656	216.66	9.2760	1.4916
16,800	16,756	216.66	9.1317	1.4683
16,900	16,855	216.66	8.9895	1.4455
17,000	16,955	216.66	$8.8496 + 3$	$1.4230 - 1$
17,100	17,054	216.66	8.7119	1.4009
17,200	17,154	216.66	8.5763	1.3791
17,300	17,253	216.66	8.4429	1.3576
17,400	17,353	216.66	8.3115	1.3365
17,500	17,452	216.66	8.1822	1.3157
17,600	17,551	216.66	8.0549	1.2952
17,700	17,651	216.66	7.9295	1.2751
17,800	17,750	216.66	7.8062	1.2552
17,900	17,850	216.66	7.6847	1.2357
18,000	17,949	216.66	$7.5652 + 3$	$1.2165 - 1$
18,100	18,049	216.66	7.4475	1.1975
18,200	18,148	216.66	7.3316	1.1789
18,300	18,247	216.66	7.2175	1.1606
18,400	18,347	216.66	7.1053	1.1425
18,500	18,446	216.66	6.9947	1.1247
18,600	18,546	216.66	6.8859	1.1072
18,700	18,645	216.66	6.7788	1.0900
18,800	18,745	216.66	6.6734	1.0731
18,900	18,844	216.66	6.5696	1.0564
19,000	18,943	216.66	$6.4674 + 3$	$1.0399 - 1$
19,100	19,043	216.66	6.3668	1.0238
19,200	19,142	216.66	6.2678	1.0079
19,300	19,242	216.66	6.1703	$9.9218 - 2$
19,400	19,341	216.66	6.0744	9.7675
19,500	19,440	216.66	5.9799	9.6156
19,600	19,540	216.66	5.8869	9.4661
19,700	19,639	216.66	5.7954	9.3189
19,800	19,739	216.66	5.7053	9.1740
19,900	19,838	216.66	5.6166	9.0313

Altitude				
h_G, m	h, m	Temperature T, K	Pressure p, N/m^2	Density ρ, kg/m^3
20,000	19,937	216.66	5.5293 + 3	8.8909 − 2
20,200	20,136	216.66	5.3587	8.6166
20,400	20,335	216.66	5.1933	8.3508
20,600	20,533	216.66	5.0331	8.0931
20,800	20,732	216.66	4.8779	7.8435
21,000	20,931	216.66	4.7274	7.6015
21,200	21,130	216.66	4.5816	7.3671
21,400	21,328	216.66	4.4403	7.1399
21,600	21,527	216.66	4.3034	6.9197
21,800	21,725	216.66	4.1706	6.7063
22,000	21,924	216.66	4.0420 + 3	6.4995 − 2
22,200	22,123	216.66	3.9174	6.2991
22,400	22,321	216.66	3.7966	6.1049
22,600	22,520	216.66	3.6796	5.9167
22,800	22,719	216.66	3.5661	5.7343
23,000	22,917	216.66	3.4562	5.5575
23,200	23,116	216.66	3.3497	5.3862
23,400	23,314	216.66	3.2464	5.2202
23,600	23,513	216.66	3.1464	5.0593
23,800	23,711	216.66	3.0494	4.9034
24,000	23,910	216.66	2.9554 + 3	4.7522 − 2
24,200	24,108	216.66	2.8644	4.6058
24,400	24,307	216.66	2.7761	4.4639
24,600	24,505	216.66	2.6906	4.3263
24,800	24,704	216.66	2.6077	4.1931
25,000	24,902	216.66	2.5273	4.0639
25,200	25,100	216.96	2.4495	3.9333
25,400	25,299	217.56	2.3742	3.8020
25,600	25,497	218.15	2.3015	3.6755
25,800	25,696	218.75	2.2312	3.5535
26,000	25,894	219.34	2.1632 + 3	3.4359 − 2
26,200	26,092	219.94	2.0975	3.3225
26,400	26,291	220.53	2.0339	3.2131
26,600	26,489	221.13	1.9725	3.1076
26,800	26,687	221.72	1.9130	3.0059
27,000	26,886	222.32	1.8555	2.9077
27,200	27,084	222.91	1.7999	2.8130
27,400	27,282	223.51	1.7461	2.7217
27,600	27,481	224.10	1.6940	2.6335
27,800	27,679	224.70	1.6437	2.5484
28,000	27,877	225.29	1.5949 + 3	2.4663 − 2
28,200	28,075	225.89	1.5477	2.3871
28,400	28,274	226.48	1.5021	2.3106
28,600	28,472	227.08	1.4579	2.2367
28,800	28,670	227.67	1.4151	2.1654
29,000	28,868	228.26	1.3737	2.0966
29,200	29,066	228.86	1.3336	2.0301
29,400	29,265	229.45	1.2948	1.9659
29,600	29,463	230.05	1.2572	1.9039
29,800	29,661	230.64	1.2208	1.8440

Altitude				
h_G, m	h, m	Temperature T, K	Pressure p, N/m^2	Density ρ, kg/m^3
30,000	29,859	231.24	1.1855 + 3	1.7861 − 2
30,200	30,057	231.83	1.1514	1.7302
30,400	30,255	232.43	1.1183	1.6762
30,600	30,453	233.02	1.0862	1.6240
30,800	30,651	233.61	1.0552	1.5734
31,000	30,850	234.21	1.0251	1.5278
31,200	31,048	234.80	9.9592 + 2	1.4777
31,400	31,246	235.40	9.6766	1.4321
31,600	31,444	235.99	9.4028	1.3881
31,800	31,642	236.59	9.1374	1.3455
32,000	31,840	237.18	8.8802 + 2	1.3044 − 2
32,200	32,038	237.77	8.6308	1.2646
32,400	32,236	238.78	8.3890	1.2261
32,600	32,434	238.96	8.1546	1.1889
32,800	32,632	239.55	7.9273	1.1529
33,000	32,830	240.15	7.7069	1.1180
33,200	33,028	240.74	7.4932	1.0844
33,400	33,225	214.34	7.2859	1.0518
33,600	33,423	241.93	7.0849	1.0202
33,800	33,621	242.52	6.8898	9.8972 − 3
34,000	33,819	243.12	6.7007 + 2	9.6020 − 3
34,200	34,017	243.71	6.5171	9.3162
34,400	34,215	244.30	6.3391	9.0396
34,600	34,413	244.90	6.1663	8.7720
34,800	34,611	245.49	5.9986	8.5128
35,000	34,808	246.09	5.8359	8.2620
35,200	35,006	246.68	5.6780	8.0191
35,400	35,204	247.27	5.5248	7.7839
35,600	35,402	247.87	5.3760	7.5562
35,800	35,600	248.46	5.2316	7.3357
36,000	35,797	249.05	5.0914 + 2	7.1221 − 3
36,200	35,995	249.65	4.9553	6.9152
36,400	36,193	250.24	4.8232	6.7149
36,600	36,390	250.83	4.6949	6.5208
36,800	36,588	251.42	4.5703	6.3328
37,000	36,786	252.02	4.4493	6.1506
37,200	36,984	252.61	4.3318	5.9741
37,400	37,181	253.20	4.2176	5.8030
37,600	37,379	253.80	4.1067	5.6373
37,800	37,577	254.39	3.9990	5.4767
38,000	37,774	254.98	3.8944 + 2	5.3210 − 3
38,200	37,972	255.58	3.7928	5.1701
38,400	38,169	256.17	3.6940	5.0238
38,600	38,367	256.76	3.5980	4.8820
38,800	38,565	257.35	3.5048	4.7445
39,000	38,762	257.95	3.4141	4.6112
39,200	38,960	258.54	3.3261	4.4819
39,400	39,157	259.13	3.2405	4.3566
39,600	39,355	259.72	3.1572	4.2350
39,800	39,552	260.32	3.0764	4.1171

Altitude				
h_G, m	h, m	Temperature T, K	Pressure p, N/m^2	Density ρ, kg/m^3
40,000	39,750	260.91	2.9977 + 2	4.0028 − 3
40,200	39,947	261.50	2.9213	3.8919
40,400	40,145	262.09	2.8470	3.7843
40,600	40,342	262.69	2.7747	3.6799
40,800	40,540	263.28	2.7044	3.5786
41,000	40,737	263.87	2.6361	3.4804
41,200	40,935	264.46	2.5696	3.3850
41,400	41,132	265.06	2.5050	3.2925
41,600	41,300	265.65	2.4421	3.2027
41,800	41,527	266.24	2.3810	3.1156
42,000	41,724	266.83	2.3215 + 2	3.0310 − 3
42,400	41,922	267.43	2.2636	2.9489
42,400	42,119	268.02	2.2073	2.8692
42,600	42,316	268.61	2.1525	2.7918
42,800	42,514	269.20	2.0992	2.7167
43,000	42,711	269.79	2.0474	2.6438
43,200	42,908	270.39	1.9969	2.5730
43,400	43,106	270.98	1.9478	2.5042
43,600	43,303	271.57	1.9000	2.4374
43,800	43,500	272.16	1.8535	2.3726
44,000	43,698	272.75	1.8082 + 2	2.3096 − 3
44,200	43,895	273.34	1.7641	2.2484
44,400	44,092	273.94	1.7212	2.1889
44,600	44,289	274.53	1.6794	2.1312
44,800	44,486	275.12	1.6387	2.0751
45,000	44,684	275.71	1.5991	2.0206
45,200	44,881	276.30	1.5606	1.9677
45,400	45,078	276.89	1.5230	1.9162
45,600	45,275	277.49	1.4865	1.8662
45,800	45,472	278.08	1.4508	1.8177
46,000	45,670	278.67	1.4162 + 2	1.7704 − 3
46,200	45,867	279.26	1.3824	1.7246
46,400	46,064	279.85	1.3495	1.6799
46,600	46,261	280.44	1.3174	1.6366
46,800	46,458	281.03	1.2862	1.5944
47,000	46,655	281.63	1.2558	1.5535
47,200	46,852	282.22	1.2261	1.5136
47,400	47,049	282.66	1.1973	1.4757
47,600	47,246	282.66	1.1691	1.4409
47,800	47,443	282.66	1.1416	1.4070
48,000	47,640	282.66	1.1147 + 2	1.3739 − 3
48,200	47,837	282.66	1.0885	1.3416
48,400	48,034	282.66	1.0629	1.3100
48,600	48,231	282.66	1.0379	1.2792
48,800	48,428	282.66	1.0135	1.2491
49,000	48,625	282.66	9.8961 + 1	1.2197
49,200	48,822	282.66	9.6633	1.1910
49,400	49,019	282.66	9.4360	1.1630
49,600	49,216	282.66	9.2141	1.1357
49,800	49,413	282.66	8.9974	1.1089

Altitude				
h_G, m	h, m	Temperature T, K	Pressure p, N/m^2	Density ρ, kg/m^3
50,000	49,610	282.66	$8.7858 + 1$	$1.0829 - 3$
50,500	50,102	282.66	8.2783	1.0203
51,000	50,594	282.66	7.8003	$9.6140 - 4$
51,500	51,086	282.66	7.3499	9.0589
52,000	51,578	282.66	6.9256	8.5360
52,500	52,070	282.66	6.5259	8.0433
53,000	52,562	282.66	6.1493	7.5791
53,500	53,053	282.42	5.7944	7.1478
54,000	53,545	280.21	5.4586	6.7867
54,500	54,037	277.99	5.1398	6.4412
55,000	54,528	275.78	$4.8373 + 1$	$6.1108 - 4$
55,500	55,020	273.57	4.5505	5.7949
56,000	55,511	271.36	4.2786	5.4931
56,500	56,002	269.15	4.0210	5.2047
57,000	56,493	266.94	3.7770	4.9293
57,500	56,985	264.73	3.5459	4.6664
58,000	57,476	262.52	3.3273	4.4156
58,500	57,967	260.31	3.1205	4.1763
59,000	58,457	258.10	2.9250	3.9482
59,500	58,948	255.89	2.7403	3.7307

APPENDIX E

Standard Atmosphere, English Engineering Units

Altitude				
h_G, ft	h, ft	Temperature T, °R	Pressure p, lb/ft^2	Density ρ, slugs/ft^3
−16,500	−16,513	577.58	3.6588 + 3	3.6905 − 3
−16,000	−16,012	575.79	3.6641	3.7074
−15,500	−15,512	574.00	3.6048	3.6587
−15,000	−15,011	572.22	3.5462	3.6105
−14,500	−14,510	570.43	3.4884	3.5628
−14,000	−14,009	568.65	3.4314	3.5155
−13,500	−13,509	566.86	3.3752	3.4688
−13,000	−13,008	565.08	3.3197	3.4225
−12,500	−12,507	563.29	3.2649	3.3768
−12,000	−12,007	561.51	3.2109	3.3314
−11,500	−11,506	559.72	3.1576 + 3	3.2866 − 3
−11,000	−11,006	557.94	3.1050	3.2422
−10,500	−10,505	556.15	3.0532	3.1983
−10,000	−10,005	554.37	3.0020	3.1548
−9,500	−9,504	552.58	2.9516	3.1118
−9,000	−9,004	550.80	2.9018	3.0693
−8,500	−8,503	549.01	2.8527	3.0272
−8,000	−8,003	547.23	2.8043	2.9855
−7,500	−7,503	545.44	2.7566	2.9443
−7,000	−7,002	543.66	2.7095	2.9035
−6,500	−6,502	541.88	2.6631 + 3	2.8632− 3
−6,000	−6,002	540.09	2.6174	2.8233
−5,500	−5,501	538.31	2.5722	2.7838
−5,000	−5,001	536.52	2.5277	2.7448
−4,500	−4,501	534.74	2.4839	2.7061
−4,000	−4,001	532.96	2.4406	2.6679
−3,500	−3,501	531.17	2.3980	2.6301
−3,000	−3,000	529.39	2.3560	2.5927

Altitude				
h_G, ft	h, ft	Temperature T, °R	Pressure p, lb/ft^2	Density ρ, slugs/ft^3
−2,500	−2,500	527.60	2.3146	2.5558
−2,000	−2,000	525.82	2.2737	2.5192
−1,500	−1,500	524.04	2.2335 + 3	2.4830 − 3
−1,000	−1,000	522.25	2.1938	2.4473
−500	−500	520.47	2.1547	2.4119
0	0	518.69	2.1162	2.3769
500	500	516.90	2.0783	2.3423
1,000	1,000	515.12	2.0409	2.3081
1,500	1,500	513.34	2.0040	2.2743
2,000	2,000	511.56	1.9677	2.2409
2,500	2,500	509.77	1.9319	2.2079
3,000	3,000	507.99	1.8967	2.1752
3,500	3,499	506.21	1.8619 + 3	2.1429− 3
4,000	3,999	504.43	1.8277	2.1110
4,500	4,499	502.64	1.7941	2.0794
5,000	4,999	500.86	1.7609	2.0482
5,500	5,499	499.08	1.7282	2.0174
6,000	5,998	497.30	1.6960	1.9869
6,500	6,498	495.52	1.6643	1.9567
7,000	6,998	493.73	1.6331	1.9270
7,500	7,497	491.95	1.6023	1.8975
8,000	7,997	490.17	1.5721	1.8685
8,500	8,497	488.39	1.5423 + 3	1.8397 − 3
9,000	8,996	486.61	1.5129	1.8113
9,500	9,496	484.82	1.4840	1.7833
10,000	9,995	483.04	1.4556	1.7556
10,500	10,495	481.26	1.4276	1.7282
11,000	10,994	479.48	1.4000	1.7011
11,500	11,494	477.70	1.3729	1.6744
12,000	11,993	475.92	1.3462	1.6480
12,500	12,493	474.14	1.3200	1.6219
13,000	12,992	472.36	1.2941	1.5961
13,500	13,491	470.58	1.2687 + 3	1.5707 − 3
14,000	13,991	468.80	1.2436	1.5455
14,500	14,490	467.01	1.2190	1.5207
15,000	14,989	465.23	1.1948	1.4962
15,500	15,488	463.45	1.1709	1.4719
16,000	15,988	461.67	1.1475	1.4480
16,500	16,487	459.89	1.1244	1.4244
17,000	16,986	458.11	1.1017	1.4011
17,500	17,485	456.33	1.0794	1.3781
18,000	17,984	454.55	1.0575	1.3553
18,500	18,484	452.77	1.0359 + 3	1.3329 − 3
19,000	18,983	450.99	1.0147	1.3107
19,500	19,482	449.21	9.9379 + 2	1.2889
20,000	19,981	447.43	9.7327	1.2673
20,500	20,480	445.65	9.5309	1.2459
21,000	20,979	443.87	9.3326	1.2249

Altitude				
h_G, ft	h, ft	Temperature T, °R	Pressure p, lb/ft^2	Density ρ, slugs/ft^3
21,500	21,478	442.09	9.1376	1.2041
22,000	21,977	440.32	8.9459	1.1836
22,500	22,476	438.54	8.7576	1.1634
23,000	22,975	436.76	8.5724	1.1435
23,500	23,474	434.98	8.3905 + 2	1.1238 − 3
24,000	23,972	433.20	8.2116	1.1043
24,500	24,471	431.42	8.0359	1.0852
25,000	24,970	429.64	7.8633	1.0663
25,500	25,469	427.86	7.6937	1.0476
26,000	25,968	426.08	7.5271	1.0292
26,500	26,466	424.30	7.3634	1.0110
27,000	26,965	422.53	7.2026	9.9311 − 4
27,500	27,464	420.75	7.0447	9.7544
28,000	27,962	418.97	6.8896	9.5801
28,500	28,461	417.19	6.7373 + 2	9.4082 − 4
29,000	28,960	415.41	6.5877	9.2387
29,500	29,458	413.63	6.4408	9.0716
30,000	29,957	411.86	6.2966	8.9068
30,500	30,455	410.08	6.1551	8.7443
31,000	30,954	408.30	6.0161	8.5841
31,500	31,452	406.52	5.8797	8.4261
32,000	31,951	404.75	5.7458	8.2704
32,500	32,449	402.97	5.6144	8.1169
33,000	32,948	401.19	5.4854	7.9656
33,500	33,446	399.41	5.3589 + 2	7.8165 − 4
34,000	33,945	397.64	5.2347	7.6696
34,500	34,443	395.86	5.1129	7.5247
35,000	34,941	394.08	4.9934	7.3820
35,500	35,440	392.30	4.8762	7.2413
36,000	35,938	390.53	4.7612	7.1028
36,500	36,436	389.99	4.6486	6.9443
37,000	36,934	389.99	4.5386	6.7800
37,500	37,433	389.99	4.4312	6.6196
38,000	37,931	389.99	4.3263	6.4629
38,500	38,429	389.99	4.2240 + 2	6.3100 − 4
39,000	38,927	389.99	4.1241	6.1608
39,500	39,425	389.99	4.0265	6.0150
40,000	39,923	389.99	3.9312	5.8727
40,500	40,422	389.99	3.8382	5.7338
41,000	40,920	389.99	3.7475	5.5982
41,500	41,418	389.99	3.6588	5.4658
42,000	41,916	389.99	3.5723	5.3365
42,500	42,414	389.99	3.4878	5.2103
43,000	42,912	389.99	3.4053	5.0871
43,500	43,409	389.99	3.3248 + 2	4.9668 − 4
44,000	43,907	389.99	3.2462	4.8493
44,500	44,405	389.99	3.1694	4.7346
45,000	44,903	389.99	3.0945	4.6227
45,500	45,401	389.99	3.0213	4.5134

Altitude				
h_G, ft	h, ft	Temperature T, °R	Pressure p, lb/ft^2	Density ρ, slugs/ft^3
46,000	45,899	389.99	2.9499	4.4067
46,500	46,397	389.99	2.8801	4.3025
47,000	46,894	389.99	2.8120	4.2008
47,500	47,392	389.99	2.7456	4.1015
48,000	47,890	389.99	2.6807	4.0045
48,500	48,387	389.99	2.2173 + 2	3.9099 − 4
49,000	48,885	389.99	2.5554	3.8175
49,500	49,383	389.99	2.4950	3.7272
50,000	49,880	389.99	2.4361	3.6391
50,500	50,378	389.99	2.3785	3.5531
51,000	50,876	389.99	2.3223	3.4692
51,500	51,373	389.99	2.2674	3.3872
52,000	51,871	389.99	2.2138	3.3072
52,500	52,368	389.99	2.1615	3.2290
53,000	52,866	389.99	2.1105	3.1527
53,500	53,363	389.99	2.0606 + 2	3.0782 − 4
54,000	53,861	389.99	2.0119	3.0055
54,500	54,358	389.99	1.9644	2.9345
55,000	54,855	389.99	1.9180	2.8652
55,500	55,353	389.99	1.8727	2.7975
56,000	55,850	389.99	1.8284	2.7314
56,500	56,347	389.99	1.7853	2.6669
57,000	56,845	389.99	1.7431	2.6039
57,500	57,342	389.99	1.7019	2.5424
58,000	57,839	389.99	1.6617	2.4824
58,500	58,336	389.99	1.6225 + 2	2.4238 − 4
59,000	58,834	389.99	1.5842	2.3665
59,500	59,331	389.99	1.5468	2.3107
60,000	59,828	389.99	1.5103	2.2561
60,500	60,325	389.99	1.4746	2.2028
61,000	60,822	389.99	1.4398	2.1508
61,500	61,319	389.99	1.4058	2.1001
62,000	61,816	389.99	1.3726	2.0505
62,500	62,313	389.99	1.3402	2.0021
63,000	62,810	389.99	1.3086	1.9548
63,500	63,307	389.99	1.2777 + 2	1.9087 − 4
64,000	63,804	389.99	1.2475	1.8636
64,500	64,301	389.99	1.2181	1.8196
65,000	64,798	389.99	1.1893	1.7767
65,500	65,295	389.99	1.1613	1.7348
66,000	65,792	389.99	1.1339	1.6938
66,500	66,289	389.99	1.1071	1.6539
67,000	66,785	389.99	1.0810	1.6148
67,500	67,282	389.99	1.0555	1.5767
68,000	67,779	389.99	1.0306	1.5395
68,500	68,276	389.99	1.0063 + 2	1.5032 − 4
69,000	68,772	389.99	9.8253 + 1	1.4678
69,500	69,269	389.99	9.5935	1.4331
70,000	69,766	389.99	9.3672	1.3993

Altitude				
h_G, ft	h, ft	Temperature T, °R	Pressure p, lb/ft^2	Density ρ, slugs/ft^3
70,500	70,262	389.99	9.1462	1.3663
71,000	70,759	389.99	8.9305	1.3341
71,500	74,256	389.99	8.7199	1.3026
72,000	71,752	389.99	8.5142	1.2719
72,500	72,249	389.99	8.3134	1.2419
73,000	72,745	389.99	8.1174	1.2126
73,500	73,242	389.99	7.9259 + 1	1.1840 − 4
74,000	73,738	389.99	7.7390	1.1561
74,500	74,235	389.99	7.5566	1.1288
75,000	74,731	389.99	7.3784	1.1022
75,500	75,228	389.99	7.2044	1.0762
76,000	75,724	389.99	7.0346	1.0509
76,500	76,220	389.99	6.8687	1.0261
77,000	76,717	389.99	6.7068	1.0019
77,500	77,213	389.99	6.5487	9.7829 − 5
78,000	77,709	389.99	6.3944	9.5523
78,500	78,206	389.99	6.2437 + 1	9.3271 − 5
79,000	78,702	389.99	6.0965	9.1073
79,500	79,198	389.99	5.9528	8.8927
80,000	79,694	389.99	5.8125	8.6831
80,500	80,190	389.99	5.6755	8.4785
81,000	80,687	389.99	5.5418	8.2787
81,500	81,183	389.99	5.4112	8.0836
82,000	81,679	389.99	5.2837	7.8931
82,500	82,175	390.24	5.1592	7.7022
83,000	82,671	391.06	5.0979	7.5053
83,500	83,167	391.87	4.9196 + 1	7.3139 − 5
84,000	83,663	392.69	4.8044	7.1277
84,500	84,159	393.51	4.6921	6.9467
85,000	84,655	394.32	4.5827	6.7706
85,500	85,151	395.14	4.4760	6.5994
86,000	85,647	395.96	4.3721	6.4328
86,500	86,143	396.77	4.2707	6.2708
87,000	86,639	397.59	4.1719	6.1132
87,500	87,134	398.40	4.0757	5.9598
88,000	87,630	399.22	3.9818	5.8106
88,500	88,126	400.04	3.8902 + 1	5.6655 − 5
89,000	88,622	400.85	3.8010	5.5243
89,500	89,118	401.67	3.7140	5.3868
90,000	89,613	402.48	3.6292	5.2531
90,500	90,109	403.30	3.5464	5.1230
91,000	90,605	404.12	3.4657	4.9963
91,500	91,100	404.93	3.3870	4.8730
92,000	91,596	405.75	3.3103	4.7530
92,500	92,092	406.56	3.2354	4.6362
93,000	92,587	407.38	3.1624	4.5525
93,500	93,083	408.19	3.0912 + 1	4.4118 − 5
94,000	93,578	409.01	3.0217	4.3041
94,500	94,074	409.83	2.9539	4.1992

Altitude				
h_G, ft	h, ft	Temperature T, °R	Pressure p, lb/ft^2	Density ρ, slugs/ft^3
95,000	94,569	410.64	2.8878	4.0970
95,500	95,065	411.46	2.8233	3.9976
96,000	95,560	412.27	2.7604	3.9007
96,500	96,056	413.09	2.6989	3.8064
97,000	96,551	413.90	2.6390	3.7145
97,500	97,046	414.72	2.5805	3.6251
98,000	97,542	415.53	2.5234	3.5379
98,500	98,037	416.35	2.4677 + 1	3.4530 − 5
99,000	98,532	417.16	2.4134	3.3704
99,500	99,028	417.98	2.3603	3.2898
100,000	99,523	418.79	2.3085	3.2114
100,500	100,018	419.61	2.2580	3.1350
101,000	100,513	420.42	2.2086	3.0605
101,500	101,008	421.24	2.1604	2.9879
102,000	101,504	422.05	2.1134	2.9172
102,500	101,999	422.87	2.0675	2.8484
103,000	102,494	423.68	2.0226	2.7812
103,500	102,989	424.50	1.9789 + 1	2.7158 − 5
104,000	103,484	425.31	1.9361	2.6520
104,500	103,979	426.13	1.8944	2.5899
105,000	104,474	426.94	1.8536	2.5293
106,000	105,464	428.57	1.7749	2.4128
107,000	106,454	430.20	1.6999	2.3050
108,000	107,444	431.83	1.6282	2.1967
109,000	108,433	433.46	1.5599	2.0966
110,000	109,423	435.09	1.4947	2.0014
111,000	110,412	436.72	1.4324	1.9109
112,000	111,402	438.35	1.3730 + 1	1.8247 − 5
113,000	112,391	439.97	1.3162	1.7428
114,000	113,380	441.60	1.2620	1.6649
115,000	114,369	443.23	1.2102	1.5907
116,000	115,358	444.86	1.1607	1.5201
117,000	116,347	446.49	1.1134	1.4528
118,000	117,336	448.11	1.0682	1.3888
119,000	118,325	449.74	1.0250	1.3278
120,000	119,313	451.37	9.8372 + 0	1.2697
121,000	120,302	453.00	9.4422	1.2143
122,000	121,290	454.62	9.0645 + 0	1.1616 − 5
123,000	122,279	456.25	8.7032	1.1113
124,000	123,267	457.88	8.3575	1.0634
125,000	124,255	459.50	8.0267	1.0177
126,000	125,243	461.13	7.7102	9.7410 − 6
127,000	126,231	462.75	7.4072	9.3253
128,000	127,219	464.38	7.1172	8.9288
129,000	128,207	466.01	6.8395	8.5505
130,000	129,195	467.63	6.5735	8.1894
131,000	130,182	469.26	6.3188	7.8449
132,000	131,170	470.88	6.0748 + 0	7.5159 − 6
133,000	132,157	472.51	5.8411	7.2019

Altitude				
h_G, ft	h, ft	Temperature T, °R	Pressure p, lb/ft^2	Density ρ, slugs/ft^3
134,000	133,145	474.13	5.6171	6.9020
135,000	134,132	475.76	5.4025	6.6156
136,000	135,119	477.38	5.1967	6.3420
137,000	136,106	479.01	4.9995	6.0806
138,000	137,093	480.63	4.8104	5.8309
139,000	138,080	482.26	4.6291	5.5922
140,000	139,066	483.88	4.4552	5.3640
141,000	140,053	485.50	4.2884	5.1460
142,000	141,040	487.13	$4.1284 + 0$	$4.9374 - 6$
143,000	142,026	488.75	3.9749	4.7380
144,000	143,013	490.38	3.8276	4.5473
145,000	143,999	492.00	3.6862	4.3649
146,000	144,985	493.62	3.5505	4.1904
147,000	145,971	495.24	3.4202	4.0234
148,000	146,957	496.87	3.2951	3.8636
149,000	147,943	498.49	3.1750	3.7106
150,000	148,929	500.11	3.0597	3.5642
151,000	149,915	501.74	2.9489	3.4241
152,000	150,900	503.36	$2.8424 + 0$	$3.2898 - 6$
153,000	151,886	504.98	2.7402	3.1613
154,000	152,871	506.60	2.6419	3.0382
155,000	153,856	508.22	2.5475	2.9202
156,000	154,842	508.79	2.4566	2.8130
157,000	155,827	508.79	2.3691	2.7127
158,000	156,812	508.79	2.2846	2.6160
159,000	157,797	508.79	2.2032	2.5228
160,000	158,782	508.79	2.1247	2.4329
161,000	159,797	508.79	2.0490	2.3462

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